



Megapack 2 XL Design and Installation Manual

Revision 1.3

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IMPORTANT SAFETY INFORMATION

SAVE THESE INSTRUCTIONS

THIS MANUAL CONTAINS IMPORTANT INFORMATION THAT MUST BE READ, UNDERSTOOD, AND FOLLOWED DURING DESIGN AND INSTALLATION OF THE MEGEPACK SYSTEM.

SYMBOLS

This manual uses the following symbols to highlight important information:

 **DANGER:** Indicates a hazardous situation which, if not avoided, could result in severe injury or death.

 **WARNING:** Indicates a hazardous situation which, if not avoided, could result in injury.

 **CAUTION:** Indicates a hazardous situation which, if not avoided, could result in minor injury or damage to the equipment.

 **NOTE:** Indicates an important step or tip that leads to best results but is not safety- or damage-related.

PRODUCT WARNINGS

 **DANGER:** Risk of electrical shock. Multiple energy sources terminate inside this equipment. Always check with a properly rated voltmeter that there is no voltage on the bus before working.

 **DANGER:** Lock out all sources of AC power at the source before servicing Megapack or removing the AC circuit breaker or bus bar access panels in the Customer Interface Bay.

 **DANGER:** During installation, all equipment must be de-energized.

 **DANGER:** Hazardous voltage can cause severe injury or death.

 **DANGER:** Electric shock could occur when touching live components.

 **DANGER:** Shutting off power to Megapack does not de-energize the battery, and thus a shock hazard may still be present.

 **DANGER:** Megapack, even in a normally discharged condition, is likely to contain substantial electrical charge and can cause injury or death if mishandled.

 **DANGER:** The battery used in this device may present a risk of fire or chemical burn if mistreated. Do not disassemble, operate above 50°C (122°F), or incinerate.

 **WARNING:** To reduce the risk of injury, read all instructions.

 **WARNING:** In order to operate, Megapack requires a solidly grounded circuit such that the line-to-ground nominal voltage does not exceed 300 V.



WARNING: All installations must conform to the laws, regulations, codes, and standards applicable in the jurisdiction of installation, such as National Electric Code (NEC) ANSI/NFPA 70 or the Canadian Electrical Code CSA C22.1.

WARNING: Do not install adjacent to or expose to any external heat source.

WARNING: Personal Protective Equipment (PPE) is required when working inside the Megapack enclosures. Service personnel must wear safety glasses and gloves with a minimum voltage rating of 1500 V DC, Class 0 per ASTM D120 and IEC EN60903 standards.

WARNING: Megapack has no user-serviceable parts. All service must be performed by Tesla-approved service personnel.

WARNING: The installation instructions are for use by qualified personnel only. To reduce the risk of electric shock, do not perform any servicing other than that specified in the operating instructions unless you are qualified to do so.

WARNING: Only use this equipment as specified by Tesla. If the equipment is used in a manner that is not specified by Tesla, the protection provided by the equipment might be impaired.

WARNING: For continued protection against risk of fire, use only replacement fuses of the same type and rating as the original fuse. Fuses must only be replaced by trained personnel.

CAUTION: Megapack input and output circuits are isolated from the enclosure. System grounding, when required by the National Electric Code, ANSI/NFPA 70, or other relevant standard, is the responsibility of the installer.

CAUTION: Do not paint any part of Megapack other than external white metal surfaces, using only Tesla-provided touch-up paint. Internal or external components such as exterior cabinets or grilles should not be painted.

CAUTION: Do not use cleaning solvents to clean Megapack or expose the system to flammable or harsh chemicals or vapors.

CAUTION: Do not use fluids, parts, or accessories other than those specified in Tesla manuals, including use of non-genuine Tesla parts or accessories, or parts or accessories not purchased directly from Tesla or a Tesla-approved party.



CAUTION: Hearing damage could occur if not wearing hearing protection while Megapack is in operation.

Refer to the *Lithium-Ion Battery Emergency Response Guide* for detailed hazard information specific to the lithium-ion battery. The guide also provides hazard information for a single Tesla Megapack unit.

Voltage Classification

This section defines voltage classification as used in this document.

The table below represents Tesla's standard voltage ranges. The defined ranges, comparable to global codes and standards, help categorize potential electrical hazards where applicable.



NOTE: Any voltage referred to in this document is low voltage unless otherwise specified.

Table 1. Voltage Classifications

Classification	Acronym	Alternating Current (AC) Range	Direct Current (DC) Range
Ultra-Low Voltage	ULV	0-48 V	0-48 V
Low Voltage	LV	50-1,000 V	50-2,000 V
Medium Voltage	MV	1,000-35,000 V (1kV-35kV)	2,000-35,000 V (2kV-35kV)
Sub-Transmission Medium Voltage	STMV	35,000-69,000 V (35kV-69kV)	35,000-69,000 V (35kV-69kV)
High Voltage	HV	Above 69,000 V (>69kV)	Above 69,000 V (>69kV)



Reference Documents

Visit the Tesla Partner Portal at <https://partners.tesla.com/> to find reference material referred to within this guide, including:

- Megapack 2 XL Installation Guide (abridged, mobile-friendly installation instructions) - <https://partners.tesla.com/home/en-us/content/view/megapack-2-xl-installation/>
- Megapack 2 XL Drawings - https://partners.tesla.com/home/en-us/content/download/megapack_2_xl_drawings.zip
- Megapack 2 XL Specification - https://partners.tesla.com/home/en-us/content/download/megapack_2_xl_specification.pdf
- Megapack 2 XL Transportation and Storage Guidelines - https://partners.tesla.com/home/en-us/content/download/megapack_2_xl_transportation_and_storage_guidelines.pdf
- Megapack 2 XL Compliance Packet - https://partners.tesla.com/home/en-US/content/download/Megapack_2_XL_Compliance_Packet.zip
- Megapack 2 XL Compliance Summary - https://partners.tesla.com/home/en-US/content/download/Megapack_2_XL_Compliance_Summary.pdf
- Megapack 2 Controls and Communications Manual - https://partners.tesla.com/home/en-us/content/download/megapack_2_controls_and_communications_manual.pdf
- SCADA Design Manual - https://partners.tesla.com/home/en-US/content/download/SCADA_Design_Manual.pdf
- Microgrid Controller Owner's Manual - https://partners.tesla.com/home/en-us/content/download/microgridcontroller_manual_owners.pdf
- Lithium-Ion Battery Emergency Response Guide - <https://www.tesla.com/firstresponders>
- Megapack Deployment Overview - https://partners.tesla.com/home/en-us/content/download/megapack_deployment_overview.pdf
- Megapack Deployment Checklists - https://partners.tesla.com/home/en-us/content/download/megapack_deployment_checklists.zip
- Megapack Design Review Checklist - https://partners.tesla.com/home/en-US/content/download/Megapack_Design_Review_Checklist.pdf
- Megapack 2 Operation and Maintenance Manual - https://partners.tesla.com/home/en-US/content/download/Megapack_2_Operation_and_Maintenance_Manual.pdf
- Tesla Energy Operations Contact List and Response Times - https://partners.tesla.com/home/en-US/content/download/Tesla_Energy_Operations_Contact_List_and_Response_Times.pdf
- Application Note: Megapack Integrated Wireway - https://partners.tesla.com/home/en-us/content/download/megapack_integrated_wireway_appnote.pdf
- Application Note: Grounding Conductor Recommendations for Megapack Applications - https://partners.tesla.com/home/en-us/content/download/megapack_grounding_conductor_recommendations_appnote.pdf
- Application Note: Electrical Protection Architecture - https://partners.tesla.com/home/en-us/content/download/electrical_protection_architecture_application_note.pdf
- Application Note: Onsite Water Supply Sizing For Exposure Protection - https://partners.tesla.com/home/en-US/content/download/Onsite_Water_Supply_Sizing_For_Exposure_Protection_Application_Note.pdf
- Application Note: On-Site Maintenance Infrastructure Requirements - https://partners.tesla.com/home/en-us/content/download/on-site_maintenance_infrastructure_requirements_appnote.pdf



1 Megapack Overview

1.1 Introduction

Tesla Megapack 2 XL (Megapack) is a modular, fully integrated, AC-coupled industrial battery energy storage system (BESS). This document provides the site designer the relevant information during the pre-construction phase and also provides the installation contractor the necessary details to install a Tesla Megapack system.

 **CAUTION:** Any deviation from what is specified in this manual must be submitted to Tesla in writing in advance for approval.

1.2 Roles and Responsibilities

 **NOTE:** Responsibilities listed are the minimum required by Tesla. Additional responsibilities may be required.

1.2.1 Site Designer

The site designer ("designer") is responsible for all technical decisions for the project. This includes performing tasks such as:

- Civil, mechanical, and electrical design, including:
 - Consulting site geotechnical report for the geotechnical design requirements
- Coordinating controls and operation modes as required, including:
 - Consulting with the SCADA engineer
 - Refer to the *SCADA Design Manual* for details
- Creating project-specific documentation and drawings as required (such as site layout, single-line, and network architecture)
- Coordinating and obtaining permits as required
- Creating site movement and crane delivery plan

1.2.2 Installation Contractor

The installation contractor ("installer") is responsible for completing the mechanical and electrical installation of the Megapack system according to appropriate local fire, electrical, and building code requirements, in addition to meeting all criteria provided in the *Megapack Deployment Checklists*.

As part of mechanical completion, the installer is responsible for coordinating the logistics with Tesla from the factory to the site, and directly responsible for moving Megapack between the truck trailer and final foundation location. Logistics-specific items may include:

- Site-specific movement plan
- Arranging for craning, including providing a site-specific crane lifting plan
- Coordination with responsible third party transporters such as truck drivers
- Providing all required documentation to responsible parties



The installer must provide the following for Tesla approval prior to completing the *Megapack Deployment Checklists*:

- Arc flash calculations showing the available energy at the Megapack AC bus
- Isolating methods plan or switching scheme identifying the order of opening and who is responsible for each step

The *Megapack Deployment Checklists* list the criteria that must be inspected and completed on-site for Tesla to consider the site completed and ready for commissioning. The list includes items such as:

- Physical installation such as proper anchoring, door swing, access, etc. of all equipment
- Site and foundation grading, drainage, and access
- Electrical termination checks for polarity and torque for all equipment
- Harness termination checks
- Ethernet cable checks
- Insulation resistance “megger” testing for all wires
- Megapack phase checking
- Meter wiring and CT polarity check
- Recording site information (serial numbers, unit and meter locations, inspection date, etc.) and emailing a scanned copy to Tesla
- Megapack and Tesla Site Controller start-up

 **DANGER:** While performing Megapack installation, always follow all [IMPORTANT SAFETY INFORMATION on page 3](#) and ensure personal safety at all times per local and national regulations. The installer is responsible for providing required personal protective equipment (PPE), including such items as safety glasses, hard hats, appropriate boots, and appropriate gloves (cut and electrical).

1.3 Megapack System Components

A Megapack system consists of the following components:

- One or more Megapacks ([Megapack on page 9](#))
 - Megapack 2 XL – Part number 1848844-XX-Y¹
- The Tesla Site Controller ([Tesla Site Controller on page 13](#))
 - Standard Tesla Site Controller Enclosure (for sites up to 18 MW) – Part number 1471208-XX-Y¹
 - Large Tesla Site Controller (for sites greater than 18 MW) – Part number: Either 1459154-XX-Y or 1700130-XX-Y¹

¹Where X is a number between 0 and 9, and Y is a letter.

A Megapack site may additionally consist of other equipment including what is outlined in [Overview of Site Components on page 20](#).



Figure 1. Example Megapack Site



1.4 Megapack

Megapack is bi-directional, supporting charge and discharge. It converts power for storage in rechargeable lithium-ion battery packs (battery modules) and is designed in a modular fashion in order to support a range of AC power.

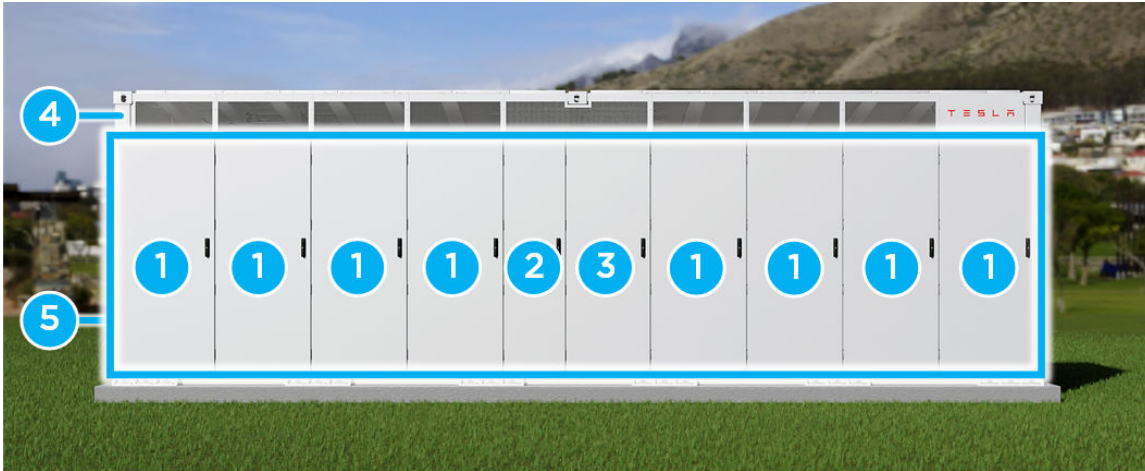
 **NOTE:** Individual Megapack specifications are detailed on the product label (see [Megapack Labels on page 15](#)).

Each Megapack contains up to 24 battery modules with inverters, a thermal bay and associated thermal roof components, an AC circuit breaker, and a set of customer interface terminals and internal controls circuit boards. An external auxiliary power supply is not required for Megapack; Megapack pulls auxiliary power for the control power and thermal management from the local AC.

Depending on the system configuration (2-hour or 4-hour), each Megapack can be configured with different quantities of battery modules which, together with the site's grid voltage, determine Megapack's nominal power rating.

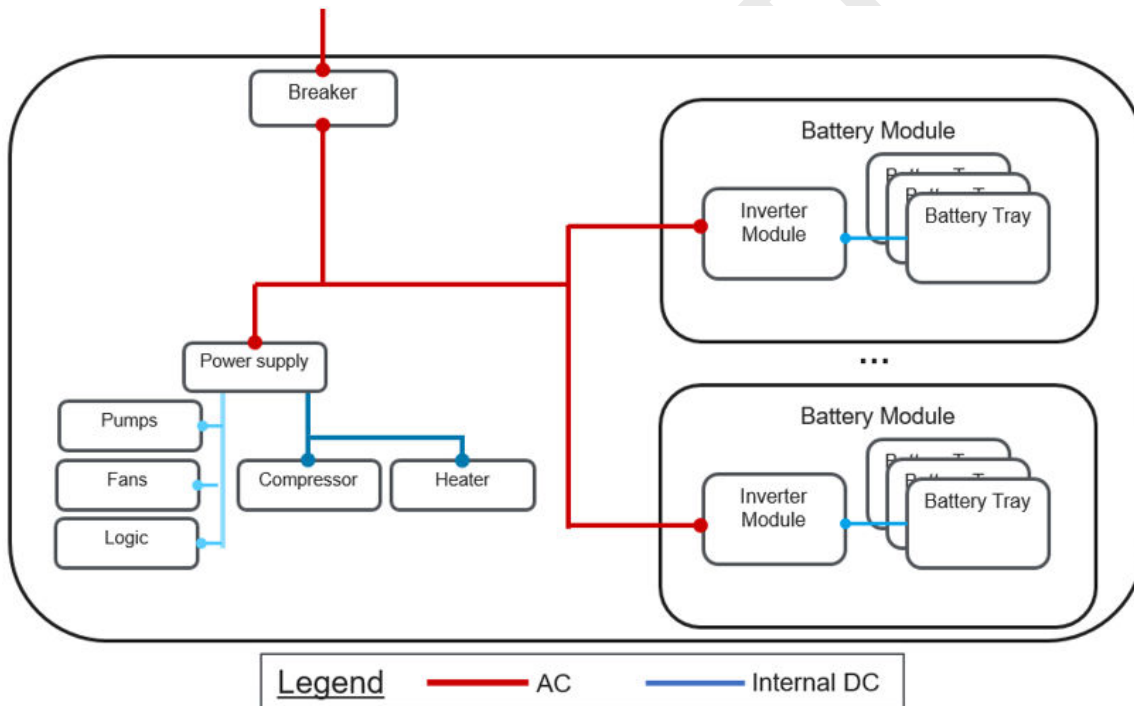


Figure 2. Megapack Overview



1. Battery module bays ([Battery Modules on page 10](#))
2. Thermal cabinet ([Thermal Components on page 11](#))
3. Customer Interface Bay ([Customer Interface Bay on page 12](#))
4. Thermal roof ([Thermal Components on page 11](#))
5. IP66 enclosure ([The Megapack IP66 Enclosure on page 12](#))

Figure 3. Megapack Internal Architecture



1.4.1 Battery Modules

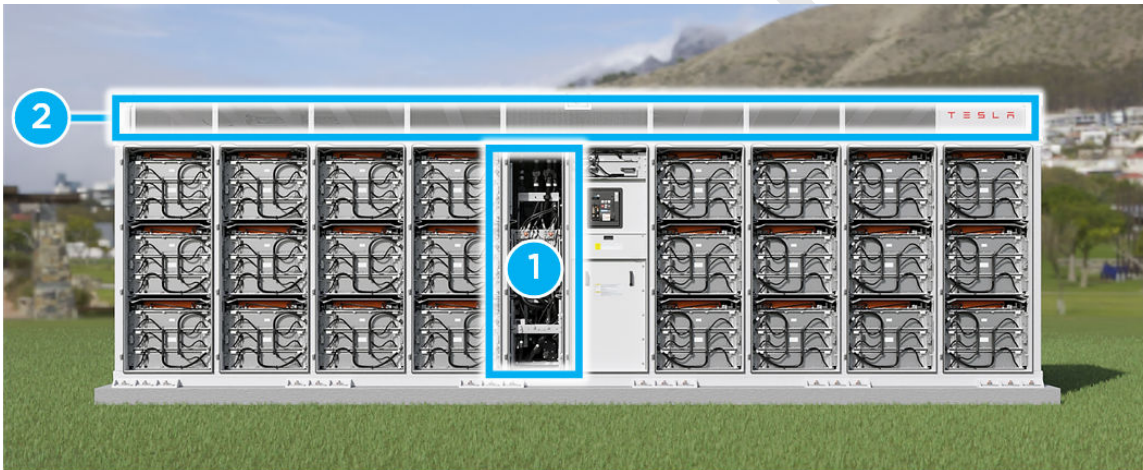
Battery modules are factory-installed into Megapack bays and contain prismatic lithium-ion battery cells, the smallest non-divisible energy storage components of the Megapack. A battery module in turn is the smallest field-replaceable battery unit. One Megapack contains up to 24 battery modules in the 2-hour or 4-hour system.

**Figure 4. Battery Module**

Each battery module includes an integrated inverter module for DC/AC power conversion. Battery modules are connected in parallel, each with an AC and communications output connection. The modules do not require any field assembly or adjustments and may only be replaced by Tesla Service or approved third-party service provider.

1.4.2 Thermal Components

Bay 5 in Megapack is called the thermal cabinet because it houses Megapack's thermal management system, which provides active cooling and heating to the internal Megapack components. The thermal bay is accessible for servicing from ground level.

Figure 5. Thermal Components

1. Thermal cabinet
2. Thermal roof

The thermal roof, or top cabinet of the enclosure, provides ventilation airspace and contains fans and radiators that cool the ethylene glycol-water coolant solution. The thermal roof is accessible for servicing with the help of a ladder or mechanical lift.

See [Thermal System on page 34](#) for more information.



WARNING: See [Loading and Unloading on page 68](#) for important precautions about working on the roof.



WARNING: The thermal management section is locked during operation. Do not open this cabinet while fans are in use, to avoid hazard from moving parts.



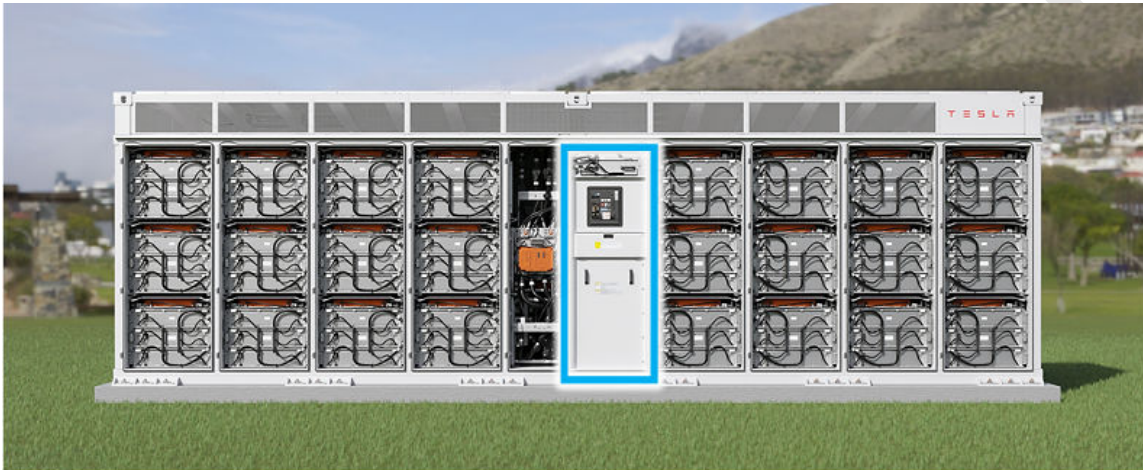
NOTE: Megapack includes an enable circuit as a safety feature. Opening the door to the thermal bay shuts down the Megapack. The thermal components located on the roof should not be serviced during operation.

WARNING: The thermal management sections are locked during operation. The thermal bay and thermal roof are only serviceable when the system is not operating. Do not open the thermal enclosures while the unit is operational to avoid hazard from moving parts.

1.4.3 Customer Interface Bay

The Customer Interface Bay is a single bay that includes all the external connections needed for initial installation (the AC bus bars and customer I/O area) and the AC circuit breaker.

Figure 6. Customer Interface Bay



For design information, see [Customer Interface on page 60](#). For installation tasks, see [Installing Customer Connections on page 82](#).

1.4.4 The Megapack IP66 Enclosure

Megapack's enclosure is rated according to the IP Code (Ingress Protection Code) to IP66. This means it affords a high protection against particle and water ingress in order to protect the enclosure contents.

This high protection rating must be maintained at all times. In particular, special precautions must be observed while installing or servicing Megapack in order to prevent particles or water from entering the enclosure.

At a minimum, this means the following conditions must be observed:

- **Megapack must remain sealed:** While actively working on Megapack, ensure that all Megapack doors are closed and conduit or wireway openings are sealed **before you leave the site each day** in order to keep the Megapack interior dry and clean.
 - Refer to [Sealing the Wireway Openings on page 77](#) for recommendations on maintaining this required seal.
- **Megapack's interior must remain dry:** Do not perform any work on Megapack that requires opening its doors when there is a possibility of moisture (from precipitation or excess humidity) entering the enclosure.
- **Megapack must remain protected from dust and debris:** At all times when doors are open, ensure that dust or debris of any kind does not enter the enclosure. For example, do not operate a leaf blower near an open door and ensure that doors are closed when wind may blow debris.



Refer to [Accessing the Megapack Enclosure on page 70](#) for more information on critical considerations about properly accessing Megapack.

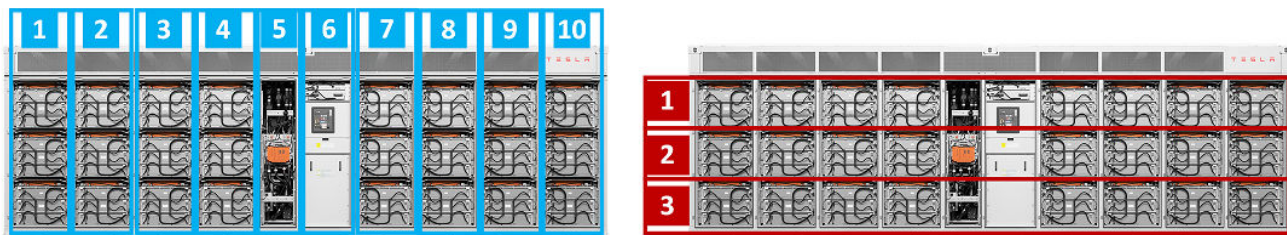
WARNING: Failure to properly seal the Megapack openings and enclosure may violate the integrity of the IP66 enclosure and allow moisture, particles, rodents or other objects to enter the enclosure and cause significant damage to equipment.

CAUTION: Some foaming agents such as plumbing foam can degrade conductor insulation and PVC conduit pipes. Ensure all sealants are compatible with site materials.

1.4.5 Megapack Numbering Key

A numbering key can help identify locations of Megapack enclosure components to assist you when referring to them during installation or service. Below are the numbering keys for **bay** (vertical tower) and **shelf** (horizontal row) locations:

Figure 7. Megapack Bay Numbering (Left) and Shelf Numbering (Right)



For example, as depicted in [Megapack Overview on page 10](#):

- The thermal bay is located in bay 5
- The Customer Interface Bay is bay 6
- Battery modules are located on shelves 1-3 in bays 1-4 and 7-10.

1.5 Tesla Site Controller

The Tesla Site Controller is the single point of interface for the utility, network operator, or customer SCADA systems to control and monitor the entire energy storage site. It hosts the control algorithm that manages the charge and discharge functions of the battery system units, aggregating real-time information and using the information to optimize the commands sent to each individual battery unit.

The battery system communicates with the Tesla Site Controller over a private TCP network. Customer systems communicate with the Tesla Site Controller via Modbus, DNP3, or REST API interfaces.

One Tesla Site Controller is typically required for each point of interconnection, however Tesla may choose to provide a second redundant Tesla Site Controller. Tesla does not include additional networking equipment that may be required for the system to operate.

The Tesla Site Controller has three network interfaces:

- LAN 2 RJ45/Ethernet port, which connects to the Tesla Network
- LAN 1 RJ45/Ethernet port, which connects to the Customer Network and can be configured for WAN access
- Integrated cellular modem, which provides cellular access for Tesla's remote connection

There are two physical variations of the Tesla Site Controller:



- **Standard Tesla Site Controller** - used in sites up to 18 MW. Delivered in an enclosure called the Standard Tesla Site Controller Enclosure, which may include one or two Standard Tesla Site Controllers.
- **Large Tesla Site Controller** - for sites larger than 18 MW. Delivered as the controller only.

Figure 8. Standard Tesla Site Controller in the Standard Tesla Site Controller Enclosure

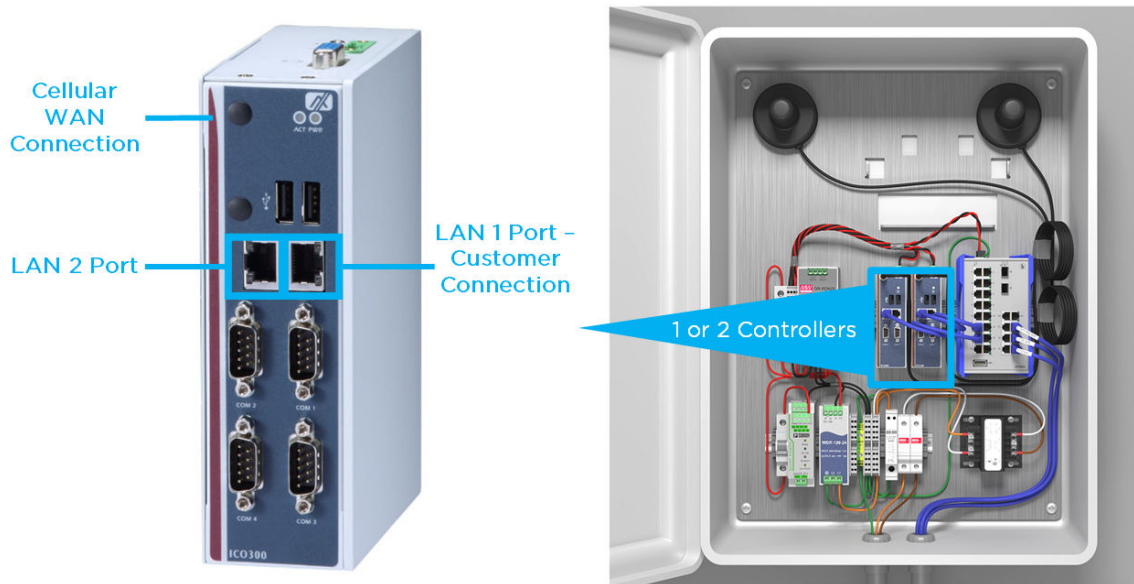
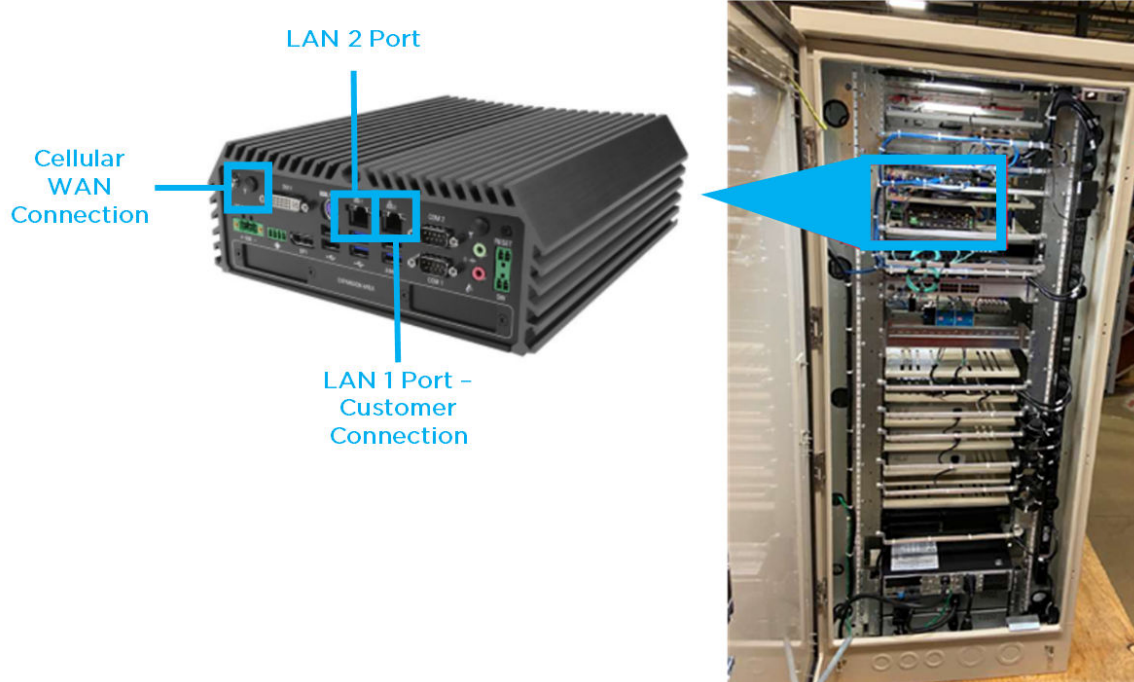


Figure 9. Large Tesla Site Controller mounted inside SCADA Rack Enclosure



NOTE: DB9 and USB ports are disabled and not used for any communication on the devices.

For more information:

- See the *SCADA Design Manual* for additional, required information when designing an energy storage site larger than 18 MW.



- See the relevant *Controls and Communications Manual* for complete instructions on how to interface with the Tesla Site Controller.
- See [Installing the Tesla Site Controller on page 87](#) for complete installation information.
- See [Standard Tesla Site Controller Enclosure Wiring on page 53](#) and [Network and Internet Considerations on page 29](#) for design considerations.

1.6 Labels and Serial Numbers

1.6.1 Megapack Labels

Megapack labels provide the specifications and product reference for each individual Megapack.

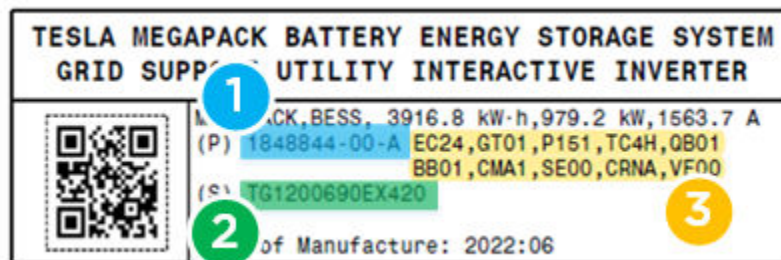
 **NOTE:** The labels pictured here are example labels. Refer to the actual labels for applicable ratings and specifications per option codes.

 **NOTE:** For more information about option codes, refer to the *Megapack 2 XL Option Codes Quick Reference Guide*

The **Megapack nameplate label** is a small metal label laser-etched onto the outside of the Customer Interface Bay door above the handle. It contains:

- Description of the product, with kWh, kW, and amp ratings
- Part number followed by option codes for battery and inverter ratings only
- Serial number
- Date of manufacture

Figure 10. Megapack Nameplate Label - Example



1. Part number
2. Serial number
3. Option codes

 **NOTE:** The nameplate label lists selected option codes only. For the complete list of option codes, refer to the main Megapack label.

The **main Megapack label** is a large sticker affixed to the inside of the Customer Interface Bay door, and contains information such as:

- AC input/output specifications including relevant power and energy ratings
- Part number followed by complete list of option codes
- Serial number
- Date of manufacture
- Weight (mass)



- Other detailed product specifics and compliance marks

Figure 11. Main Megapack Label - Example

TESLA MEGAPACK BATTERY ENERGY STORAGE SYSTEM GRID-SIDE UTILITY INTERACTIVE INVERTER	
1 2 3	
PACK, BESS, 3916.8 kW·h, 979.2 kW, 1563.7 A (P) 1848844-00-A EC24, GT01, P151, TC4H, QB01 (S) 161200690EX420 BB01, CMA1, SE00, CRNA, VF00 Manufacture: 2022:06	
Protective Class	Class I
Enclosure Type	Type 3R/IP66
Ingress Protection (Enclosure/Thermal Roof)	IP66 / IP2X
Operating Temperature Range	-30°C to +50°C
Inverter Topology	Non-Isolated
Nominal Battery Energy (AC) at 979.2 kW	3916.8 kW·h
Battery Type	Li-Ion
Mass	36346 kg
AC Input/Output	
Nominal Voltage (3-Phase)	480 VAC
Maximum Continuous Power	1512 kVA
Voltage Range	422.4-552 V
Maximum Continuous Current	1818.7 A
Frequency	45-66 Hz
Power Factor Range	-1 to 1
Maximum Output Fault Current	2182.4 A
Maximum Utility Backfeed Current	4200 A
Maximum Supply Fault Current	85 kA AC
Refrigerant	R-134a 1.5 kg
Refrigerant Oil	POE 0.22 L
Coolant Volume	EGW 360 L
Certified to:	CAN/CSA STD C22.2 No. 107.1
Conforms to:	ANSI/CAN/UL 1973, ANSI/CAN/UL 9540 UL STD. 1741
CAUTION: RISK OF ELECTRIC SHOCK. WARNING: POWER FED FROM MORE THAN ONE SOURCE. DISCONNECT ALL SOURCES OF SUPPLY BEFORE SERVICING. ENERGY STORED IN CAPACITOR. DO NOT REMOVE COVER UNTIL 5 MINUTES AFTER DISCONNECTING THE EQUIPMENT. REFER SERVICING TO QUALIFIED SERVICE PERSONNEL.	
ATTENTION: RISQUE DE CHOC ÉLECTRIQUE. AVERTISSEMENT: L'ALIMENTATION PROVIENT DE PLUS D'UNE SOURCE. COUPER TOUTES LES SOURCES D'ALIMENTATION AVANT UN SERVICE ÉNERGIE STOCKÉE DANS DES CONDENSATEURS. ATTENDRE 5 MINUTES AVANT DE RETIRER LE COUVERCLE APRÈS AVOIR COUPÉ TOUTES LES SOURCES D'ALIMENTATION. CONFIER L'ENTRETIEN À DU PERSONNEL QUALIFIÉ.	
5 Minutes	
	Tesla, Inc. 700 D'Arcy Pkwy Lathrop, CA 95330, USA Tel: 1(877)798-3762 www.tesla.com

1. Part number
2. Serial number
3. Option codes

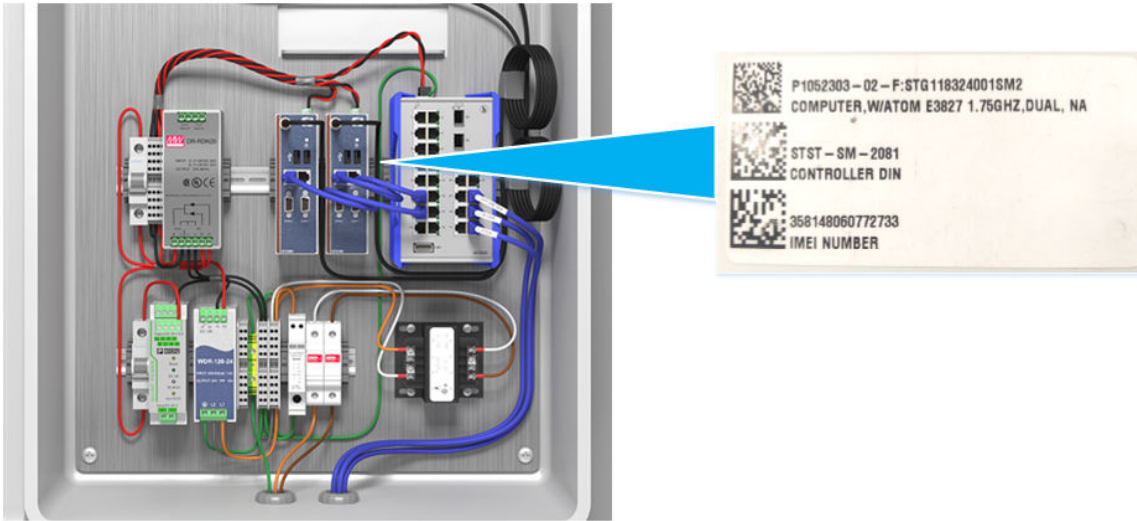
1.6.2 Tesla Site Controller Serial Number

For both variants of the Tesla Site Controller, look on the controller computer or computers to identify the serial number.

The Standard Tesla Site Controller serial number and VIN are located on a label on the side of the Standard Tesla Site Controller computer in the Standard Tesla Site Controller Enclosure:



Figure 12. Standard Tesla Site Controller Serial Number (Example) - Location



The VIN is typically labeled *CONTROLLER DIN* and takes the format *STST-SM-####*.



NOTE: There may be two Standard Tesla Site Controllers in the Standard Tesla Site Controller Enclosure. Each has its own identifiers.

The Large Tesla Site Controller serial number is located on the side of the Large Tesla Site Controller computer.

1.7 Emergency Response Guide

The *Lithium-Ion Battery Emergency Response Guide* (ERG) provides an overview of the product materials, handling and use precautions, hazards, emergency response procedures, and storage and transportation instructions. The document may be used in place of traditional material safety data sheets (MSDS) or safety data sheets (SDS) commonly associated with the health and safety of a chemical product.

Tesla BESSs, as described in the ERG, meet the OSHA definition of “articles” and are therefore exempt from requiring SDS format. Refrigerant and coolant SDSs are however also available on the Tesla Partner Portal.



NOTE: Tesla BESS coolant is not a regulated substance according to the United States Department of Transportation (USDOT). Refer to the specific safety data sheet (SDS) for battery coolant.



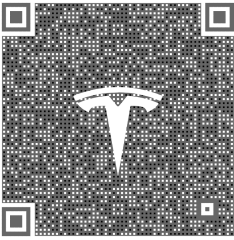
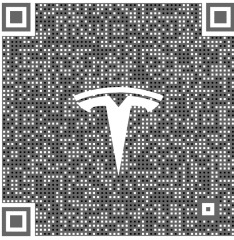
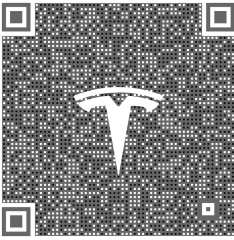
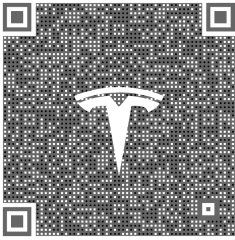
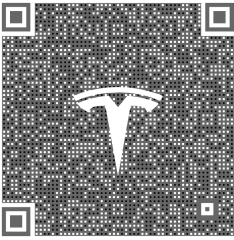
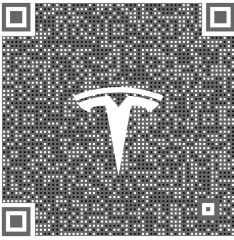
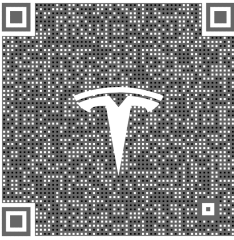
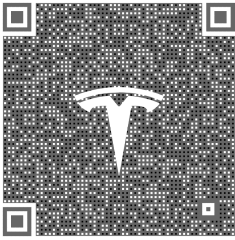
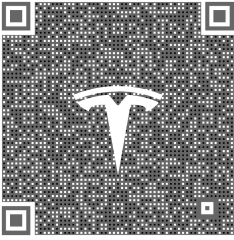
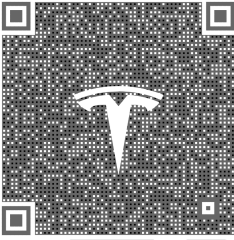
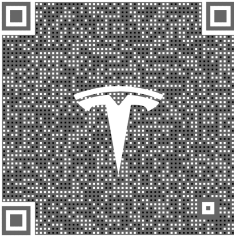
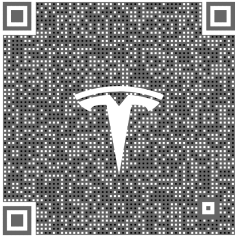
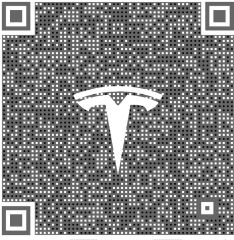
NOTE: Tesla BESS refrigerant is a regulated substance according to the USDOT. Refer to the specific SDS for R-134a.



NOTE: For projects in Australia or New Zealand, contact your Tesla representative for the product's safety data sheet.

Tesla recommends that a physical copy of the ERG is transported along with Megapack, and subsequently remains on site and accessible at all times, for the life of the product. Download the guide from <https://tesla.com/firstresponders>, or scan the code for a given language below:



English	Deutsch	Español	Français
			
עברית	Italiano	日本語	한국어
			
Nederlands	简体中文	繁體中文	Português
			
Slovenščina			
			

1.8 Getting Support

This guide, in addition to other required and supportive documentation, is distributed via the Tesla Partner Portal at <https://partners.tesla.com/>.

 **NOTE:** Ensure that the installation team has the most recent revisions of this document and the *Megapack Deployment Checklists* before beginning work. If you are an installation contractor and are not sure if you have the required documents, check with the site manager or Tesla support.

For Megapack support or to provide product feedback:



- Email IndustrialStorageSupport@tesla.com (responses can take 24-48 hours)
- Refer to [Urgent Contact Information on page 19](#) with urgent or time-sensitive requests

1.8.1 Urgent Contact Information

In case of critical performance issues (for example, if the Tesla site is 100% non-operational), call the appropriate Tesla telephone number listed below and select **Urgent** for 24/7 support.

For other inquiries, refer to *Tesla Energy Operations Contact List and Response Times* or email and telephone support contacts provided below.

Online Support and Ticketing Portal

<https://energysupport.tesla.com>

Email Support

IndustrialStorageSupport@tesla.com

Urgent Support via Telephone (24x7)

North America: +1 650-681-6060

Asia/Australia/New Zealand: +61 2 432 802 81

Europe/Middle East/Africa: +31 2 08 88 53 32

Technical Support via Telephone

Americas: +1 650-681-6060	Asia/Pacific: +61 2 432 802 81	France: +33 173218702
Japan: +0120 312-441	Netherlands: +31 208885332	Slovenia: +38 617778699
South Africa: +27 213004878	Switzerland: +41 445155607	United Kingdom: +44 1628450645



2 Site Architecture

2.1 Overview of Site Components

In addition to [Megapack System Components on page 8](#), a Megapack site may require other equipment. Typical components include:

- Meters (battery meter required, other meters optional) ([Energy Meter Design on page 20](#))
- Transformers ([Transformer Design on page 24](#))
- Line reactors ([Resonance of Devices Connected to Common 480 V AC Bus on page 28](#))
- Customer communications networking ([Network and Internet Considerations on page 29](#))
- SCADA components ([SCADA on page 30](#))
- Provisions for CMA units ([CMA Overview on page 30](#))
- Water supply ([Water Supply on page 30](#))
- Maintenance infrastructure ([On-Site Maintenance Infrastructure on page 39](#))
- Switchgear

2.2 Product Configurations

Megapack may be configured to meet specific site and project requirements. A fully populated Megapack has the following nominal ratings at 480 V AC:

Table 2. Nominal Fully Populated Ratings

System Duration	Max. Configurable AC Apparent Power (kVA)	AC-Rated Real Power (kW)	AC Minimum Initial Energy (kWh)
2-hour*	2400.0	1927.2	3854.4
4-hour	1512.0**	979.2	3916.8

* Specifications are preliminary and subject to change.

** The apparent power rating of the inverter is expected to increase in 2023 manufacturing from 63 kVA to 68 kVA. This will increase the total Max. Configurable AC Apparent Power to 1632.0 kVA. Refer to the [Megapack 2 XL Specification](#) for more information.

The AC power (kVA) of a fully populated Megapack may be factory-configured to a lower value than the above.

2.3 Energy Meter Design

The Tesla Site Controller uses various meter inputs for different control functions. Energy metering in various capacities as described in this section is therefore required in order for the system to operate.

Energy meters must be provided, installed, and configured by the contractor. Tesla will confirm communication with these energy meters but is not responsible for their accuracy.

Energy meters must abide by the minimum accuracy requirements as listed in the table below per region as appropriate. Equipment with greater accuracy may be installed if needed.



Table 3. Minimum Accuracy Requirements

Equipment	United States	Non-United States
Current Transformer (CT)	Class 1.2% Per ANSI/IEEE C57.13	Class 0.5S Per IEC 61869-2
Voltage Transformer (VT)	Class 1.0% Per ANSI/IEEE C57.13	Class 1.0 Per IEC 61869-3



NOTE: See [Connecting Energy Meters on page 100](#) for instructions on connecting meters.

The Tesla Site Controller supports only the following meter hardware, using TCP/IP communication:

- **SEL-735 Power Quality and Revenue Meter**

- Any model that supports Form 9 (Wye) or Form 5 (Delta) and has an Ethernet port (copper or fiber)
- For projects with specific response time requirements, Tesla requires an **Advanced SEL-735** model. Contact Tesla for more information.



NOTE: The system requires Ethernet to connect meters to the Tesla Site Controller. RS-485 is not supported.



NOTE: Tesla is not responsible for the accuracy of the energy meters provided, installed and configured by the customer. Please consult with the energy meter manufacturer in the event of faulty communications from energy meters.

2.3.1 Energy Meter Functions

Meters can be used in several capacities, not all of which are required.

Every site requires a battery meter that measures the AC energy output of the battery system. Some meters, such as a site meter (which measures the net load of the site with the battery system included), solar meter, generation meter, or others, are optional depending on application.

Multiple meters of the same function can be used. When two or more meters with the same function are deployed on a site, the Tesla Site Controller will automatically aggregate the values reported by the meters and use the result as if it was a single meter.

If a meter is indicated as optional in this guide, it means the meter may not be required in order for the battery to receive and execute charge or discharge commands. However, depending on the use case, these meters may be required in order to send the correct commands. Therefore, **consider the intended operation of the system before deciding which meters to include in the design.**

For example:

- Systems that intend to collect the Federal Investment Tax Credit for a combined PV and storage system may be required to charge only from the PV, in which case a revenue meter is required.
- Systems connected to a utility may be forbidden from exporting battery power to the grid, in which case a site meter, and possibly a solar meter (if there's onsite generation) may be required.



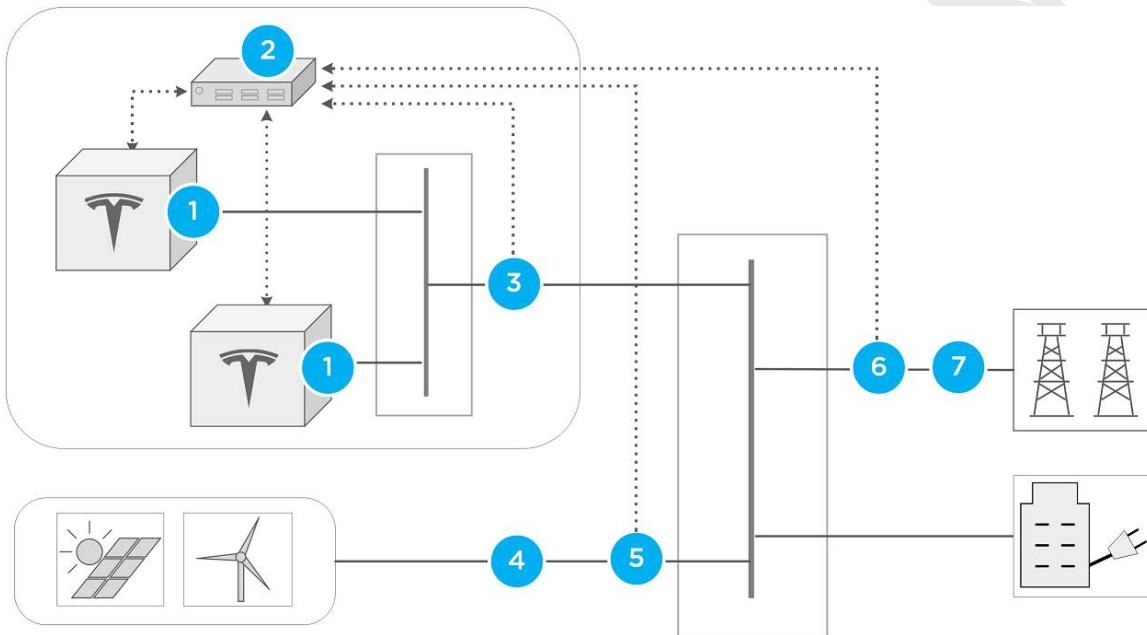
- Sites involving PV installations or wind or other power sources may require an additional generation meter.

Meters are required or optional as described below:

- Battery meter – required
- Site meter – required for certain applications
- Solar meter – required for certain applications
- Generator meter – optional
- Revenue / Utility meter - optional site-level requirement
- Bus meter – optional
- Load meter – calculated or optional

Your site may have additional meters as required per design. An example metering overview appears below:

Figure 13. Metering Overview - Example



(Dashed lines indicate communication with Tesla Site Controller)

1. Tesla battery system
2. Tesla Site Controller
3. Battery meter
4. Revenue meter
5. Solar (or wind) meter
6. Site meter
7. Utility meter



2.3.2 Battery Meter

All systems require a battery meter. The battery meter is bidirectional and is used to measure the AC power and energy charge and discharge of the system. The Tesla Site Controller uses the battery meter for closed-loop controls to ensure high accuracy on the power setpoint. Check with Tesla for configuration options if more than one battery meter may be needed.

2.3.3 Site Meter

The site meter monitors the feed from the utility or the point of common coupling (PCC). It measures the site import (negative for any export). The site meter is required for the following control functions described in the *Controls and Communications Manual*:

- *Site control*
- *Opticaster*
- *Voltage control* (where the voltage reference is not supplied by the battery meter)
- *Non-export control* (where required by utilities)
- Powerhub UI/reporting (to show energy production in the Tesla Powerhub platform)

2.3.4 Solar Meter

A solar meter is required in order to configure the site to charge only from solar.

A solar meter is recommended for:

- Opticaster sites that have solar (to improve tariff optimization and forecasting)
- Microgrid sites (to aid in troubleshooting/diagnostics with waveform capture)
- Powerhub UI/reporting (to show energy production in the Tesla Powerhub platform)

2.3.5 Generator Meter

A generator meter is recommended for:

- Microgrid sites (to aid in troubleshooting/diagnostics with waveform capture)
- Powerhub UI/reporting (to show energy production in the Tesla Powerhub platform)

2.3.6 Bus Meter

The bus meter is optional and can be used to ensure that the battery system does not overload an intermediate bus or transformer on the system.

2.3.7 Load Meter

The load data provided by the Tesla Site Controller is a calculated value based on the site, battery, and generation values. The system cannot have both a load and site meter. If a system does not have a site meter (for example, in an off-grid microgrid), a load meter can be installed to provide a direct rather than a calculated value for load data.



2.4 Transformer Design

In order to operate, Megapack requires a solidly grounded circuit to ensure that the line-to-ground nominal voltage does not exceed 300 V AC during steady-state conditions. The circuit is solidly grounded by connecting the Megapack terminals to the wye side of a transformer. The transformer may be:

- A step-up/step-down transformer
- A grounding transformer
- An isolation transformer

Customers can parallel multiple Megapacks on a single secondary (low voltage side) winding. One transformer grouped with several Megapacks is called a “transformer block.”

Megapacks do not require galvanic isolation from each other; therefore, a standard single winding primary single winding secondary transformer is sufficient for connecting multiple Megapacks. Megapack also does not require any additional transformer features such as k-factor or shielding requirements.

For operations and maintenance needs, the line side bus bars and cables within the Megapack must be able to be isolated and locked out using a visually verifiable means. This can be generally performed with a no-load disconnecting means at the secondary of the transformer (due to the circuit breaker in the Megapack performing the load break functionality), or via a load break disconnect at the primary winding. Arc flash, personal safety and operational procedures should be considered when selecting the means of disconnection.

The customer’s engineer should ensure that the requirements listed in this section are met alongside other technical considerations such as utility compliance, protection, maintenance, certification and load requirements. For transformer configurations not discussed here, contact Tesla for review.



NOTE: For more information on transformer design considerations, refer to the *Application Note: Grounding Conductor Recommendations for Megapack Applications*.



NOTE: For detailed information about islanding and the Islanding Controller, refer to the *Microgrid Islanding Controller Specification* and the *Controls and Communications Manual*.

For more information or with specific questions, contact your Tesla representative.

2.4.1 High-Voltage Ride Through

High-Voltage Ride Through (HVRT) refers to the capability of electric generators to remain connected (“ride through”) during short periods of high voltage from the electricity network (grid). In the Megapack system, HVRT is achieved by introducing power transformers between Megapacks and the grid. These transformers are required if Megapack will be exposed to grid voltage greater than 1.2 per unit for any period of time. In this case, high voltage on primary side of the transformer injects more magnetizing current into the transformer primary, which does not increase secondary-side voltage.

If Megapack will be exposed to grid voltage greater than 1.2 per unit for any amount of time, the transformer must be designed such that its secondary-side impedance is **at least 2.875%**.

Example

- Two 1 MVA Megapacks connected to a 4 MVA-rated transformer with 5% impedance
- Transformer impedance on primary side (where grid is connected) = 5%
- Secondary-side impedance



$$= \text{Primary side impedance} * \frac{\text{Megapack VA rating}}{\text{Transformer VA rating}}$$

$$= 5 * \frac{2 * 10^6}{4 * 10^6}$$

$$= 2.5\% < 2.875\% \text{ (Required minimum value)}$$

In this example, the transformer impedance is not high enough to meet the required minimum value, so we use the calculation below to determine the minimum required impedance value:

- Transformer impedance on primary side = ?
- Transformer impedance on secondary side = 2.875%
- Transformer impedance on primary side

$$= \text{Secondary side impedance} * \frac{\text{Transformer VA rating}}{\text{Megapack VA rating}}$$

$$= 2.875 * \frac{4 * 10^6}{2 * 10^6}$$

$$= 5.75\%$$

Result

Transformer impedance should be at least 5.75%

2.4.2 Short Circuit Current Withstand

The total short-circuit withstand rating of the Megapack is 85 kAIC. Ensure that your transformer is designed to provide less than this fault current at inverter terminals by following the guidance in this section.

To ensure that the maximum current flowing through Megapack does not exceed 85 kA, maintain a minimum impedance value on the secondary side of the transformer such that in the event of a fault the current does not exceed 85 kA.

Even if there are multiple connected Megapacks, assume a single Megapack is subject to full fault current. In some circumstances, this increases the minimum allowable impedance of the transformer.

 **NOTE:** Transformers manufactured to ANSI standards will always have +7.5% to -7.5% of impedance tolerance which must be taken into consideration.

Example

- Two 2,200 kVA Megapacks connected to a transformer at 480 V
- Transformer has a rating of 4,400 kVA, or 5,292 amps at 480 V [$I = VA / (\sqrt{3} * V)$]
- Then, the max allowable short circuit current is given by:



$$I_{short\ circuit} = I_{rated} * \frac{100}{Z\%}$$

- From the above expression, base impedance is calculated as

$$Z\% = I_{rated} * \frac{100}{I_{short\ circuit}}$$

$$= 5292 * \frac{100}{85 * 10^3}$$

- The base impedance must be selected such that the available short circuit current is less than 85 kA.

Result

The percentage transformer impedance value is calculated to be 6.2%. However, considering impedance tolerances, the base impedance of the transformer procured must be **at least 6.7% in accordance with ANSI standards**.

Per the example in [High-Voltage Ride Through on page 24](#), the transformer impedance should be > 5.75%, and,

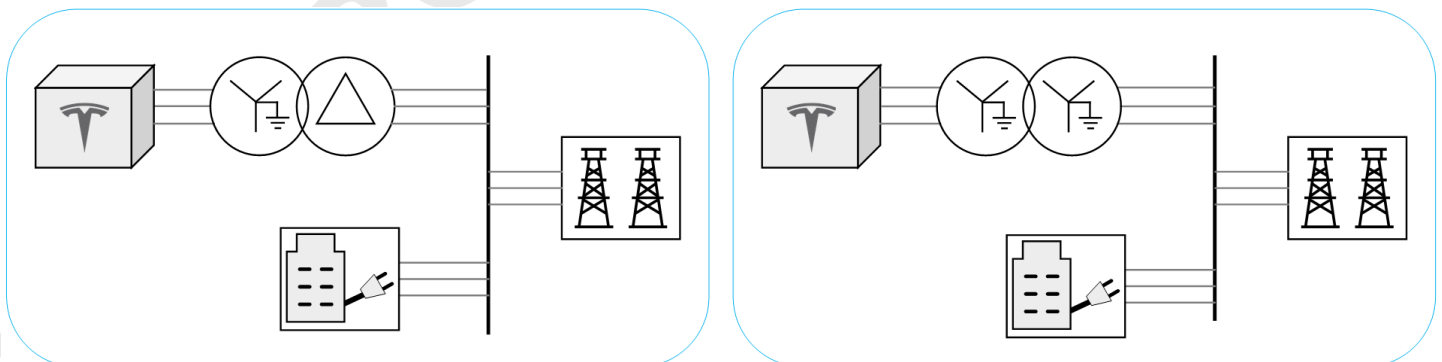
Per this example, for short-circuit current withstand, the transformer impedance should be > 6.7%,

Therefore, the final impedance of transformer should be at least 6.7%.

2.4.3 Approved On-Grid (Grid-Following) Configurations

The side of the transformer connected to Megapack must be a wye-grounded connection. The other side of the transformer may be a delta or wye connection for on-grid (grid-following) applications.

Figure 14. On-Grid (Grid-Following) Transformer Configurations



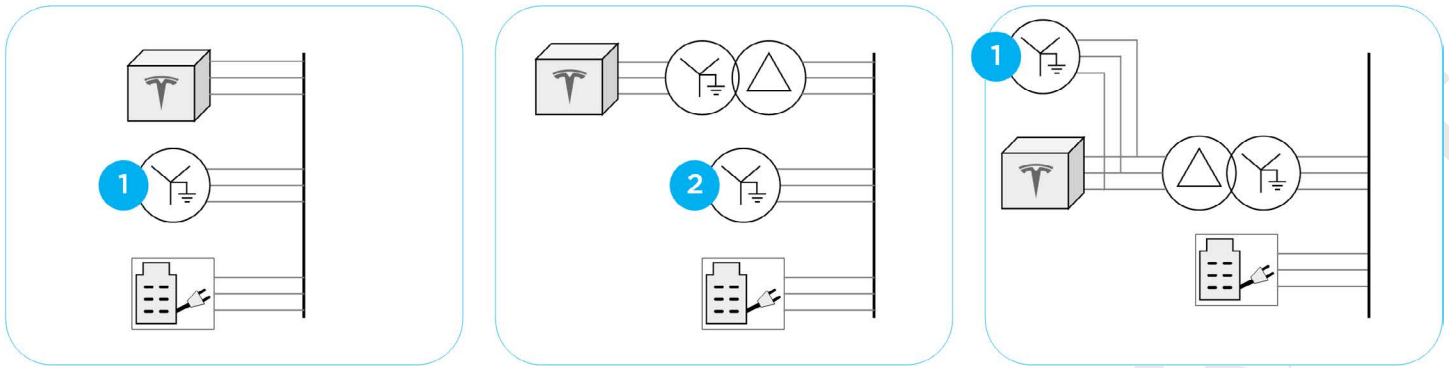
NOTE: For grid-following applications, the solidly grounded circuit may be provided by the utility transformer.

2.4.4 Approved Off-Grid (Grid-Forming) Configurations

The side of the transformer connected to Megapack must be a wye-grounded connection. The other side of the transformer must be a delta connection for off-grid (grid-forming) applications.



Figure 15. Off-Grid (Grid-Forming) Transformer Configurations

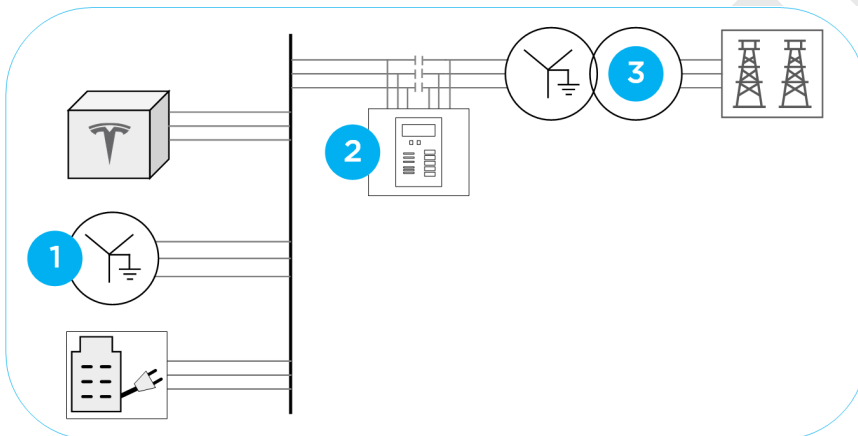


1. Grounding Transformer
2. Optional Grounding Transformer: An example of a potential system ground source and neutral source for line-to-neutral loads if present. This is not required for Megapack's proper operation in this example.

2.4.5 Approved Islanding Configurations

For islanding applications, ensure that a solidly grounded circuit exists during both on-grid and off-grid operation, as shown in the figures below. Depending on the project requirements, the grounding transformer may require disconnection when on-grid.

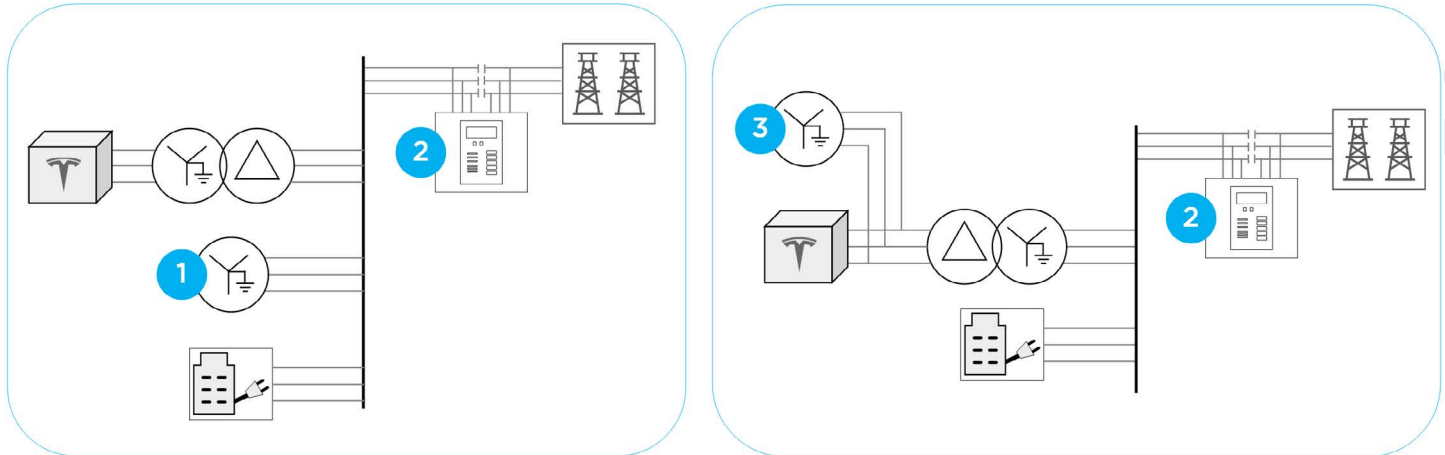
Figure 16. Islanding with a Utility Transformer and a Grounding Transformer



1. Grounding Transformer
2. Islanding Controller
3. Delta or wye



Figure 17. Islanding with a Step-up/Step-down Transformer



1. Optional Grounding Transformer: An example of a potential system ground source and neutral source for line-to-neutral loads if present. This is not required for Megapack's proper operation in this example.
2. Islanding Controller
3. Grounding Transformer

2.5 Resonance of Devices Connected to Common 480 V AC Bus

Excessive resonance is expected when the Megapack system is connected to a common 480 V AC bus with power electronics inverters (such as solar inverters) or loads (such as variable frequency drives (VFDs) or data center power supplies) that fulfill both conditions:

1. The total apparent power rating of the power electronics exceeds 20% of the apparent power rating of the Megapack system.
2. The equivalent switching frequency at the AC terminals of the power electronics is less than 35 kHz.

In order to mitigate the risk of resonance, you must design for one of the following:

- Relocate the equipment
- Provide a line reactor
- Provide an isolation transformer

2.5.1 Line Reactor

If using a line reactor, it must meet the minimum specifications below:

Table 4. Minimum Reactor Specifications

Specifications	Minimum Requirement
Inductance	0.022 mH per phase
Current Capacity	Exceeds continuous current rating of the Megapack(s) connected to this reactor
Location	In series between Megapack(s) and common AC bus

2.5.2 Isolation Transformer

If using an isolation transformer, it must meet the minimum specifications below. Refer to [Transformer Design on page 24](#) for additional details.



Table 5. Minimum Transformer Specifications

Specifications	Minimum Requirement
Impedance	Between 3% and 7%
Current Capacity	Exceeds continuous current rating of the Megapack(s) connected to this transformer
Location	In series between Megapack(s) and common AC bus

2.6 Network and Internet Considerations

The Tesla Site Controller and meters communicate using TCP/IP with Modbus and C37.118 protocols, therefore connectivity requirements must be part of the design.

Tesla requires communication between the Tesla Site Controller and all meters and Megapacks onsite. Every Megapack requires a wired Ethernet connection to the Tesla Site Controller. In multi-Megapack sites, this often requires the use of customer-supplied Ethernet switches and may require the use of fiber.

The Tesla Site Controller requires an internet connection. It has a built-in cellular connection that can provide a communication link between the Tesla Site Controller and Tesla via the internet. If the cellular network is not sufficient, a hardwired internet connection is mandatory. Additionally, if the customer wants a network connection for a control interface, the Tesla Site Controller is also the customer point of connection to the Megapack.

The Tesla Site Controller requires network connectivity to control the Megapack. Ensure adequate connectivity by installing a permanent external antenna at sites where the Tesla Site Controller is installed inside metal enclosures such as switchgear cabinets, in concrete buildings such as parking structures, or in areas with poor coverage.

See [Relocating the Antenna on page 96](#) for instructions on external antenna installation, and see [Terminating Auxiliary DC Power and Communications Wiring on page 82](#) for required installation procedures.



NOTE: Refer to the *Controls and Communications Manual* for complete control details.



NOTE: Some projects may use external cell antennas different from the included external antenna. The Standard Tesla Site Controller computer has an RP-SMA female connector (outer threading and pin) and requires an antenna with an RP-SMA male connector (inner threading and hole).



NOTE: Removing any component from the Standard Tesla Site Controller Enclosure other than the antenna (for example, to mount the backplane in a different enclosure or control room) must meet the environmental requirements on the Tesla datasheet and is not covered in the certified installation configuration. Discuss any modifications with Tesla prior to installation.



NOTE: For more information or with specific questions, contact your Tesla representative.

2.6.1 Computer Networking

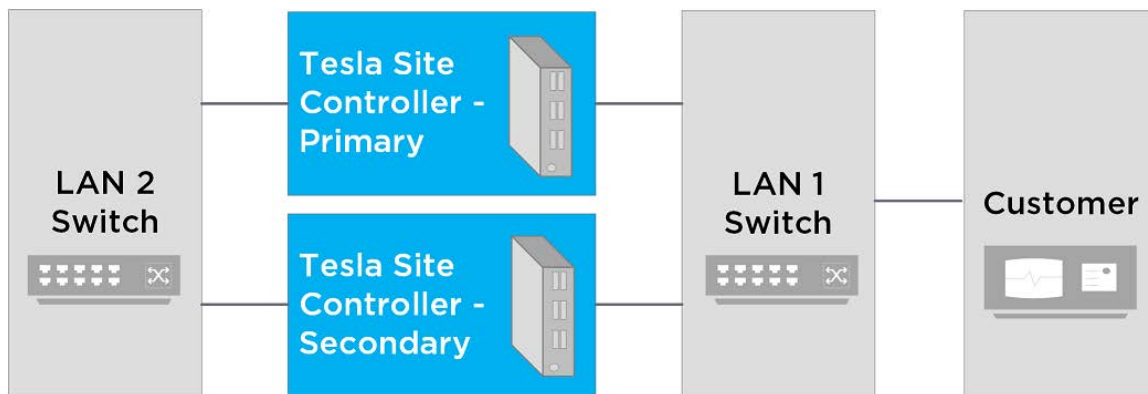
The Tesla Site Controller ([Tesla Site Controller on page 13](#)) has a single LAN 1 RJ45 port, a single LAN 2 RJ45 port, and an integrated cellular modem.

The Standard Tesla Site Controller Enclosure may have one or two controllers (Standard Tesla Site Controllers), hence one or two LAN 1 RJ45 ports and one or two LAN 2 RJ45 ports. Each controller has an integrated cellular modem. Large-scale sites typically ship with two controllers (Large Tesla Site Controllers).



For Standard Tesla Site Controller Enclosures including two Standard Tesla Site Controllers or for sites including dual Large Tesla Site Controllers, both LAN 1 and LAN 2 of each computer need to be connected in parallel, as indicated in the diagram below:

Figure 18. Dual Computer Networking



NOTE: If you are connecting through a firewall, see the *Firewall Rules* table in the *SCADA Design Manual* for important connection information.

2.7 SCADA

For many applications, Megapack can be operated as a standalone system as detailed in the *Controls and Communications Manual*. However, for custom solutions, Megapack can be integrated with a supervisory control and data acquisition (SCADA) system to expand its functionality.

The *SCADA Design Manual* further outlines what items to consider when designing a SCADA system, what additional features may be supported by Tesla, and how to integrate them with the Tesla Site Controller.

2.8 CMA Overview

A Capacity Maintenance Agreement (CMA) guarantees power and energy capacity over the life of the product. A capacity-maintained system adds CMA units (battery enclosures) over time to maintain power and energy. To support this, the site and foundation designs and the electrical infrastructure must pre-allocate space for the additional units during initial construction. This prevents the need to re-permit the entire site later in its lifecycle. A CMA must be contracted at the beginning of the project.

Guidance must be given to installation partners to ensure they understand the purpose of the space left for future CMA units, and they do not attempt to optimize layout during initial construction. Access must also be provided for a crane so that the supplementary CMA units can be added or removed as needed.

Upon request, Tesla can provide specifications for CMA lengths of 10, 15, or 20 years. If a site intends to use a CMA, contact Tesla for additional site design assistance.

Refer to [Capacity Maintenance Agreement on page 46](#) for clearance requirements.

2.9 Water Supply

If required by the AHJ, the site may need a water supply per NFPA 855 Section 4.13 or applicable local codes and regulations. For more information, refer to *Application Note: Onsite Water Supply Sizing For Exposure Protection*.



3 Mechanical Design

3.1 Mechanical Specifications

While Megapack's dimensions remain fixed, its mass and center of gravity vary depending on its energy rating, corresponding to the number of battery modules installed. The maximum mass is displayed below. For precise values per configuration, refer to [Megapack Center of Gravity on page 32](#) and [Megapack Mass on page 32](#).

Table 6. Megapack Dimensions and Mass (Weight)

Width	Depth	Height	Max. Shipping Mass
8800 mm	1650 mm	2785 mm	38,100 kg
(346 ½ in)	(65 in)	(110 in)	(84,000 lb)



CAUTION:

- Enclosure dimensions are provided for nominal design guidance. Do not use the dimensions above as anchoring specifications. For precise dimensions and structural and anchoring details, see the *Megapack 2 XL Drawings* on the Partner Portal.
- Dimensions as listed are as measured for the enclosure envelope exclusive of anchor brackets.
- Mass (weight) as listed is maximum shipping mass. Mass changes depending on product configuration and can be configured lighter based on project-specific requirements.

Table 7. Standard Tesla Site Controller Enclosure Dimensions and Mass

Width	Depth	Height	Max. Mass (Weight)
255 mm	560 mm	724 mm	22.4 kg
(10 in)	(22 in)	(29.2 in)	(49.4 lb)

Table 8. Additional Enclosure Specifications

Impact Rating	IK09
Ingress Rating	IP66/NEMA 3R (Megapack enclosure) IP20 (Thermal system) IP67/NEMA 3R (Standard Tesla Site Controller Enclosure)
Roof Live Load	Can accept a person of up to 300 lbs (136 kg) gross weight in clearly designated areas.
Roof Snow Load	732.3 kg/m ² (150 psf, or 7.18 kN _{Earth} /m ²) If the site may experience greater snow loading, contact Tesla to discuss.
Thermal System	Each Megapack contains coolant and refrigerant in capacities as specified in Thermal System on page 34 .



3.1.1 Noise Guidance

Each Megapack has rooftop fans dedicated to cooling power electronics and battery modules. Megapack's main source of noise comes from these fans. Noise emissions depend on the number of fans and the fan duty cycle, which depend on the configuration of Megapacks on your site.

Megapack will emit a maximum of 75 dBA (SPL sound pressure) at a 10-meter distance from any side surface of the enclosure, while operating at full power. Maximum values may generally only be reached under specific operating conditions. The majority of the time, noise emitted by Megapack will be much lower.

Tesla can provide more details around sound power and sound pressure if required for the project site.

3.2 Megapack Mass

To calculate Megapack's precise mass, use its energy rating (number of battery modules) together with a formula as described below.

1. Determine Megapack's number of battery modules using the two numbers in the EC option code (see [Megapack Labels on page 15](#)). For example, the option code **EC14** indicates a Megapack with 14 battery modules.
2. Apply that number in place of *nn* in the formula as indicated:

$$38,100 - 1,270 \cdot (24 - nn)$$

The solution indicates Megapack mass in kg.

For example, a Megapack with option code **EC14** has **14 battery modules**, and its mass is calculated as follows:

$$38,100 - 1,270 \cdot (24 - 14) = 38,100 - 1,270 \cdot 10 = 38,100 - 12,700 = \mathbf{25,400 \text{ kg}}$$

3.3 Megapack Center of Gravity

Megapack's center of gravity is located close to the geometric center of the enclosure on the X and Y axis for all product configurations. For the precise location for a fully populated Megapack, see below.

Figure 19. Center of Gravity Dimensions

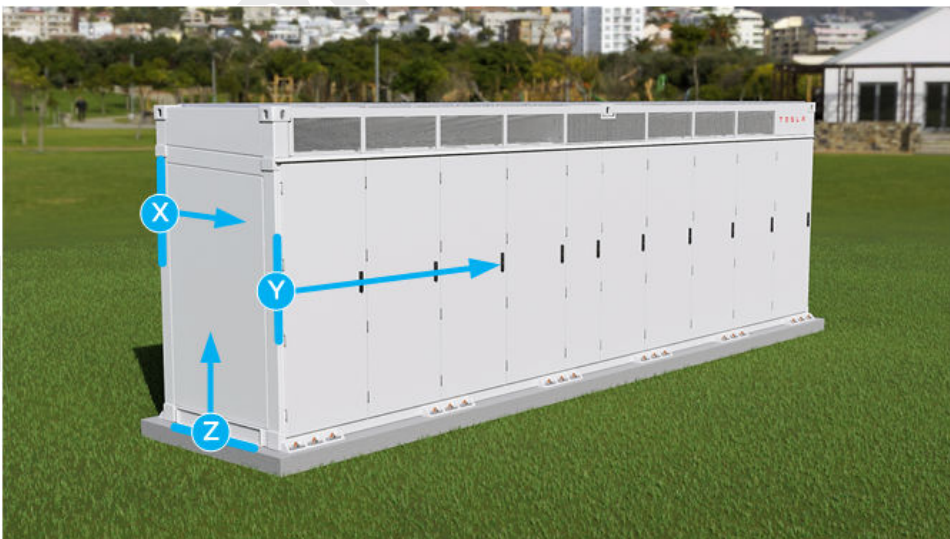




Table 9. Center of Gravity Location

System Duration	X (Distance from Rear Panel)	Y (Distance from Left Panel)	Z (Distance from Base)
2-Hour	791 mm (31 in)	4395 mm (173 in)	1202 mm (47 ¼ in)
4-Hour	791 mm (31 in)	4395 mm (173 in)	1199 mm (47 ¼ in)

3.4 Enclosure Colors

Megapack's exterior paint color code is *RAL 9016 Traffic White*. Touch-up paint ships as a Megapack accessory (see [Touch-Up Paint on page 76](#)).

Figure 20. Megapack Enclosure Color



The Standard Tesla Site Controller Enclosure is unpainted and is a flat grey color.



Figure 21. Standard Tesla Site Controller Enclosure Color



3.5 Thermal System

Megapack includes a thermal management system that provides active cooling and heating to the internal Megapack components. An external HVAC or thermal system is therefore not required for Megapack to operate.

The thermal system includes radiators and pumps that circulate coolant through the battery to maintain thermal control, in addition to an in-line heater that can warm the coolant. The thermal subsystem also includes refrigerant in a sealed system. All Megapacks ship with the necessary coolant and refrigerant included as specified below.

NOTE: Depending on the number of Megapacks installed on a site, storage, use and handling of these substances may require reporting, hazard management plans, or containment procedures as required by local codes and regulations.

Table 10. Thermal Capacities

	Composition	Max Quantity (Approx)
Coolant	50-50 ethylene glycol-water	400 L / 106 gal
Refrigerant	R-134a (1,1,1,2-Tetrafluoroethane)	3 kg / 6.6 lb

Since the thermal subsystem is a fully closed-loop system with a compressor, the refrigerant line includes a pressure relief valve that can activate if incorrect maintenance or operation creates excessive pressure. The system operates autonomously and does not require user feedback.

Megapack is designed to be installed outdoors within the rated operating temperature specifications. Indoor installations are not allowed without consulting Tesla.

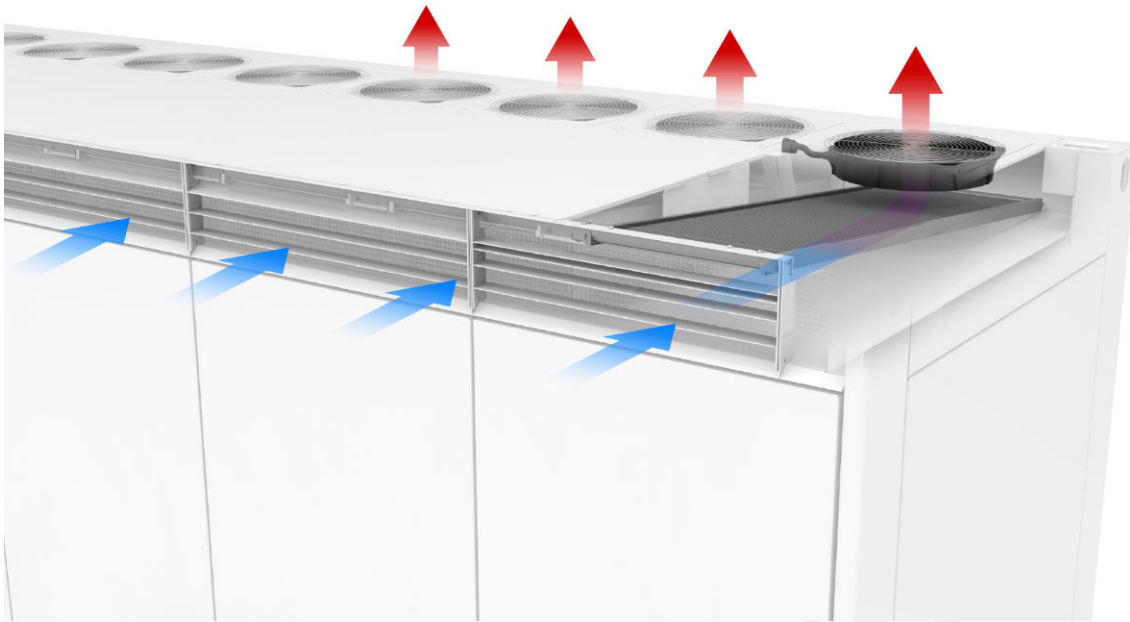
CAUTION: Do not install batteries in areas where temperatures routinely approach or exceed 50°C (122°F).

NOTE: The physical characteristics of the project site may cause localized heating effects (rise above ambient) adjacent to Megapack – for example, an installation in a location with a strong heat island effect, black pavement (parking lots), or in full direct sunlight. Although not required by Tesla, shade structures or canopies may help mitigate local environmental heating effects. Ultimately, the customer must understand the risk to performance of localized heating effects and take responsibility for any mitigation. Any shade structures must conform to the distances detailed in [Clearance on page 43](#).



Air flows through the enclosures as shown below:

Figure 22. Megapack Airflow



3.6 Hazard Mitigation and Safety

Tesla conducts extensive analysis and testing to assess hazardous conditions related to Tesla products. Megapack includes multiple layers of protection to mitigate hazardous electrical and fire conditions.

Megapack neither contains nor needs built-in smoke, gas, or fire detection or suppression devices. When required by the AHJ, third-party multi-spectrum IR heat or flame detectors can be installed externally at the site-level.

Refer to the Tesla *Lithium-Ion Battery Emergency Response Guide* for detailed hazard and response information.



4 Civil Design

4.1 Site Civil Design

4.1.1 Delivery Access Plan

The Megapack enclosure is a large, overweight load. In order to prepare for successful delivery, access roads must be capable of supporting the loaded delivery vehicle. These sections discuss road and site design conditions and requirements such as road dimensions, construction material, slopes, and grade in order to prepare for successful delivery.

Tesla recommends that a **site movement and crane unloading plan** be created as part of the design process prior to construction. Reach out to Tesla for more information about creating this plan.



NOTE: Tesla is not responsible for unloading Megapack. The installation contractor together in consultation with the site designer shall provide a site movement and crane unloading plan.



NOTE: Failure to meet these requirements can lead to project delays.



CAUTION: Failure to follow the recommended guidelines may result in damage to delivery vehicle or Megapack.

4.1.1.1 Access Route

The delivery access route must be able to support a loaded, 7-axle delivery vehicle. See [Mechanical Design on page 31](#) for details on the estimated equipment gross weight.

The plan must account for means of accessing the site such as illustrated below.

Figure 23. Access Route: Example

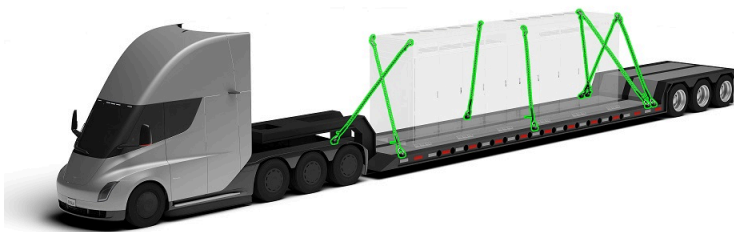




4.1.1.2 Delivery Equipment

The plan must also allow for access by appropriate transportation equipment including cranes and delivery vehicles.

Figure 24. Megapack Delivery Vehicle: Example



4.1.1.3 Access Road Requirements

The plan must ensure that the site roads meet the following requirements:

- Linear driving width: Minimum 3660 mm (144 in)
- Turn width: Minimum 6100 mm (240 in)
- Intersection width: Minimum turning radius of 14 m (45 ft), free of obstructions
- Slope: Maximum 10%, including drainage ditches and bumps
- Site road surface must be made of materials approved by the regional regulations.
 - Tesla recommends compacted aggregate surface specified by Regional Department of Transportation or equivalent agency.

Figure 25. Minimum Roadway Width: Linear Driving

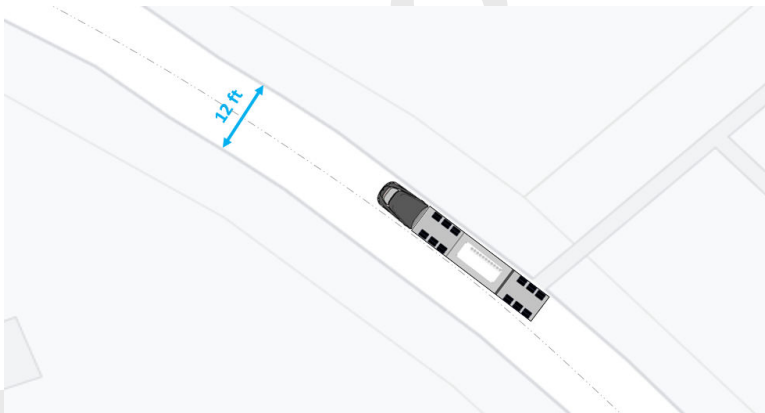




Figure 26. Minimum Roadway Width: Around Turns

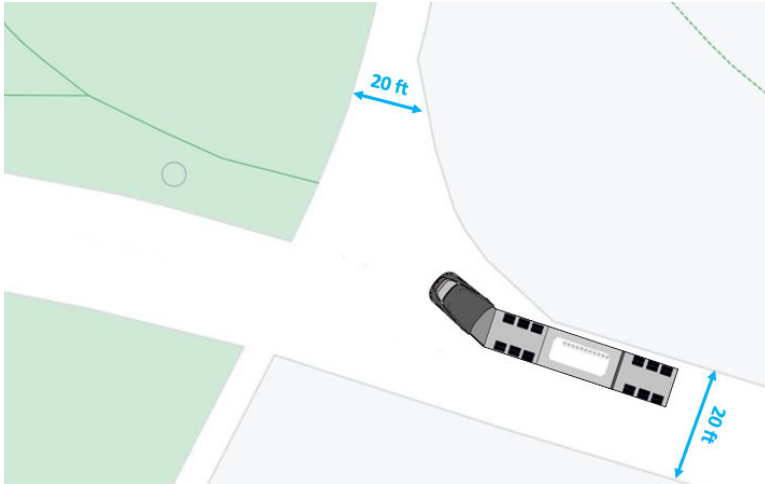


Figure 27. Minimum Intersection Turn Radius

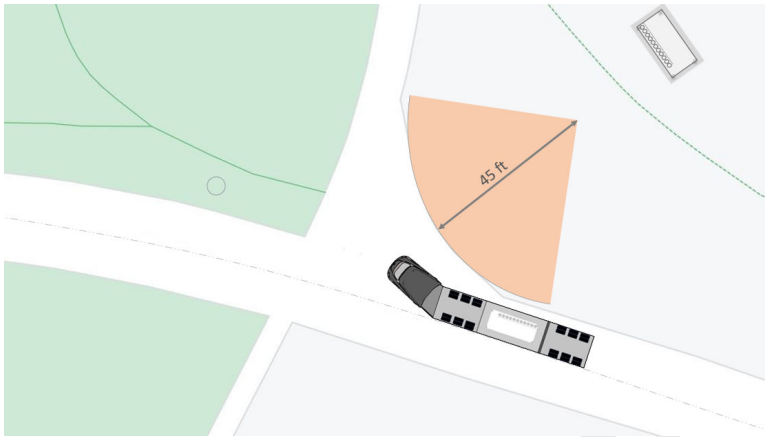
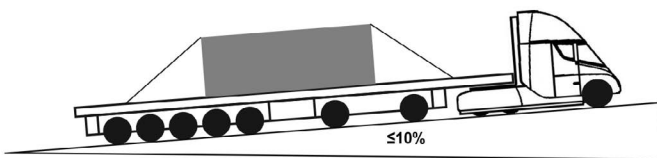


Figure 28. Maximum Roadway Slope



4.1.1.4 Day of Delivery Conditions

Ensure that the installation contractor prepares for day-of-delivery access as specified in [Preparing for Installation on page 75](#).





4.1.2 Fencing Requirements

Megapack shall be installed in a manner that deters access by persons who are not qualified. When deterring access, fences, screens, walls, or barriers no shorter than 2.1 m (7 ft) in height are suggested. The distance from any fence to the equipment shall match the clearance requirements listed in [Clearance on page 43](#), or as noted per the exceptions below. Fencing shall be locked and posted with a placard stating “Authorized Users Only,” or similar. If applicable, see 2018 IFC 1206.2.8.7.3.

Exception: If the installation is located within a property that already contains perimeter fencing to prevent unauthorized public access, additional fencing might not be required.

4.1.3 Access Requirements

Tesla must retain access to the site and all Tesla equipment over the lifetime of the site.



NOTE: For installations not at grade, contact Tesla for additional guidance.

4.1.3.1 Accessing the Site

- The site shall provide and maintain a clear access route from the public right-of-way to the front of each Megapack for delivery, installation, replacement, and removal of Megapack and its components.
- The access route must support Megapack and its transportation equipment, including crane and forklift as required. Ensure that access accommodates required vehicle turn radii of at least 4725 mm (186 in) and gross vehicle weight (including Megapack mass). For details on Megapack mass, see [Mechanical Specifications on page 31](#)

4.1.3.2 Accessing Megapack

- Tesla must have the ability to remove any locks preventing access. Refer to [Megapack Door Security on page 73](#) for information on securing and locking doors while retaining Tesla access.
- Keep the doors of all enclosures free of all obstructions such as snow, sand, and blown debris during system operation. With any concerns, contact Tesla during the site design phase.
- Area in front of Megapack must be capable of supporting a vehicle mass of at least 6100 kg (13450 lb).

4.1.4 On-Site Maintenance Infrastructure

Maintenance infrastructure is a requirement for the rapid and effective maintenance of Megapack sites. The required infrastructure should be incorporated into site design, made available to Tesla during construction and for the life of the project, and should be completed no later than 2 weeks prior to commissioning.

For details, refer to the *Application Note: On-Site Maintenance Infrastructure Requirements*.

4.1.5 Site-Level Power

To ensure that sites can be commissioned safely and serviced effectively, the site design shall include at least 1 (one) receptacle socket positioned such that the outlet is not more than 15 m (50 ft) from all Megapacks at the site. The receptacle should be designed to local requirements for human use & safety and should include a ground fault circuit interrupter (GFCI). Common access points can include the block-level transformer, lighting, or network enclosure.

The receptacle socket shall be rated to at least 2,400 VA at the local voltage. For current sizing, assume no more than one receptacle will be used at full power simultaneously at any given time. For North American sites, the recommendation is to provide NEMA 5-20 receptacles.



4.2 Megapack Civil Specifications

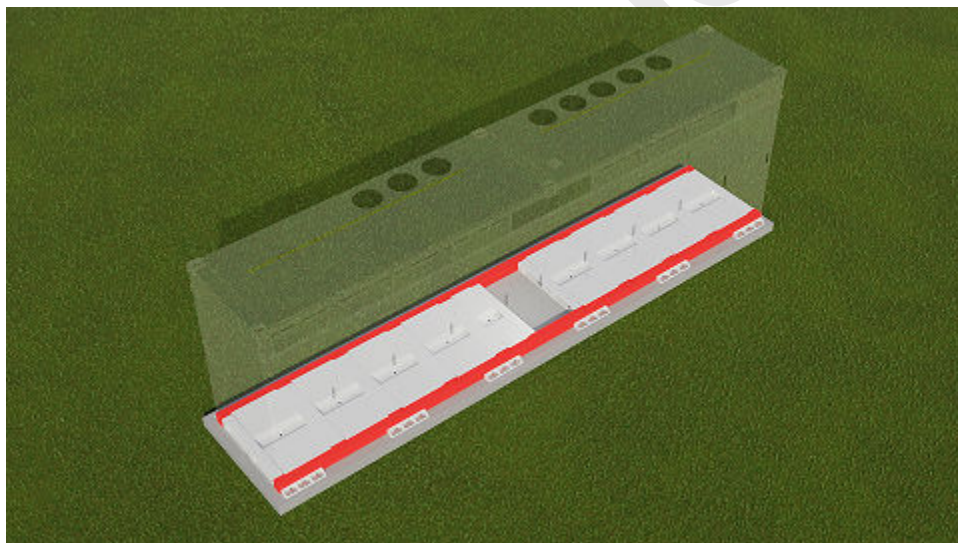
Table 11. Civil Specifications

Operating Temperature	-30°C to 50°C (-22°F to 122°F) Megapack provides full rated power in all ambient temperature conditions and at maximum elevation under up to 1200 W/m ² solar loading, in any direction.
Humidity	Up to 100% condensing
Storage	-30°C to 50°C (-22°F to 122°F) See <i>Megapack 2 XL Transportation and Storage Guidelines</i> for more information.
Maximum Altitude	3000 m (9840 ft) above sea level
Wind	Category 5 hurricane sustained wind speeds of up to 157 mph (252 km/h)
Seismic Rating	IEEE 693-2018 High PL: $ZPA=1.0$ g 5% damping ICC-ES AC 156-2018 $S_{DS}=2.50$ g $z/h=0$ $I_p=1.5$

4.3 Equipment Bearing Areas and Anchoring

Megapack requires support along the length of the long sides to a width of 200 mm (8 in). The foundational sections that must be supported are shown below:

Figure 29. Megapack Ground Support



Megapacks may be anchored to their foundations using their anchor brackets, which are steel flanges welded to the base of Megapack. The use, quantity, and location of anchors ([Anchor Requirements on page 41](#)) is determined by the engineer of record and must conform to the laws, regulations, codes, and standards applicable in the jurisdiction of installation. Refer to clearance requirements in [Clearance on page 43](#).

Anchors are needed if either of the following conditions are true:

- Uplift due to lateral forces such as wind or seismic cannot be effectively resisted by the Megapack's weight



- Sliding due to lateral forces such as wind or seismic cannot be effectively resisted by friction



NOTE: For anchor locations per Megapack installation type and complete anchor installation instructions, refer to [Anchoring Megapack on page 78](#).

Figure 30. Megapack Anchor Brackets

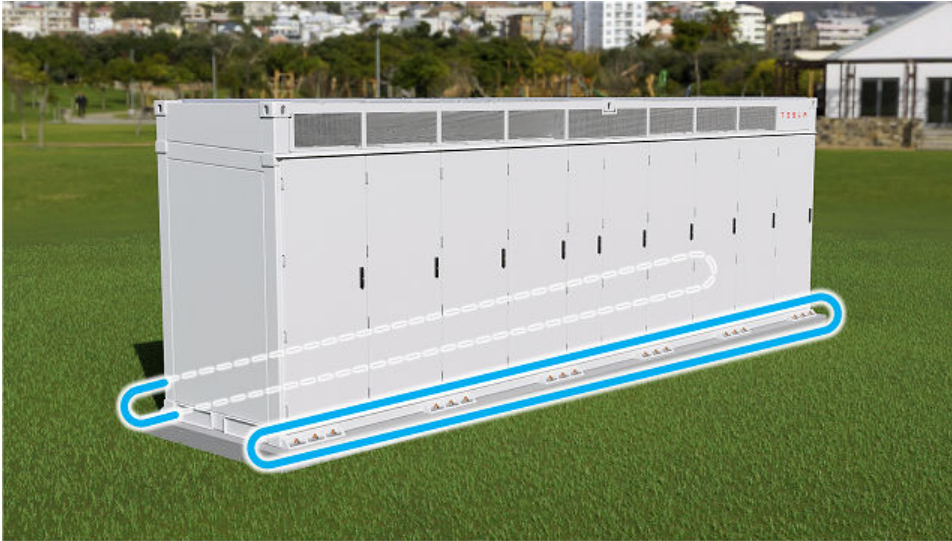


Table 12. Anchor Bracket Specifications

Description	Length	Depth	Thickness
Steel flanges welded to base of Megapack	438 mm (17 ¼ in)	102 mm (4 in)	16 mm (5/8 in)

4.3.1 Anchor Requirements

Anchors must be provisioned by the customer as selected by the structural engineer of record per site- and foundation-specific requirements according to the guidelines in this section.



NOTE: Refer to [Anchoring Megapack on page 78](#) for additional information.

When provisioning anchor hardware for Megapack, consider the following:

Table 13. Enclosure Anchoring Specifications

Anchor brackets	12 (6 in front; 6 in rear)
Anchor provisions	36 (3 per bracket)
Material	Stainless steel
Max allowable anchor diameter	27 mm (1 in)
Anchor hole diameter	40 mm (1 1/2 in)
Anchor position tolerance	± 11 mm (7/16 in)



CAUTION: If using a concrete foundation, the structural engineer must provide the minimum allowable distance from the installed anchor to the edge of the foundation.



NOTE: Anchor hardware must not protrude above foundation surface greater than specified in [Anchor Protrusion on page 81](#).



CAUTION: Modification of anchor brackets is not permitted in any way.

4.3.1.1 Additional Anchoring Considerations

- If installing back-to-back Megapacks and rear anchors are required, Tesla recommends that Megapacks are spaced with enough distance between them in order to install and access each Megapack's rear anchors. Refer to [Equipment Clearance Requirements - Side View / Front View on page 44](#) for back-to-back spacing requirements. Contact Tesla for more information if these requirements cannot be met.
- Hilti KB-TZ SS wedge anchors or HAS-R 304/316 SS and Hilti HIT-HY 200 epoxy anchor system are recommended.
- The final anchor selection must be made by the structural engineer of record.
- Refer to the seismic and wind specification values in [Megapack Civil Specifications on page 40](#).

4.3.1.2 Additional Considerations for Concrete

If using concrete foundations, follow these guidelines in addition to the above:

- Concrete rebar reinforcement location must be coordinated with equipment anchor locations to avoid overlap.
- The pad thickness and size shall be sufficient to accommodate Megapack anchoring as required per ACI 318 or equivalent local standards.
- Specify rebar with a minimum and maximum allowed spacing, to allow the contractor to adjust rebar spacing to avoid conflicts with equipment anchoring.
- Alternately, the minimum concrete cover over the rebar can be increased to exceed anchor embedment depth, to avoid rebar conflicts with equipment anchoring.

4.4 Foundation Design

4.4.1 Foundation Construction

Megapacks must be installed on a foundation or base strong enough to support the weight of the equipment listed in [Mechanical Specifications on page 31](#) and to resist all anchor loads. Observe equipment clearances per [Clearance on page 43](#).

Foundation or base examples include, but are not limited to, concrete pad, grade beams, structural steel deck or skid. The following considerations must be respected:

- Equipment must all be installed on a single level surface.
- If using non-concrete pad designs, approve first with Tesla before work begins.

4.4.2 Foundation Installation Requirements



NOTE: Foundation requirements apply to Megapack units as well as any CMA units ([CMA Overview on page 30](#)). CMA foundations must be installed at the same time as Megapack foundations.

- Foundation finish must have a smooth, even surface of uniform texture and appearance, free from bulges, depressions, and other imperfections that would impact equipment anchorage or foundation/base drainage.
 - If the foundation top surface is concrete, it must have a Class B finish.



- The top of the foundation must be above adjacent grade, 305 mm (12 in) maximum, with the edge of the foundation a maximum of 305 mm (12 in) from the front of Megapack. If the site does not allow this foundation height, contact Tesla. The foundation may not be below grade and must conform to local construction standards and regulations.
- A maximum of 13 mm (0.5 in) differential settlement is permitted.
- Ensure that the foundation is designed to prevent animals and insects from accessing Megapack's underside or debris from accumulating under Megapack.
- The area around the foundation must be designed to prevent standing water.
 - Any walls must be designed with sufficient clearance and be provisioned with features such as drains or weep holes to allow for proper drainage.
 - Megapack can withstand no more than 13 cm (5 in) of standing water for up to 30 minutes.
- Ensure that the foundation observes all clearance requirements as listed in [Clearance on page 43](#).
 - This includes designing for door clearance. Ensure that the doors of each Megapack are not constrained by any adjacent cabinet or structure. Refer to specific dimensions for door swing in [Door Clearance and Maximum Door Swing Requirements on page 46](#).
- For anchoring requirements, see [Anchor Requirements on page 41](#).
 - Concrete pad and equipment anchorage design shall meet ACI 318 and ASCE 7 requirements or the local building code requirements as required by the Authority Having Jurisdiction (AHJ).
 - Post-installed anchors (anchors installed into existing hardened concrete) must be approved to resist seismic loads in the installation jurisdiction and have a current ICC-ES or IAPMO report. Where anchoring conditions require an engineered design, it must be performed in accordance with the ICC-ES or IAPMO report and the cracked concrete provisions of ACI 318-11, Appendix D.



CAUTION: Modification of anchor brackets is not permitted in any way.

If the foundation has areas of unevenness that violate any conditions above:

- For concrete foundations, use grout to even the surface
- For other foundation materials, use shims to even the surface
- In all cases, have a structural engineer approve the modification before implementing

If any aspect of the foundation inspection does not pass after remediation, contact Tesla before proceeding.

4.5 Clearance



NOTE: The clearances listed are as required by the product. Additional clearances to non-Megapack equipment may be required per local codes and regulations.

4.5.1 Equipment Clearances



NOTE: All clearances listed must be observed from Megapack to any equipment including other Tesla-provided components such as the Standard Tesla Site Controller Enclosure.



Figure 31. Equipment Clearance Requirements - Side View / Front View

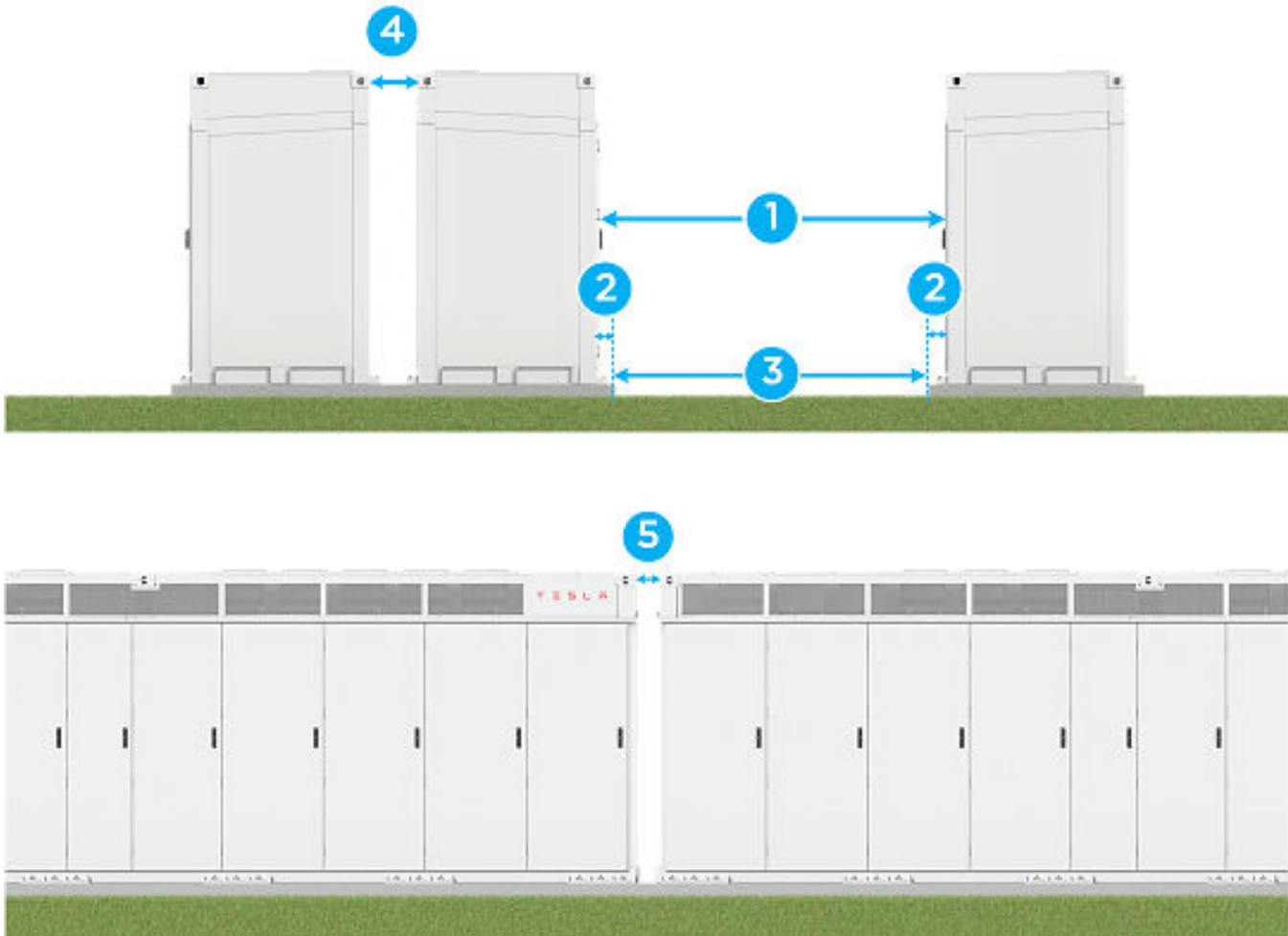


Table 14. Equipment Clearance Requirements

Callout	Type	Minimum	Maximum	Notes
1	Front	2440 mm (96 in)	None	Measured from face of doors. Tesla-required clearance for maintenance access.
2	Foundation overhang	100 mm (4 in)	305 mm (12 in)	Varies depending on anchor and site design but must fall within this range. See Equipment Bearing Areas and Anchoring on page 40 for more information.
3	Drive aisle	1960 mm (77 in)	None	Measured from foundation. Tesla-required clearance for maintenance access.
4	Back-to-back	230 mm (9 in) <i>minimum</i> . 460 mm (18 in) <i>recommended</i> for access purposes.	None	Measured from the outside faces of the ISO corners at top of Megapack. See also Additional Anchoring Considerations on page 42 .
5	Side	150 mm (6 in)	None	Measured from the outside faces of the ISO corners at top of Megapack.



4.5.2 Exposures and Fire Clearances

Figure 32. Exposure Clearances - Isometric View / Side View

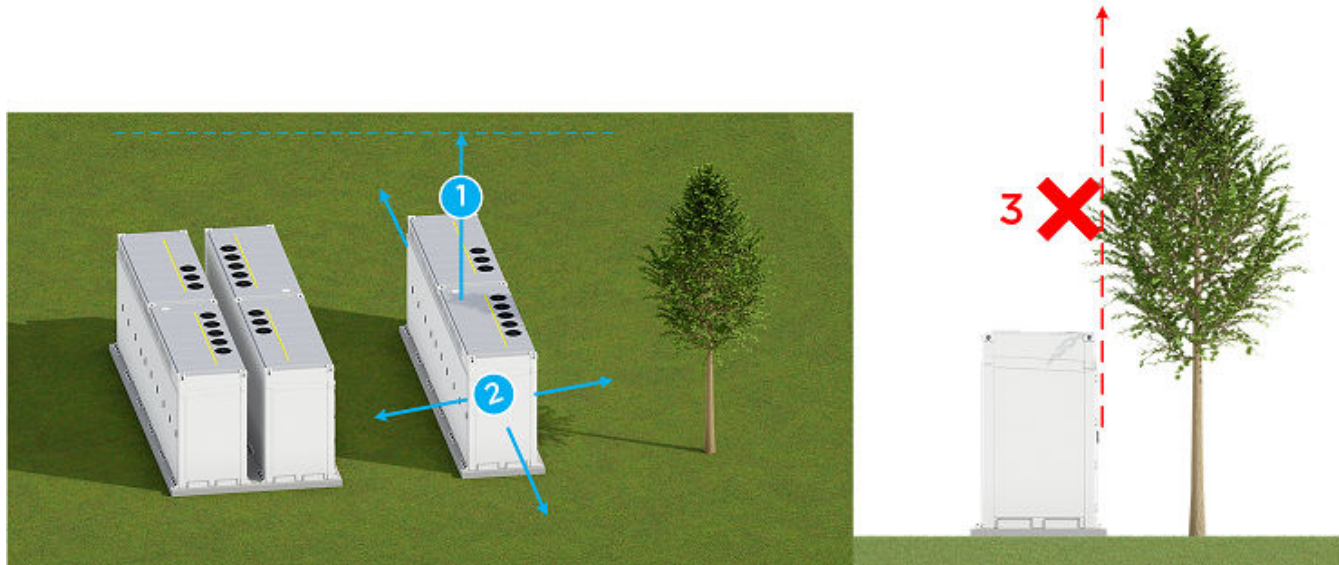


Table 15. Exposure Clearance Requirements

Callout	Type	Minimum	Maximum	Notes
1	Vertical	2440 mm (96 in)	None	Must extend the entire area of the service clearance. Some service equipment extends beyond the roof of the enclosure. Megapack may only be installed by a crane, thus actual clearance during installation will be greater.
2	Combustible	1530 mm (60 in) <i>Do not install Megapack under combustible objects.</i>	None	Minimum clearance as noted is required on all sides from combustible objects including trees, wooden fences, and other combustible structures. Megapack must not be installed under combustible objects.
3	Vertical combustible	NA	NA	Do not install Megapack under combustible objects, at any distance.

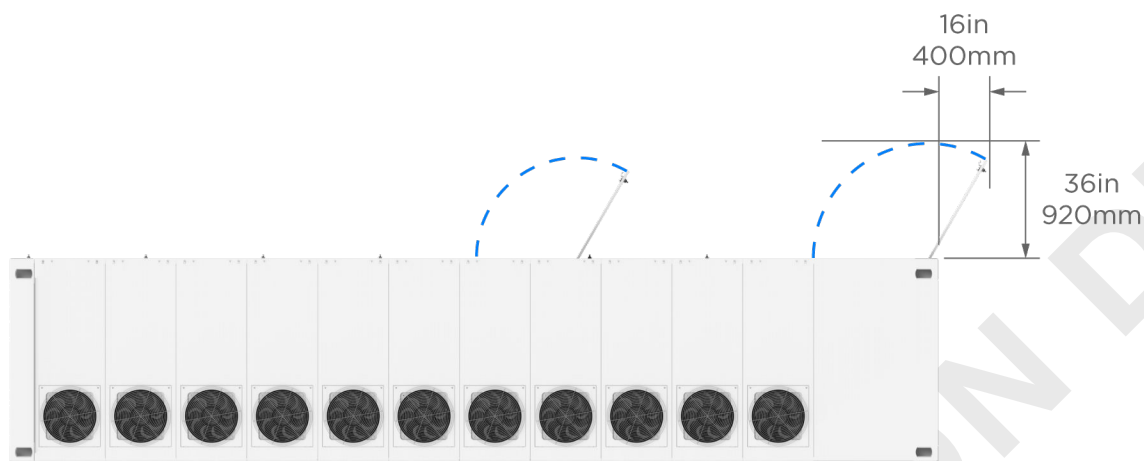
- NOTE:**
- Megapack is not intended to be installed within 3050 mm (120 in) from accessible means of egress and exposures (such as buildings, public ways, and hazards not associated with electrical grid infrastructure as defined by the clearance requirements in the International Fire Code and NFPA 855).
 - Any installation that requires clearances of less than 3050 mm (120 in) to accessible means of egress or exposures may require a freestanding fire barrier per requirements in the International Fire Code and NFPA 855.



4.5.3 Door Swing

To allow full door clearance and access to all components, sites must be laid out to allow 400 mm (16 in) of clearance past the enclosure footprint for the last door on the left. The site layout must ensure that no wall or other structure interferes with any door opening fully:

Figure 33. Door Clearance and Maximum Door Swing Requirements



4.5.4 Capacity Maintenance Agreement

For any site that has signed a Capacity Maintenance Agreement (CMA), the site layout must provide adequate space to allow for the addition of CMA units for the duration of the contract. Up to three CMA units may be required per Megapack as determined by Tesla.

Clearance requirements include:

- Clearance upon the foundation to place 1-3 CMA units at either side of each Megapack
- 150 mm (6 in) side clearance between enclosures (including Megapack, other CMA units, or other exposures)
- Clearance for an overhead lift to access the site

Refer to the dimensions below and contact your Tesla representative for more information about the space needed for CMA units for your particular site.

Table 16. CMA Unit Dimensions*

Width (Max)	Depth (Min)**	Height	Side Clearance
1100 mm (43 in)	1600 mm (63 in)	2500 mm (96 in)	150 mm (6 in)

*Dimensions are preliminary and subject to change.

**Overall depth could be greater depending on servicing needs and aisle space.

Figure 34. Example Layout for 1-3 CMA Units



Table 17. Additional Width Requirements for 1-3 CMA Units

Number of Units	Unit Width	Side Clearance	Total Additional Width
1	1100 mm (43 in)	150 mm (6 in)	1250 mm (49 in)
2	1 x 2 2200 mm (86 in)	1 x 2 300 mm (12 in)	2500 mm (98 in)
3	1 x 3 3300 mm (129 in)	1 x 3 450 mm (18 in)	3750 mm (147 in)



5 Electrical Design

5.1 Electrical Specifications

The inverter modules in Megapack are current-limited sources.



NOTE: Refer to the *Megapack Interconnection Datasheet* for additional information.

Table 18. Inverter Specifications

Max Continuous Output Current	Factory-configurable
Overload Capability	120% of rated current (10 sec max)
Nominal AC Voltage	480 V
Output Voltage Range	422-552 V AC
Nominal Frequency (configurable)	50 or 60 Hz
Frequency Range	45-66 Hz
Phases	3
System Configuration	3-wire, Wye Note: Grounded Wye required at transformer secondary
Full Load Efficiency	98.3%
California Energy Commission Weighted Efficiency	98.5%
Power Factor at Full Load	> 99%
Adjustable Power Factor (Controller Feature)	-1 to +1
Total Current Harmonic Distortion (THD)	< 5%
Power Regulation Accuracy	< 2%
Overvoltage Category	Category III up to 3000 m
Maximum Short Circuit Current Withstand	85 kAIC



WARNING: In order to operate, Megapack requires a solidly grounded circuit such that the line-to-ground nominal voltage does not exceed 300 V AC.



NOTE: The per-unit short circuit current in the case of a three-phase-to-ground bolted fault is 1.2 per unit.

Table 19. Grid-Forming Additional Electrical Specifications

Total Voltage Harmonic Distortion (THD)	< 8% (Individual Harmonic: Max 6%)
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Table 20. Tesla Site Controller Specifications

Standard Tesla Site Controller Enclosure Input Voltage Range	120 V AC +/- 12 V AC
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	240 V AC +/- 24 V AC 480 V AC +/- 48 V AC
Large Tesla Site Controller Input Voltage Range	9-48 V DC
Operating Power Consumption	100 W maximum

5.2 Megapack Wiring

5.2.1 Wiring Interface Summary

Megapack requires power and communications wiring between system components and to the site's main AC panel. The table below summarizes the wiring interfaces needed. All materials are supplied by the contractor.

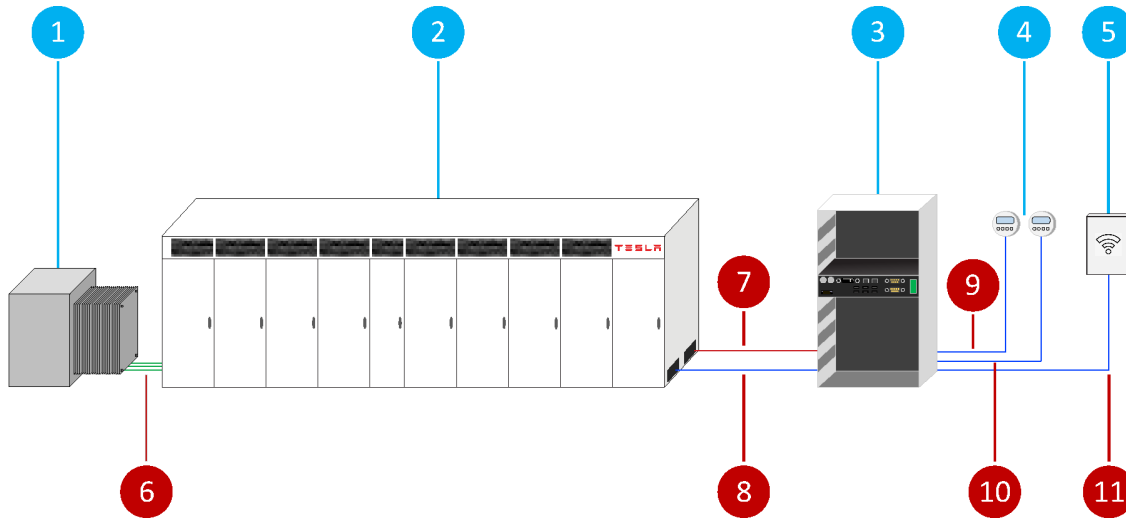
Table 21. System Wiring Interface Summary

From Equipment	To Equipment	Wiring Interface	Minimum Conduit Size
Megapack	Controller or field network enclosure	Communications cable (see Communications Cable Requirements on page 50)	N/A
		DC conductors (for communication control power or microgrid applications)	N/A
	Grid interconnection	AC conductors (3-phase and ground)	N/A
Controller	AC power supply	AC conductors (2-phase and ground)	25 mm (1 in)
	Meters	Communications cable (see Communications Cable Requirements on page 50)	25 mm (1 in)

An example interface plan is shown below:



Figure 35. Example Wiring Interface Plan



1. Transformer
2. Megapack 2 XL
3. Tesla Site Controller
4. Meters
5. Customer Communications
6. AC circuits between Megapack to Transformer
7. DC Power circuit between Megapack and Tesla Site Controller
8. Communications circuit between Megapack and Tesla Site Controller
9. Communications circuit from Tesla Site Controller to site meter as applicable
10. Communications circuit from Tesla Site Controller to battery meter
11. Communications circuit from Tesla Site Controller to customer communication interface

Each Megapack requires a 3-wire circuit (3 phases, ground) connection. Conductors enter Megapack via the above-ground wireway, and terminate on the AC bus bars ([AC Bus Bar Area on page 67](#)) in the Customer Interface Bay:

Megapack is designed with two above-ground wireways through which to route AC cabling to LV transformer termination and communications cabling. The wireways do not require any site trenching construction. Each wireway is 410 mm x 95.6 mm (16 1/8 in x 3 3/4 in) and runs from each side of the Megapack into the AC bus bar area in the Customer Interface Bay. Circuits using wireways must be designed and installed in accordance with local code. For more information on wireway design, refer to *Application Note: Megapack Integrated Wireway*.

5.2.2 Communications Cable Requirements

Communications wiring for the battery system requires special attention to network cabling. Follow the requirements in this section when wiring communications between any components of the system.

5.2.2.1 Cable Specification

Observe the following minimum required or recommended cable specifications:

Table 22. Communications Cable Specifications

Specification	Required	Recommended
Ethernet Standard	Cat 5e	Cat 6a

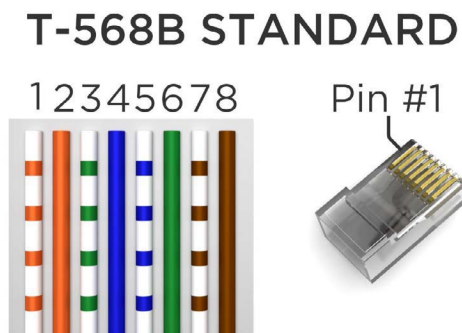


Specification	Required	Recommended
Outer Jacket	Outdoor-rated (UV stable)	Direct burial
Shielding	Yes	Yes: F/UTP
Voltage Rating	600 V	600 V
Maximum Circuit Distance (If exceeded, consult the <i>SCADA Design Manual</i> or project designer.)	100 m (330 ft)	75 m (250 ft)

5.2.2.2 Connection Specification

Communications cables must be terminated according to the T568B Wiring Standard.

Figure 36. T568B Wiring Standard



In addition, ensure the following for proper cable termination:

- Crimp with the tool and forming die that match the connector used
 - Tesla recommends pass-through style RJ45 connectors (such as EZ-38 or VDV826)
- Use **metal** connectors into the Tesla Site Controller network switch
- Use **plastic** connectors into the battery system
- Use strain relief on both ends
- Test all cables with a LAN tester prior to connecting

Figure 37. Metal (1) and Plastic (2) Ethernet Connectors





Figure 38. Ethernet Cables with Metal Connectors into the Tesla Site Controller Network Switch



5.2.3 AC Power Requirements

Dedicate a separate 3-phase circuit for each Megapack. Use 90°C copper or aluminum wire (the internal ambient temperature can reach 75°C). For details on the bus bars provided to connect the circuit to the Megapack, see [AC Bus Bar Area on page 67](#).

 **NOTE:** Each Megapack's power rating is displayed on its label (see [Megapack Labels on page 15](#)). The contractor and engineer of record are responsible for sizing conductors accordingly.

Provide an equipment grounding connection for each power circuit. The grounding conductor size depends on local requirements and the Megapack breaker rating.

 **CAUTION:** If an equipment grounding connection is not provided as part of the power circuit, an appropriate low-resistance ground fault path must be supplied. This solution must be compliant with local requirements and approved by Tesla.

Auxiliary power is not required. Megapack pulls auxiliary power for the control power and thermal management from the local AC, and therefore requires no field work.

5.2.4 Backup Power Design

There are three options when designing backup power for a microgrid or for the Standard Tesla Site Controller Enclosure (see [Standard Tesla Site Controller Enclosure Wiring on page 53](#)).

1. **Megapack-provided power** - Up to 2 A at 24 V power is available as long as the SOC is above 1% and the power supply board is functional. In this state, the energy is provided from the battery modules and is available regardless of the incoming AC power to Megapack.
2. **Customer-provided DC power** - As an alternative to the 24 V DC from Megapack, customers can provide a 24 V DC UPS that terminates into the 24 V DC redundancy module in the Standard Tesla Site Controller Enclosure.



3. **Customer-provided AC power** - Customers can supply power from a UPS or other backed-up power supply (strongly recommended for microgrids and high-reliability sites). This supply must terminate upstream of the AC transformer in the Standard Tesla Site Controller Enclosure. Do not add components to or modify the Standard Tesla Site Controller Enclosure

5.2.4.1 DC Power Requirements

Megapack provides a pair of customer interface terminals to supply 24 V DC power to external devices including the Standard Site Controller Enclosure. The wiring runs from the Standard Site Controller Enclosure to the Megapack customer I/O area. Design the wiring circuit as specified below.

Wiring Specification:

- Wire length of 45 m or less
- 2 mm² [14 AWG]
- Max continuous current: 2 A

Customer Interface Board Terminals:

- AUX 24 V OUT (+)
- AUX 24 V OUT (-)

Refer to [Customer Interface Board on page 64](#) for terminal locations.

5.3 Standard Tesla Site Controller Enclosure Wiring

The Standard Tesla Site Controller Enclosure must be powered from a dedicated 120-480 V circuit. Its internal transformer adjusts voltage as required. The engineer of record must design the means and methods of installing this circuit (see also [Tesla Site Controller Specifications on page 48](#)).

To provide an uninterruptible power supply (UPS) to the Standard Tesla Site Controller Enclosure, the same three types of power supply are available as described in [Backup Power Design on page 52](#).

5.4 Megapack Settings

Megapack settings must be configured in advance by Tesla. Talk to your Tesla representative if any of the default settings should be modified. The defaults for the Megapack follow each region's grid code.

 **NOTE:** Megapack meets the requirements of normal performance category B and abnormal performance category II & III of IEEE 1547-2018 IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces.

5.4.1 Overload

Megapack is capable of providing an additional 20% current overload for up to 10 seconds when followed by and preceded by a minimum of 10 minutes of operation at rated power or lower. In other words, Megapack can handle 1 equivalent full overload every 10 minutes if running at 100% current. This is purely a feature of the Megapack hardware.

Overload capability can be configured by Tesla at the customer's request, with an allowed percentage of overload from 0-20%.



5.4.2 Voltage Ride Through

Megapack High-Voltage Ride Through (HVRT) and Low-Voltage Ride Through (LVRT) settings are adjustable and programmed during the commissioning process to meet IEEE 1547-2018 requirements and/or comply with applicable grid code. All per-unit (pu) voltages are defined with respect to 480 V AC L-L.*

Table 23. Default HVRT Settings

Voltage Range (% of Base Value)	Response Time
Category II	
117.5<V<=120	0.2s
115<V<=117.5	0.5s
110<V<=115	1s
Category III	
110<V<=120	0.083s-12s

Table 24. Default LVRT Settings

Voltage Range (% of Base Value)	Response Time
Category II	
65<=V<88	Linear slope of 8.7s/1 p.u voltage starting at 3s @0.65 p.u
45<=V<65	0.32s
30<=V<45	0.16s
Category III	
70<=V<88	20s
50<=V<70	10s
V<50	0.083s - 1s

 **CAUTION:** Programming to values outside of the default ranges above falls outside the scope of IEEE 1547-2018.

Table 25. Setting Ranges

Parameter	Setting Range	Resolution
Voltages	0.00-Maximum HVRT Allowed	0.01 per unit
Times	0.00-60.00 sec*	0.01 sec

* Time allowed a specific per-unit voltage depends on the nominal voltage rating. Typically, default settings and acceptable ranges are tested as part of a specific certification. For deviations from certified settings in a region, contact your Tesla sales engineer for more information.

The table below specifies the maximum per-unit Overvoltage (OV) withstand and High-Voltage Ride Through (HVRT) value that can be set if Megapack is installed per Tesla's recommendations set in [High-Voltage Ride Through on page 24](#).

Table 26. Maximum HVRT and Overvoltage Withstand Values

Parameter	Per-Unit Maximum Value, Nominal Voltage of 480 V AC
Maximum HVRT Allowed	130% @ 0.2 sec



Parameter	Per-Unit Maximum Value, Nominal Voltage of 480 V AC
Maximum OV Withstand Allowed	130% @ 0.2 sec

Table 27. Trip Accuracy

Trip Setting	Accuracy
Voltage	+/- 1% of nominal voltage
Frequency	+/- 0.01 Hz
Reconnection Time	1% of set point
Trip Time	.16 seconds for <5 seconds 1% of set point for >5 seconds
Active Power	+/- 5% of rated apparent power
Reactive Power	+/- 5% of rated apparent power

Megapack has adjustable Abnormal Voltage settings parameters and ships with the following pre-defined settings:

Table 28. Interconnection System Response to Abnormal Voltages

Default Voltage Range (% of base value)	Default Clearing time (s)	Adjustable Voltage Range (% of base value)	Adjustable Clearing time range (s)
V<45	0.16	0-50	0.16-2 Response faster than 1 second
45<V<50	2	0-50	2-21 Response faster than 1 second
50<V<70	10	0-88	2-21
70<V<88	21	0-88	21-50
110<V<120	2	110-120	1-13
V>120	0.16	Fixed at 120	Fixed at 0.16

In addition to ride through capability, Megapack is capable of adding or removing VARs during VRT events to help support voltage regulation during a fault event. K-factor (reactive current support coefficient) adjusts the amount of reactive current supplied to support grid voltage during a fault. This coefficient is multiplied by the per-unit voltage sag/swell to determine the amount of reactive current (up to rated current) supplied to support the voltage. There are separate coefficients for sag and swell. Setting a K-factor to 0 disables reactive current support

During VRT (grid fault) events, reactive current is automatically prioritized over real current. Megapack can supply negative sequence reactive current (up to rated current per phase) to support voltage imbalance due to grid faults under the same algorithm listed above.



5.4.3 Frequency Ride Through

Megapack Frequency Ride Through settings are adjustable and programmed during the commissioning process to comply with IEEE 1547-2018 requirements per the settings in this section.

Table 29. Default Frequency Ride Through Settings

Default Frequency (Hz)	Default Clearing Time(s)	Adjustable Frequency (Hz)	Adjustable Clearing Time Range(s)
>62	0.16	61.8-66	.16-1000
>61.2	300	61-66	180-1000
<58.5	300	50-59	180-1000
<56.6	0.16	50-57	.16-1000

Megapack has three under-frequency (UF) and three over-frequency (OF) trip points and times, as well as one under-frequency instantaneous trip point and one over-frequency instantaneous trip point. These parameters are configurable to the ranges listed in the table below:



CAUTION: Programming to values outside of the default settings above falls outside the scope of IEEE 1547-2018.

Table 30. Frequency Trip Points

Trip Point	Frequency Range	Time (sec)	Notes
Instantaneous UF Trip	40 Hz-70 Hz	Instantaneous	0.1 Hz resolution, filtering as necessary
UF Trip Time 3	40 Hz-70 Hz	0-600	0.1 Hz and 0.01 second resolution
UF Trip Time 2	40 Hz-70 Hz	0-600	0.1 Hz and 0.01 second resolution
UF Trip Time 1	40 Hz-70 Hz	0-600	0.1 Hz and 0.01 second resolution
OF Trip Time 1	40 Hz-70 Hz	0-600	0.1 Hz and 0.01 second resolution
OF Trip Time 2	40 Hz-70 Hz	0-600	0.1 Hz and 0.01 second resolution
OF Trip Time 3	40 Hz-70 Hz	0-600	0.1 Hz and 0.01 second resolution
Instantaneous OF Trip	40 Hz-70 Hz	Instantaneous	0.1 Hz resolution, filtering as necessary

5.4.4 Harmonics

When in grid-connected mode, Megapack complies with the IEEE 1547 harmonic current distortion requirements.

5.4.5 Grid-Connected Features

5.4.5.1 Active Anti-Islanding

All systems are equipped with active anti-islanding using the Sandia Frequency Shift methodology. Megapack can autonomously and automatically detect an island condition, in which case it isolates from the grid and reports a trip to the user. Megapack can detect the island condition and trip within 2 seconds of island creation. An island condition is detected even if multiple Megapacks are connected on the same island area.



5.4.5.2 Passive Anti-Islanding

Megapack also optionally includes a rate of change of frequency (ROCOF) trip, which is configurable to site and user requirements. ROCOF is disabled by default, but Megapack detects an island condition and trips regardless of whether or not this feature is enabled. The ROCOF parameters available include:

Table 31. ROCOF Settings

Feature Name	Effect	Setting Range	Default
ROCOF Enable	Turns ROCOF on or off	n/a	Off
ROCOF Fault Limit	Sets the rate of change required for a trip	0.1-100.0 Hz/sec	1 Hz/sec
ROCOF Time Delay	Sets how long the rate of change has to be present for the inverter to trip	0-1 second	1 second



NOTE: Anti-islanding can be disabled by a qualified Tesla technician or personnel only.

5.4.5.3 Automatic Grid Reconnection

Megapack automatically reconnects to the grid after a serious grid fault event which causes inverter trip. Megapack's configurable settings determine when and under what conditions it automatically reconnects to the grid.

Table 32. Reconnection Delay Timer Default Settings

Feature Name	Effect	Setting Range	Default
Reconnect Time Delay	The amount of time Megapack waits before reconnection, after the grid returns within the Frequency and Voltage Ride Through windows defined above	0-600 sec	300 sec
Reconnect Min. Voltage	The minimum voltage at which Megapack interprets the grid is within tolerable conditions	0-150%	88.33%
Reconnect Max. Voltage	The maximum voltage at which Megapack interprets the grid is within tolerable conditions	0-150%	105.83%
Reconnect Min. Frequency	The minimum frequency at which Megapack interprets the grid is within tolerable conditions	40-70 Hz	59.3 Hz
Reconnect Max. Frequency	The maximum frequency at which Megapack interprets the grid is within tolerable conditions	40-70 Hz	60.5 Hz

The Reconnect voltage and frequency parameters are also used to determine whether it is safe for Megapack to synchronize to the grid upon initial start-up. Megapack has no time delay to connect to the grid after initial start-up.

5.4.6 Island Grid Controls

Megapack is capable of generating an island grid voltage. Its algorithm has been designed to be compatible with generators, renewable energy sources, and other microgrid assets. There are practical and sizing restrictions with microgrids that must be considered during site design.

Islanding may require additional hardware. For further design questions on this feature, talk to your Tesla representative.

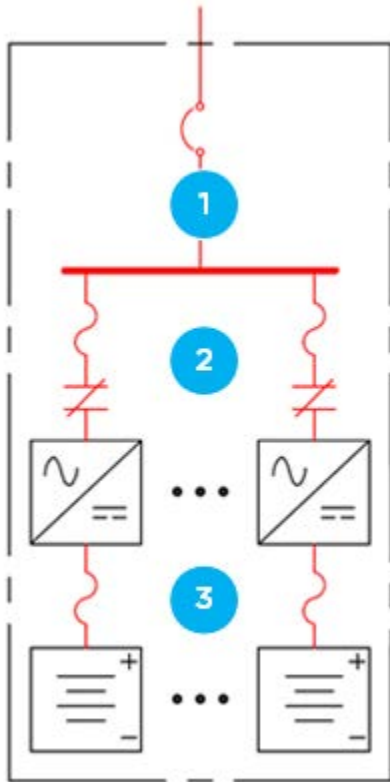


5.5 System Protection Features

Megapack has both hardware and software system protection features as described in this section.

At a high level, the system is protected as illustrated below:

Figure 39. System Protection Features



1. AC circuit breaker
2. AC fuse and contactor
3. Battery module fuse

Protection features include:

- **Battery module overcurrent protection:** The battery modules contain DC single-use fusible links mounted on the battery modules. Fuses must only be serviced by Tesla-approved personnel.
- **Inverter DC protection:** Each inverter module is equipped with its own high-speed DC disconnect.
- **Inverter AC protection:** Each inverter module is equipped with its own AC contactor and AC fuses.
- **Ground fault protection:** Megapack is provided with a DC ground fault detection system. Megapack measures insulation resistance prior to operation and looks for excessive leakage current during operation. Megapack also contains an AC circuit breaker with ground-fault trip settings. See [Megapack AC Circuit Breaker on page 61](#) for more information.

5.5.1 Required Protective Studies

While Megapack provides system protection features as outlined above, the system design engineer is responsible for providing Tesla the following information based on site conditions and protection studies for Tesla's review and approval:



- Arc flash calculations showing the available energy at the Megapack AC bus
- Isolating methods plan or switching scheme identifying the order of opening and who is responsible for each step
- See the *Megapack Deployment Checklists* for more information on submitting switching scheme information

5.5.2 Lightning Protection Design

Protection from direct lightning strike: Megapack enclosures have internal frames that act like a Faraday cage, diverting currents to flow around the internal components but not through them to ground. There is no expected significant impact to Megapack functionality due to a direct lightning strike. Hence, air terminations or lightning masts are not required to protect Megapack enclosures.

Protection from indirect lightning strike: Megapacks are designed with adequate power electronic component sizing and creepage clearance in addition to monitored surge protection on the load side of the Megapack breaker. Tests in accordance with IEEE C62.41.2 and IEEE C62.45 demonstrate that internal circuits and components of Megapack are protected from induced overvoltage from indirect lightning strikes. There is no expected impact on Megapack functionality due to an indirect lightning strike.

5.6 Safety Disconnect Features

5.6.1 Megapack AC Circuit Breaker

The Customer Interface Bay contains an AC circuit breaker that can be locked in the open position.



WARNING: Operating the Megapack AC circuit breaker does not completely de-energize the Megapack. See [Shutting the System Down on page 105](#) for more information.

5.6.2 Enable Circuit

Megapack includes an internal safety circuit called an *enable circuit* (also known as *HVIL* - *high-voltage interlock loop*) that shuts down all major power components whenever the circuit is opened. Certain factors trigger opening this circuit, including opening bay doors or turning the enable switch ([Enable Switch on page 60](#)) off. The Customer Interface Bay door is not part of the enable circuit, as it does not provide direct access to high-voltage equipment. The enable circuit does not trip the Megapack AC circuit breaker.

The Megapack internal monitoring system uses the enable circuit to monitor for critical system faults and de-energize the system if needed. It is not recommended to open a Megapack door during operation as a means of shutting the system down. When deliberately de-energizing the system, when possible, always use the Tesla Site Controller to command a soft shutdown. See [De-Energizing Megapack on page 109](#) for more information.



Figure 40. Enable Circuit Examples



1. The enable circuit is closed. Megapack is able to operate.
2. The enable circuit is open due to an open bay door that is not the Customer Interface Bay door. Megapack is unable to operate.
3. The enable circuit is open because the enable switch in the off position. Megapack is unable to operate.

5.6.3 Enable Switch

The customer I/O area in the Customer Interface Bay includes an enable switch that can interrupt the enable circuit and prevent the system from energizing. This switch is part of the enable circuit, and provides a lockable isolation point for procedures that involve actively working inside Megapack. It is the recommended customer interface to the enable circuit. When the switch is turned off, the enable circuit is open and the product will not actively operate.

WARNING: The enable switch must be turned off at any time when working inside Megapack. See [De-Energizing Megapack on page 109](#) for more information.

5.6.4 Remote Shutdown Terminals

The customer I/O area in the Customer Interface Bay contains a pair of terminals used to perform a remote shutdown, which commands the inverter to cease operation (isolating the Megapack AC bus from the battery DC source), and opens the Megapack AC circuit breaker. This may be performed when placing Megapack into storage, for example.

NOTE: Refer to [Shutting Down for Storage on page 107](#) for more information.

5.7 Customer Interface

All of the interface customers require for installation, operation, and maintenance is located in the Customer Interface Bay, which contains:



Figure 41. Customer Interface Bay



1. [Megapack AC Circuit Breaker on page 61](#)
2. [Customer I/O Area on page 64](#) - where all terminations aside from AC bus bars are made
3. [AC Bus Bar Area on page 67](#) - where terminations to the site distribution transformer or AC distribution panel are made
4. Wireway openings

Installation procedures within the Customer Interface Bay are described in [Installing Customer Connections on page 82](#). Safety disconnect features are described in [Safety Disconnect Features on page 59](#).

5.7.1 Megapack AC Circuit Breaker

The Customer Interface Bay contains a pre-installed AC circuit breaker to provide distribution system protection.

Figure 42. Megapack AC Circuit Breaker





5.7.1.1 Breaker Specification and Design

NOTE: The breaker is not part of the internal Megapack protection scheme and must be programmed on-site according to site-specific plans or engineering guidance.

The circuit breaker is a Siemens UL489 model with an ETU745 programmable protection unit. The unit includes LSIG trip functionality (Long, Short, Instantaneous, Ground Fault) with defaults as described below. Consult the manufacturer's documentation for the full AC circuit breaker specification for *Siemens Low Voltage WL Circuit Breakers*:

<https://new.siemens.com/us/en/products/energy/low-voltage/low-voltage-circuit-breakers/wl-power-circuit-breakers.html> → UL489 content

NOTE: The breaker trip power supply is fed from multiple sources in the customer I/O area. While operating normally (above 0% state-of-charge and not faulted) power is provided by the battery modules. Jumpstart power (if provided) can also be used.

NOTE: The breaker open/close buttons are located 1818 mm (71.6 in) from the base of the Megapack. If the chosen foundation will place these buttons outside of local regulatory height requirements, ensure another method of compliance is provided (raised workstation, remote open/close terminals, etc.).

NOTE: Additional protection or switching means at the output of the Megapack may be required depending on the jurisdiction. Protection and switching philosophy must be verified by the customer's engineer of record and should comply with regional and local codes.

5.7.1.2 Default Breaker Settings

The circuit breaker is shipped with pre-configured general trip settings that must be changed upon arrival through the Electronic Trip Unit (ETU). The ETU has configurable settings for Long-time overcurrent protection (L), Short-time delayed overcurrent protection (S), Instantaneous Overcurrent Protection (I), Ground Fault Protection (G), and other features provided below. The Engineer of Record is responsible for the configuration of the ETU settings for system protection and proper operation. The breaker is pre-programmed with defaults as described below:

Table 33. Default Circuit Breaker Settings

Megapack Pre-Configured Trip Settings	Option Code BB02	Option Code BB01
Frame Size	II - (3000A)	II - (1600A)
I_n (Plug Rating)	3000	1600
I_r (Long-time Current)	1 (3000A)	1 (1600A)
t_r^* (Long-time Delay)	2 sec (18000A)	2 sec (9600A)
I_{sd} (Short-time Current)	1.25 (3750A)	1.25 (2000A)
t_{sd} (Short-time Delay)	0.1 sec (I^2t)	0.1 sec (I^2t)
I_i (Instantaneous Current)	1.5 (4500A)	1.5 (2400A)
I_g (Ground-fault Current)	C (600A)	B (300A)
t_g (Ground-fault Delay)	0.2 sec (I^2t)	0.2 sec (I^2t)
Toggle Switch	I^2t , I_n =off	I^2t , I_n =off



* I_n uses the same delay as t_r , long time delay.

5.7.1.3 Interfacing with Megapack

Megapack interfaces directly with the breaker, and customer connections to the system are detailed in [Customer I/O Area on page 64](#).

Terminals for external control are accessible through the Customer Interface Bay for external closing (Breaker Close) and opening (Remote Shutdown). The Customer Interface Bay allows for the use of external relays to command the breaker however, Megapack will only close the breaker if the system has been deemed safe from the Megapack logic. The Trip command takes precedence over the Breaker Close command.

Upon fault detection, the circuit breaker will trip to open state to isolate the Megapack system. Default settings are described below:

Table 34. AC Circuit Breaker Control Summary

Breaker State	Trip Command	Breaker Close Command	Resulting Megapack Behavior
Open	Not Asserted	Asserted	Depending on desired behavior, the Megapack will perform internal checks and may close the breaker autonomously.
Open	Not Asserted	Not Asserted	The breaker will remain open until the Breaker Close command is asserted.
Open	Asserted	Asserted	The breaker will remain open; the Trip command takes precedence over the Breaker Close command and the internal Megapack breaker operations.
Open	Asserted	Not Asserted	The breaker will remain open; the Trip command takes precedence over the internal Megapack breaker operations.
Closed	Not Asserted	Asserted	Megapack will operate per its internal logic, which may open the breaker.
Closed	Not Asserted	Not Asserted	Megapack will operate per its internal logic, which may open the breaker.
Closed	Asserted	Asserted	The breaker will be tripped.
Closed	Asserted	Not Asserted	The breaker will be tripped.

5.7.1.4 Megapack Breaker Control

In addition to physically actuating the Megapack AC circuit breaker, it is possible to remotely control the breaker over the Tesla Site Controller's DNP3 interface. Work with your project engineer while designing the site to understand the ramifications for your project.

When enabled by Tesla at commissioning, customers can control the Megapack AC circuit breaker on each unit. This feature is only available on the DNP3 interface, not on REST API and Modbus. The Tesla Site Controller uses the DNP3 Latch model.

You can also view the status of the Megapack AC circuit breaker over DNP3. For more information, see the *Megapack 2 Controls and Communications Manual*.



Table 35. Breaker Control Points

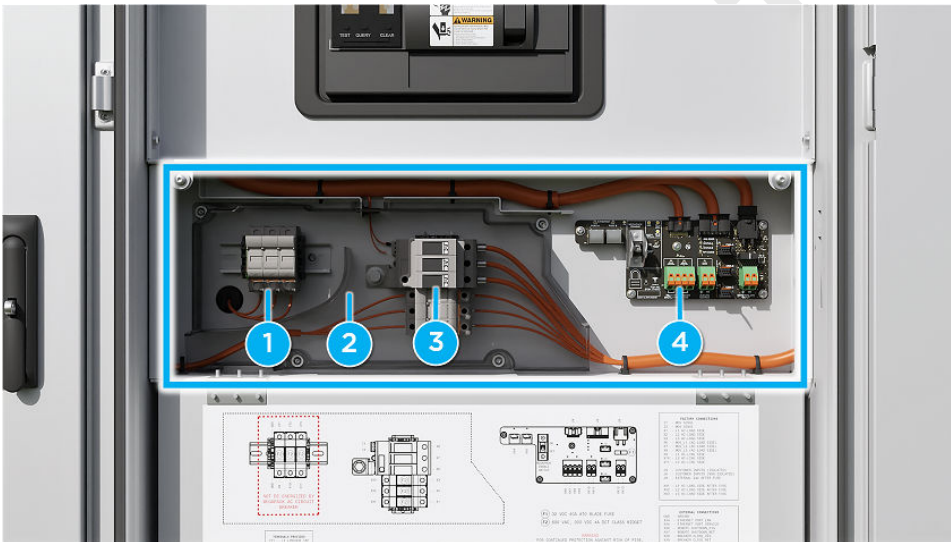
Point Label	Usage
Open Megapack Breaker	Write to 1: Opens the Megapack AC circuit breaker. Must be configured by Tesla at commissioning.
Close Megapack Breaker	Write to 1: Closes the Megapack AC circuit breaker.
Disable Megapack	Write to 1: Brings all battery modules into standby state and brings the thermal HVDC bus down. Warning: This does not replace the need for operators to follow proper de-energization procedures. Refer to De-Energizing Megapack on page 109 for more information.

5.7.2 Customer I/O Area

Each Megapack includes a single customer interface (I/O) area consisting of a wiring compartment inside the Customer Interface Bay. This compartment contains all the necessary non-power connections to be made to the Megapack. The compartment is behind a tool-accessible door and shall only be accessed during operation while following site safety procedures. See [Terminating Auxiliary DC Power and Communications Wiring on page 82](#) to terminate wiring in this area.

The customer I/O area is partially pre-wired. Only some on-site wiring is needed depending on the project needs as referenced below.

Figure 43. Customer I/O Area Components



1. Line-side (grid-side) tap (see [De-Energizing Megapack on page 109](#))
2. Touch-safe cover
3. Load-side (inverter-side) tap (see [De-Energizing Megapack on page 109](#))
4. Customer interface board (see [Customer Interface Board on page 64](#))

5.7.2.1 Customer Interface Board

The customer interface board contains terminals for Tesla use, terminals for external connections, and the Megapack enable switch.

CAUTION: All terminals in this area should be limited to 24 V. **Do not wire higher voltage in this area.** Wiring higher voltage into this area risks damage to terminals and internal components.

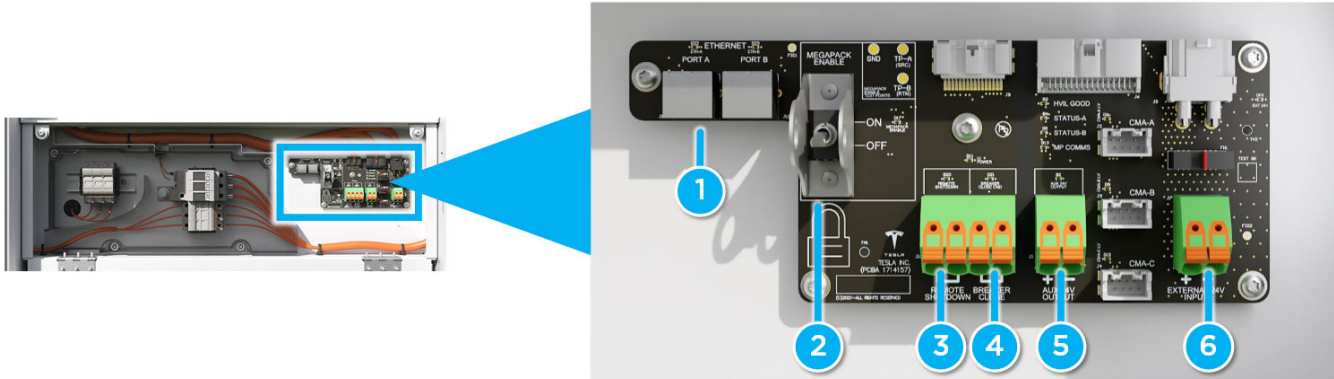


CAUTION: Certain terminals have polarity as indicated on the board. Failure to observe polarity when wiring risks damage to terminals and internal components.

PRE-PRODUCTION DRAFT



Figure 44. Customer Interface Board Detail



1. **Ethernet Ports A (communications) and B (service):** See [Network and Internet Considerations on page 29](#) and [Terminating Auxiliary DC Power and Communications Wiring on page 82](#).
2. **Megapack Enable Switch:** See [Enable Switch on page 60](#) and [De-Energizing Megapack on page 109](#).
3. **Remote Shutdown:** A pair of terminals used to enable remote shutdown capability. See [Safety Disconnect Features on page 59](#) and [Shutting Down for Storage on page 107](#).

Remote shutdown terminals ship with a jumper installed. Removing the jumper commands a remote shutdown by interrupting the integrated 20 mA DC signal circuit to the Megapack bus controller. Closing the circuit must be done in a manner that mimics a jumper.



NOTE: Only remove the jumper if remote shutdown will be configured. If remote shutdown will not be configured, ensure a jumper is present.



NOTE: Do not apply a wetting voltage, as the remote shutdown circuit cannot share a common/return with other circuits.

4. **Breaker Close:** A pair of terminals used to close the Megapack breaker.

Closing the integrated 20 mA DC signal circuit to the Megapack bus controller will close the Megapack breaker if the Megapack enable circuit is closed. If the Megapack enable circuit is open, the circuit breaker will not close.

Closing the circuit must be done in a manner that mimics a jumper.



NOTE: Do not apply a wetting voltage, as the remote shutdown circuit cannot share a common/return with other circuits.

Opening the integrated 20 mA DC signal circuit will NOT trip the breaker.

Megapack comes standard with motor to allow automatic charging of the spring-loaded breaker, otherwise the circuit breaker spring may be manually charged for actuation.



NOTE: Removing the close breaker signal voltage will prevent remote close actuation; it will not trip the breaker.

See [Megapack AC Circuit Breaker on page 61](#) for defaults and additional details regarding the breaker.

5. **Auxiliary 24 V DC Power Output:** A pair of terminals used to supply 24 V DC power to external devices. See [DC Power Requirements on page 53](#) for wiring design.
6. **External 24 V Input:** A pair of terminals that accept input of 24 V. Fused at 40A. Wiring specification: Maximum 10 AWG (6 mm²); minimum 60 W power supply.



5.7.3 AC Bus Bar Area

Megapack is provided with three tin-plated aluminum AC bus bars to connect three phases with conductors with a maximum of 1000 MCM (500 mm²) per connection using NEMA 1.75 in, 2-hole studs. Quantities of connections differ per configuration:

- 2-Hour Megapack: Up to 8 connections per phase
- 4-Hour Megapack: Up to 6 connections per phase



NOTE: For detailed drawings, refer to the *Megapack 2 XL Drawings* on the Partner Portal.

The procedure to connect conductors is described in [Wiring the AC Conductors on page 84](#).



6 Transporting Megapack

6.1 Transportation Reference

These sections provide high-level information and warnings about transportation. Refer and adhere to instructions in *Megapack 2 XL Transportation and Storage Guidelines* to ensure safety of personnel and equipment during transit, staging and storage.

6.2 Logistics

The Megapack enclosure is similar to an ISO standard shipping container, despite its custom dimensions. Megapack has six compliant ISO 1161-type lifting points for effective rigging and to aid transportation and logistics (see *Megapack 2 XL Drawings* for reference). Megapack arrives on-site on a flatbed trailer and can be offloaded by lifting from these six lifting points. Depending on site constraints and lifting equipment, additional rigging (spreader bars, shackles, etc.) may be required. Megapacks delivered outside of the continental U.S. may be delivered within a 40 foot High Cube (HC) shipping container if the as-configured weight permits, or on a flat rack if not.

See *Megapack 2 XL Transportation and Storage Guidelines* for additional details.

6.3 Shipping Guidance

The document *Megapack 2 XL Transportation and Storage Guidelines* provides guidance on shipping and transportation, including the packaged dimensions and weights, allowed storage conditions, and shipping guidance for land and sea. Air transportation is not currently possible with Megapack.

Refer to [Emergency Response Guide on page 17](#) for additional critical shipping information.

 **CAUTION:** Tesla battery systems are subject to specific transportation and storage conditions. Failure to adhere to the conditions listed in the system-specific transportation and storage guidelines may void the warranty.

6.4 Loading and Unloading

The installation contractor is responsible for arranging for unloading Megapack at the site ([Roles and Responsibilities on page 7](#)). See *Megapack 2 XL Transportation and Storage Guidelines* for full instructions on loading and unloading Megapack.

 **DANGER:** Do not attempt to load or unload Megapack without reading, understanding, and following all instructions listed in the *Megapack 2 XL Transportation and Storage Guidelines*. Failure to use proper equipment and procedures, as listed in the *Megapack 2 XL Transportation and Storage Guidelines*, may result in improper loading, severe damage to equipment, and injury or death to personnel.

 **WARNING:** Ensure proper working at heights procedures are followed if personnel need to work on the Megapack roof.

 **WARNING:** When loading and unloading equipment, use suitable lifting equipment and lifting techniques for the weights specified in *Megapack 2 XL Transportation and Storage Guidelines*.



 **WARNING:** Prior to installation, inspect the unit to ensure the absence of transport or handling damage which could affect the integrity of the product. Failure to do so could result in safety hazards. Unauthorized removal of necessary protection features, improper use, or incorrect installation or operation may lead to serious safety and shock hazards and/or equipment damage.

 **CAUTION:** Megapacks must be kept upright at all times. Equipment cabinets contain coolant that could leak and sensitive equipment that could become damaged if not positioned upright. For information on maximum allowable tilt during transportation, see *Megapack 2 XL Transportation and Storage Guidelines*.

6.5 Staging

Schedule Megapack delivery to minimize long-term storage on-site. Any Megapack storage must adhere to the requirements listed in *Megapack 2 XL Transportation and Storage Guidelines*.

 **WARNING:** Do not store or operate Megapack in temperatures over 50C (122F).



7 Accessing the Megapack Enclosure

Megapack is an IP66-rated enclosure ([The Megapack IP66 Enclosure on page 12](#)) and affords high protection against particle and water ingress. This section contains critical information about when and how to properly access Megapack in order to maintain its high protection rating.

After preparing site infrastructure and before beginning work on installing Megapack that involves accessing its enclosure, or any time you need to access the enclosure, ensure that all information in this section is taken into account.

- [Critical Door-Opening Considerations on page 71](#)
- [Accessing the AC Bus Bar Area on page 72](#)
- [Megapack Door Security on page 73](#)



7.1 Critical Door-Opening Considerations

CAUTION: Do not open any Megapack doors or other openings for 48 hours after delivery to site **and** until instructed to do so by Tesla Support per below.

Before opening any door or other opening in the Megapack enclosure:

1. Measure and record the following information:
 - Site temperature
 - Relative humidity
2. Call Tesla Support ([Getting Support on page 18](#)) and provide the temperature and humidity measurements.
3. Proceed only if instructed to do so by Tesla Support.

CAUTION: When opening Megapack doors, do not force door handles. Excessive torque could damage the handle. If the handle does not readily rotate, contact Tesla for assistance.

Do NOT Open a Megapack's doors in any of the conditions below.

CAUTION: Megapack doors should never be opened when it is raining, with no exceptions, to prevent damage to Megapack.

Conditions Under Which Megapack Doors Should NOT Be Opened

Actively raining (see below for details on snowy conditions)



Fog is visibly present



Humidity is above 95% on site (use a humidity sensor that is temperature-compensated and accurate to within 1%)



Minimum cell temperature is **below** the dew point (contact your Project Engineer for calculation support)





In snowy conditions, review the chart below before opening Megapack doors.

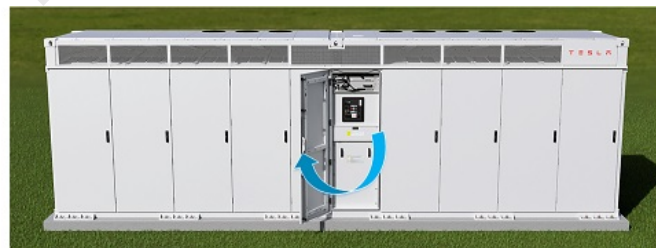
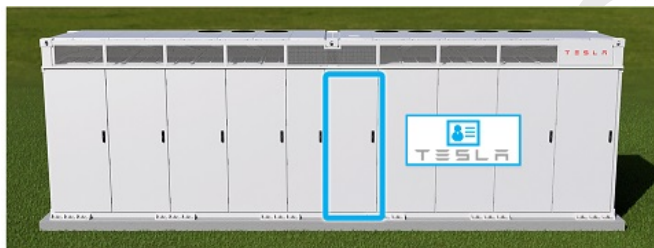
Condition	Action
If it is snowing and relative humidity is above 95%.	Do not open Megapack doors.
If it is snowing, wind speed is below 5 miles per hour, and relative humidity is below 95%.	Megapack doors may be opened only if a tarp/awning is extended above the opened door(s) to prevent any snow from entering the enclosure.
If it is snowing and wind speed is above 5 miles per hour.	Megapack doors may be opened if physical barriers are installed adjacent to opened doors to prevent wind-blown snow from entering the enclosure.

7.2 Accessing the AC Bus Bar Area

⚠ DANGER: BATTERIES ARE ALWAYS LIVE. As a battery, Megapack is always energized, even before you install it at your site. Any time you are actively working on Megapack, you must take the proper precautions to ensure that you have first de-energized it.

To access the AC bus bar area:

1. Refer to the instructions in [Critical Door-Opening Considerations on page 71](#) and open the Customer Interface Bay door if allowed.



2. De-energize Megapack ([De-Energizing Megapack on page 109](#)).
3. Use a 13 mm socket ratchet to remove the 10 screws securing the bus bar access panel to the enclosure. Remove the bus bar access panel.



4. Wait at least five minutes before entering the AC bus bar area.



CAUTION: At all times when performing work inside the AC bus bar area, leave the access panel off and bay door open.

7.3 Handling Desiccant

The AC bus bar area ships with several large bags of desiccant. Remove the bags while working in the area. After work is complete, return as many bags to the area as possible, and dispose of excess bags normally.



CAUTION: Do not place desiccant bags in direct contact with or secured to the AC bus bars.

7.4 Megapack Door Security

All Megapack doors have the ability to be locked. The Customer Interface Bay ([Customer Interface Bay on page 12](#)), as the primary customer interface for Megapack, is not required to be locked. The other doors do not need regular access and should remain closed.



NOTE: Access to the Customer Interface Bay may be subject to local codes and regulations.

Installing Locks

Combination locks are shipped with Megapack, either pre-installed on Megapack doors or delivered in the Accessory Kit (see [Accessory Kit on page 76](#)). Install these locks to ensure doors are not left open unnecessarily:

- **Set combination to 4585** for coordinated access with Tesla field service personnel. Tesla must have ability to unlock doors. If you choose a different combination, advise your Tesla contact in writing.
- If Tesla field service personnel arrives on site and observes locks are not installed, they will proactively install and coordinate with on-site personnel.
- For keyed locks, a double hasp is required to allow Tesla access by unlocking Tesla's lock.



8 Installing Megapack

CAUTION: Do not modify the outer enclosures of any Megapack component. Modification of any sort voids the warranty, as well as the certification and UL/NRTL listing provided with the product.

NOTE: For an abridged, mobile-friendly version of the installation instructions, refer to the *Megapack 2 XL Installation Guide*.

8.1 Preparing for Installation

Before beginning installation procedures, the installation crew must validate aspects of site design and ensure that the site is prepared, minimally as described in this section. Always refer to your engineer of record and plans for each specific site.

Table 36. Site Validation

Site Consideration	Task
IP66 Enclosure Integrity	Ensure that installation crew is advised of critical enclosure access considerations (Accessing the Megapack Enclosure on page 70)
Tools and Accessories	Ensure that required tools and accessories are available including Installation Accessories on page 75
Foundation	Validate foundation per Foundation Design on page 42
Clearance	Validate clearance requirements per Clearance on page 43
Wiring Infrastructure	Inspect the wiring infrastructure as per Wiring Interface Summary on page 49 . If connections cannot be provided as planned, contact Tesla to discuss alternatives. Tesla must approve the non-standard installation before work begins.
Access	Validate accessibility per Access Requirements on page 39
Road Condition Requirements	Verify that site roads along the delivery route are in good working order and free of potholes, rutting, washouts, vehicle traffic, and any other surface hazards of concern (Delivery Access Plan on page 36)
Fencing	Inspect fencing per Fencing Requirements on page 39
Emergency Response Guide	Ensure that the <i>Lithium-Ion Battery Emergency Response Guide</i> is available at the site (Emergency Response Guide on page 17)
Compliance	 CAUTION: Ensure that all installations comply with the appropriate local fire, electrical, and building code requirements.

8.2 Installation Accessories

These accessories are shipped with Megapack and used in the installation process.

8.2.1 Installation Template

This template indicates the anchoring points of a single Megapack for easier anchor hole drilling for either stand-alone or back-to-back configurations. Refer to [Anchoring Megapack on page 78](#) for further instructions.



The template part number is 1807226-00-A.

8.2.2 Accessory Kit

Each Megapack ships with an accessory kit in a box inside the Customer Interface Bay (see [Preparing the Enclosure on page 76](#)). The items inside the kit and what they are used for are detailed below:

Table 37. Accessory Kit Contents

Part Number	Description	Use Case	2hr QTY	4hr QTY
1531223-00-A	M12 DKEPs Nut	Nuts for attaching lugs to AC bus bars	70	52
1008644-00-A	M6 SEMs Bolt	Extra bolts for Customer Interface Bay door	2	2
1023878-00-A	Padlock	Padlocks for all doors	10	10
1075744-00-A	40A Blade Fuse	Spare fuses for customer interface board	2	2
1493272-00-A	Push Mount Cable Tie	Cable ties for harness routing	10	10
1553361-00-C	M8 SEMs Bolt	Extra bolts for bus bar access panel	2	2

8.2.3 Anchoring Hardware

Anchor washers ship in an additional box inside the Customer Interface Bay (see [Anchoring Megapack on page 78](#)).

Table 38. Anchoring Hardware Part Numbers

Part Number	Description
1531294-00-B	Anchor washers

8.2.4 Touch-Up Paint

Each site will be provided with touch-up paint in case of cosmetic damage during the installation process. The paint should only be used on external metal surfaces – do not apply paint to internal components or non-metal components. If damage is observed beyond what the touch-up paint can fix, please contact Tesla.

Table 39. Paint Part Numbers

Part Number	Description
1509391-00-A	Primer
1117378-00-A	Paint

See color information in [Enclosure Colors on page 33](#).

8.3 Preparing the Foundation

After preparing the site infrastructure including inspecting the foundation, prepare the foundation for installation, using the Megapack installation template as a guide (see [Installation Template on page 75](#)).

Many common foundation anchors require a hole to be drilled in the foundation. Depending on the tool used, this may need to be done prior to lifting and placing Megapack (see [Anchoring Megapack on page 78](#)).

8.4 Preparing the Enclosure

Prepare the enclosure before lifting, positioning, and anchoring Megapack.



NOTE: Refer to *Megapack 2 XL Drawings* on the Partner Portal for exact dimensions.



CAUTION: Megapacks are available in many configurations and can differ based on characteristics such as power or energy rating and weight.

To prepare the enclosure, first locate and validate the enclosure configuration to be installed at the given site location.

The Megapack enclosure is now ready to be lifted and set onto the foundation.

8.5 Lifting and Positioning the Enclosure

Use these instructions to lift and position the enclosure. Follow the procedures in *Megapack 2 XL Transportation and Storage Guidelines* for additional details on lifting and positioning.



CAUTION: Modification of ISO corners in any way is not permitted.

To lift and position the enclosure:

1. Using a lifting rig and crane, lift Megapack using the top six ISO corners and locate it over the intended foundation in its designated area.



CAUTION: Ensure that the lifting hardware has enough clearance to be removed once Megapack is properly positioned. Shackle pins may bind on neighboring Megapack if installed pin-side down.

2. Position the enclosure, taking any required anchor locations into account (see [Anchoring Megapack on page 78](#)).

8.6 Sealing the Wireway Openings

Depending on the site condition, one or more of the options below may be appropriate to properly seal Megapack wireway openings. Always defer to project-specific sealing details or the engineer of record.

- Reinstall the wireway cover plates, if they were previously removed.
- If the wireway covers have been modified, ensure that the IP66 rating is maintained using appropriate products before reinstalling.
- If reinstalling the wireway covers is not possible due to construction conditions, temporarily seal the openings using an impermeable membrane and tape until a permanent seal is installed.
- If wireway cover cannot be permanently reinstalled after Megapack construction is completed, install an alternate permanent seal:
 - Seal the wireway openings using a material identified by the sealant manufacturer as compatible with conductors.
 - Ensure that the wireway openings are completely sealed from moisture, dirt, and pest ingress.
- After commissioning, maintain this seal for the life of system.
- **If any of the above is not possible or you have any questions about maintaining Megapack's required IP66 rating, contact your Tesla project engineer.**



WARNING: It is common for there to be a lag in the time between when Megapack is placed on the foundation and when conductors are pulled and terminated. It is critical for the installer to ensure that wireway openings are sealed whenever work is not actively being performed per requirements above.



NOTE: Use a T30 Torx bit for the M6 screws that secure the wireway covers.

NOTE: Photos will be required to be sent to Tesla when completing the *Megapack Deployment Checklists* to verify all Megapack openings are sealed correctly.

8.7 Anchoring Megapack

Megapack may require anchoring to its foundation. If required by site design or the engineer of record, follow these instructions to anchor Megapack.

- NOTE:**
- Anchors can differ by site and are therefore are not provided by Tesla. Stainless steel is required. Hilti KB-TZ SS wedge anchors or HAS-R 304/316 SS and Hilti HIT-HY 200 epoxy anchor system are recommended.
 - Anchor washers are Tesla-provided, and are shipped in a box inside the Customer Interface Bay.
 - Many common anchors can require holes to be pre-drilled into the foundation before anchoring. Ensure that the foundation has been properly prepared, if so (see [Preparing the Foundation on page 76](#)).
 - Use the *Installation Template* to assist in this process (see [Installation Template on page 75](#)).
 - Refer to the *Megapack 2 XL Drawings* on the Partner Portal for exact anchor and bracket specifications and dimensions.

8.7.1 Anchoring to Foundation

Anchor brackets are flanges welded on to the base of Megapack that can be used to hold it to its foundation. To anchor Megapack to its foundation:

1. Identify the required anchors and anchor brackets per site design, using the Installation Template and images below as a guide.

CAUTION: Ensure that the Installation Template is thoroughly extended, with its ends pulled tight and the surface smoothed and wrinkle-free.

Figure 45. Available Anchor Brackets (plan view - as seen from top)

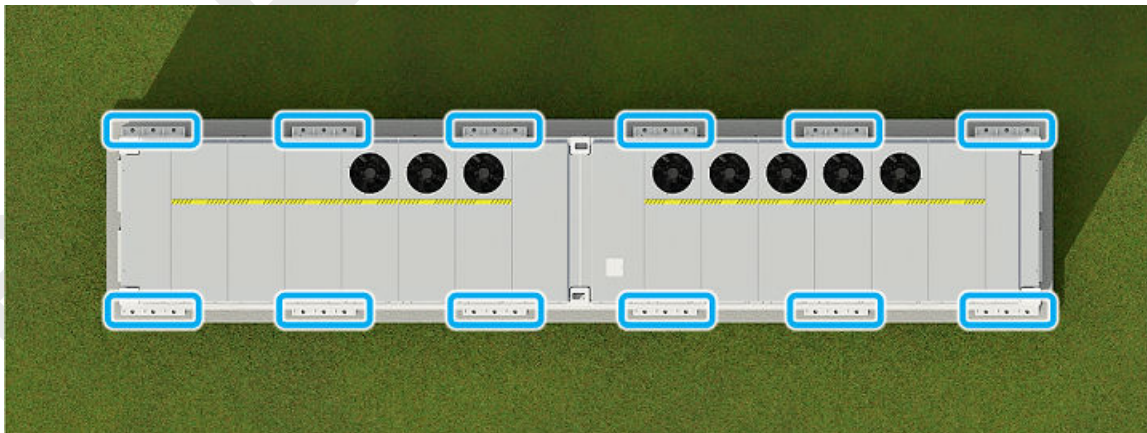
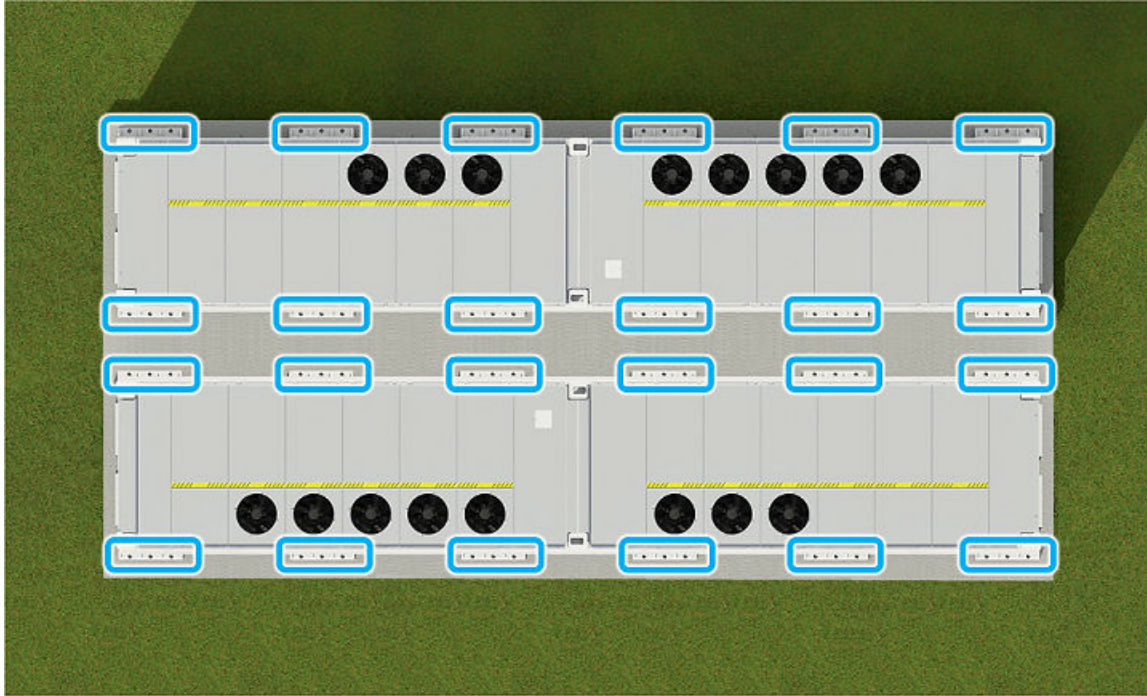


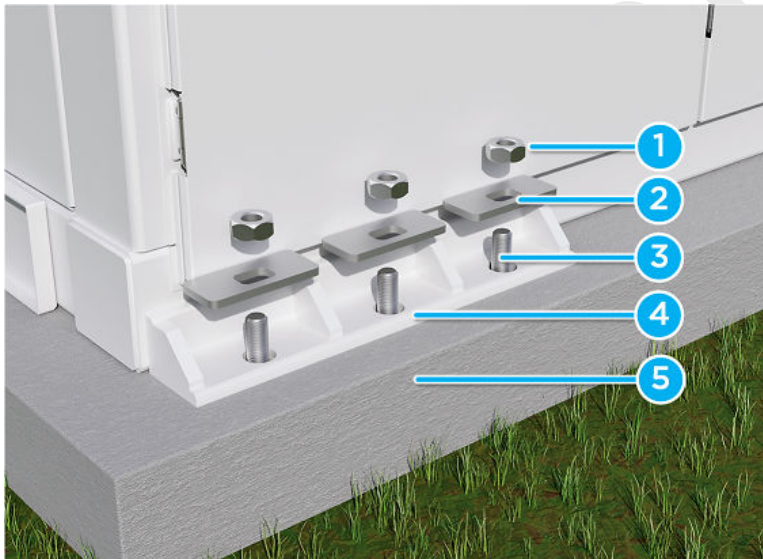


Figure 46. Available Anchor Brackets: Back-to-Back Installation



2. Secure the required anchors through the washer, the bracket, and into the foundation:

Figure 47. Megapack Anchoring Detail



1 - Anchor nut (customer-provided); 2 - Anchor washer (Tesla-provided); 3 - Anchor (customer-provided); 4 - Anchor bracket (steel, welded to Megapack); 5 - Megapack foundation. For anchor specifications, refer to [Anchor Requirements on page 41](#).

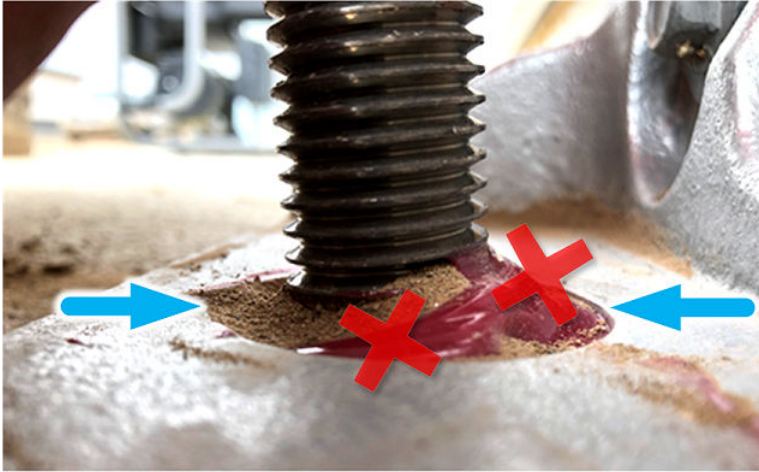


NOTE: Megapack can support up to three anchors per anchor bracket depending on site design.

- For each anchor bracket, ensure that the anchor and anchor washer are secure such that the bracket touches the foundation with no gaps. If a gap is present, the bracket **MUST** be shimmed to ensure there is no gap.
- Epoxy overflow or other debris can prevent the anchor washer from sitting flush on the anchor bracket. Ensure any debris is chipped away so that the washer sits flush to the bracket:

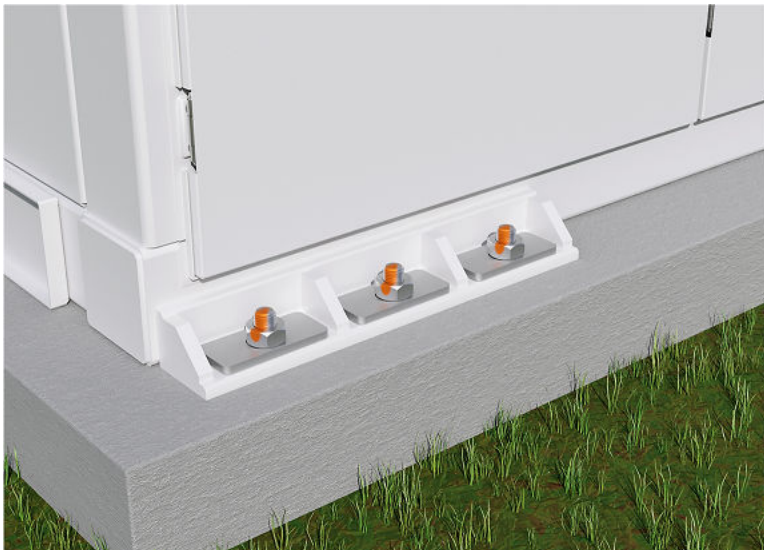


Figure 48. Epoxy Overflow



3. Using the anchor washer (note this washer is eccentric to provide as much mounting tolerance as possible), install and torque the anchor, then leave a paint mark:

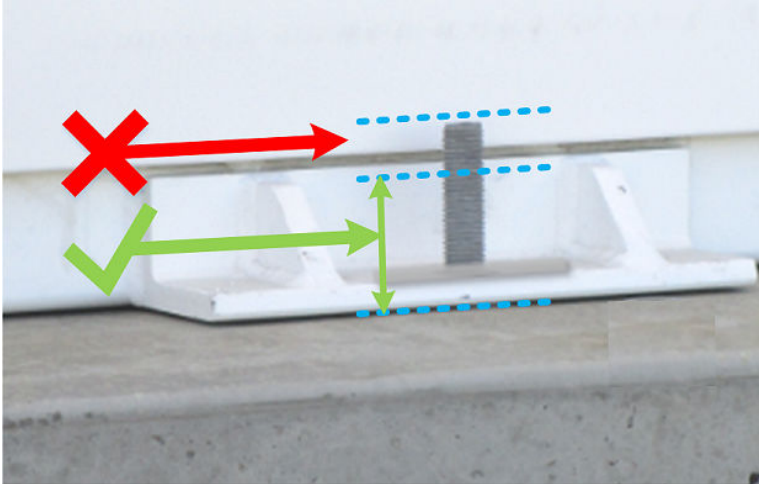
Figure 49. Torque and Leave Paint Mark



- Ensure that anchor hardware does not protrude greater than 76 mm (3 in) above the foundation surface:



Figure 50. Anchor Protrusion



4. Repeat this process for all required anchors.



9 Installing Customer Connections

All of the wiring and customer connection tasks after installing Megapack take place in the Customer Interface Bay.



9.1 Terminating Auxiliary DC Power and Communications Wiring

Terminate auxiliary DC power and communications cables by following a routing path into the customer I/O area as described below.

Figure 51. Customer I/O Routing



1. Customer I/O area panel
2. Screws securing the panel
3. Customer I/O area
4. Cable routing path
5. AC bus bar area
6. Wireway openings

To begin these procedures, access the AC bus bar area and customer I/O area:

1. De-energize Megapack ([De-Energizing Megapack on page 109](#)).
2. Access the AC bus bar area ([Accessing the AC Bus Bar Area on page 72](#)).



3. Using a T30 Torx bit, remove the two screws (callout #2) that secure the panel covering the customer I/O area, and open the panel.

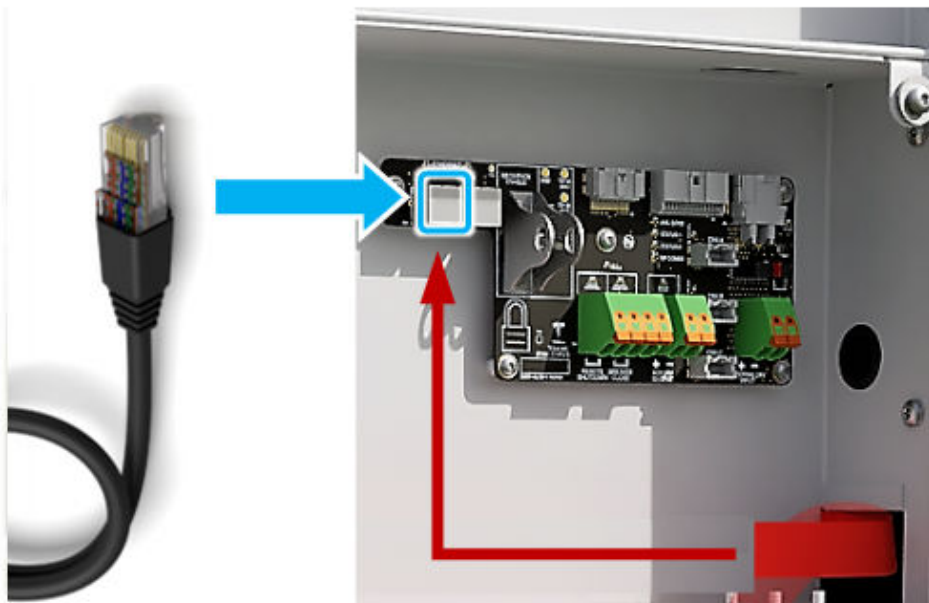
To terminate auxiliary DC power wiring:

1. Route the cable from the AC bus bar area along the wall as indicated (callout #4).
2. Pull the cable into the customer I/O area (callout #3).
3. Connect it to the desired terminal.

To terminate communications wiring:

Tesla requires Megapack to be connected to the Tesla Network along with the LAN 2 port of the Tesla Site Controller.

1. Route the Tesla Network Ethernet cable into the AC bus bar area and along the wall as indicated (callout #4).
2. Pull the cable into the customer I/O area.
3. Terminate the cable with a plastic RJ45 connector using an EZ-RJ45 crimp tool or equivalent (see [Communications Cable Requirements on page 50](#) for more information).
4. Test the connection using a Fluke CIQ-100 tool or equivalent. Leave a paint pen mark on the cable once it passes.
5. Install the cable into the communications port (*Port A*, or the **left** Ethernet port) on the customer interface board ([Customer Interface Board Detail on page 66](#)).



6. Secure (close and reattach) the bus bar access panel, secure the customer I/O area panel, and close the Customer Interface Bay door.



9.2 Wiring the AC Conductors

⚠ DANGER: Each Megapack requires a 3-wire circuit (3 phases, ground) connection, and the transformer secondary is required to be a grounded wye. Conductors enter the AC bus bar area via the wireway openings and terminate on the AC bus bars.

Prepare the area:

1. De-energize Megapack ([De-Energizing Megapack on page 109](#)).
2. Verify that the conduit or conductor wireway is prepared for the circuit conductors, including ensuring that nothing can damage the conductors during work.

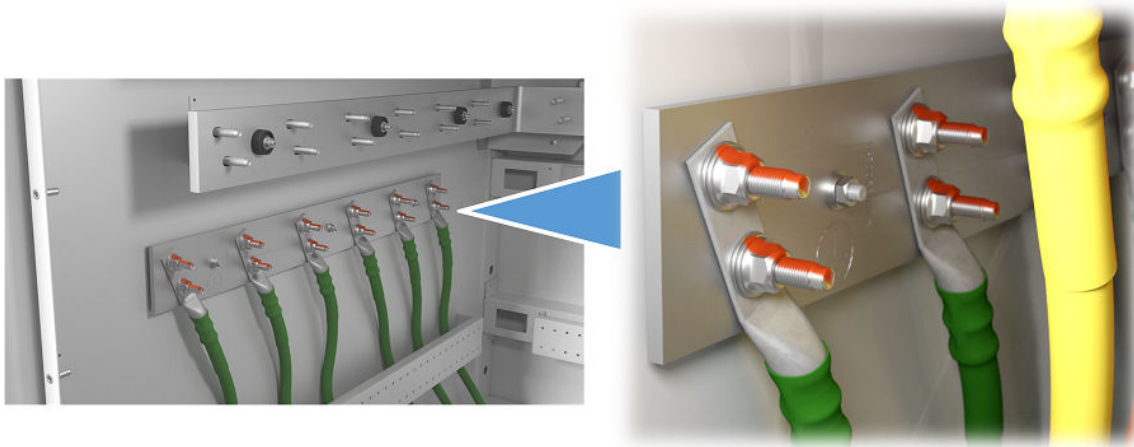
Terminate the conductors:

1. Access the AC bus bar area ([Accessing the AC Bus Bar Area on page 72](#)).
2. Pull conductors into the AC bus bar area.
3. Perform insulation resistance (Megger) testing to at least 500 V on the AC conductors with the Megapack breaker in the open position:
 - Use the insulation resistance procedure described by the engineer of record if available.
 - If there is no guidance available from the engineer of record, values under 1 MOhm are considered a failure and must be investigated and rectified.

Document the insulation resistance results in the *Megapack Deployment Checklists*.

4. Trim, strip, and dress the conductors to prepare them for termination. Apply anti-oxide coating to each lug in the Megapack and switchgear if required by the AHJ.
5. Use an 18 mm socket to connect all AC conductors to the Megapack bus bars using an approved connection method (compression lugs are recommended). Each Megapack accessory kit includes the M12 hardware for attaching the AC lugs to the AC bus (see [Installation Accessories on page 75](#)).

Figure 52. AC Conductors and AC Ground Lugs: Example



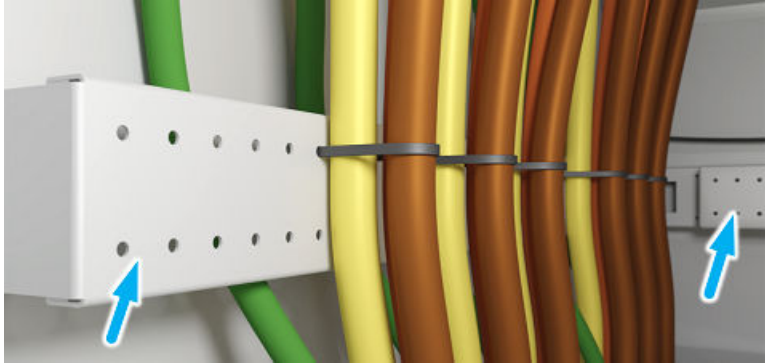
NOTE:

- The minimum electrical clearance between one bus bar and the next bus bar, including fasteners, is 13 mm (0.51 in).
- Where possible, arrange the AC conductors to minimize bend radii.
- Where metal conduits are used, conduits must have insulated ground bushings connected to equipment grounding conductors.

6. Torque all AC conductor lugs to 55 Nm (40 ft lb) and mark hardware (threads and nut) with paint pen.
7. If necessary, fasten the conductors to the strain relief bar with cable ties as illustrated below:



Figure 53. AC Conductors Fastened to Strain Relief Bars in the Customer Interface Bay: Example



CAUTION: Megapack output AC voltage is phase-rotation sensitive. All Megapacks at a given site must have the same phase rotation, either clockwise or counter-clockwise.

Check and test:

1. Check that the phases of the Megapack are connected to the corresponding utility phases.
2. Perform a grounding resistance test. The system should show:
 - < 1 Ohm between each earth bar and the main system earth bar
 - < 1 Ohm between the system's exposed metal and the local earth bar
 - Only one neutral earth bond per system, at the transformer secondary.



CAUTION: Once all power and communication wires are installed, **ensure that there is no debris in the Customer Interface Bay, and ensure that the wireway openings are sealed** (see [Sealing the Wireway Openings on page 77](#)). Secure (close and reattach) the bus bar access panel, secure the customer I/O area panel, and close the Customer Interface Bay door.

9.2.1 Providing the Arc Flash Label

An arc flash label must be affixed on the outside of each Megapack. The label must be applied to the outside of the Customer Interface Bay door (on the exterior of the enclosure), where it must be clearly visible to personnel prior to opening Megapack doors. The label and the information it records are to be provided by the engineer of record and must reflect the conditions of the AC bus bar area. The label must conform to local requirements and at a minimum must designate:

- Working distance
- Expected incident energy
- Recommended PPE
- Nominal voltage



Figure 54. Example Location of Arc Flash Label



-  **NOTE:** Arc flash labels are required for Tesla field service technicians to perform any work on Megapack. If the label is not present, Tesla will require Megapack and upstream devices to be completely de-energized before any work on Megapack can take place.
-  **NOTE:** Labels must be provided in accordance with local codes and standards, examples of which include:
- NFPA70E & IEEE 1584
 - CSA Z462-12
 - AS -1319



10 Installing the Tesla Site Controller

This section describes how to install the Tesla Site Controller ([Tesla Site Controller on page 13](#)):

- For the Standard Tesla Site Controller, follow all guidance.
- For the Large Tesla Site Controller, the customer can choose to follow the guidelines of the *SCADA Design Manual* to design and install their enclosure, then also observe the guidance in [Communication Wiring on page 98](#).

10.1 Installing the Standard Tesla Site Controller Enclosure

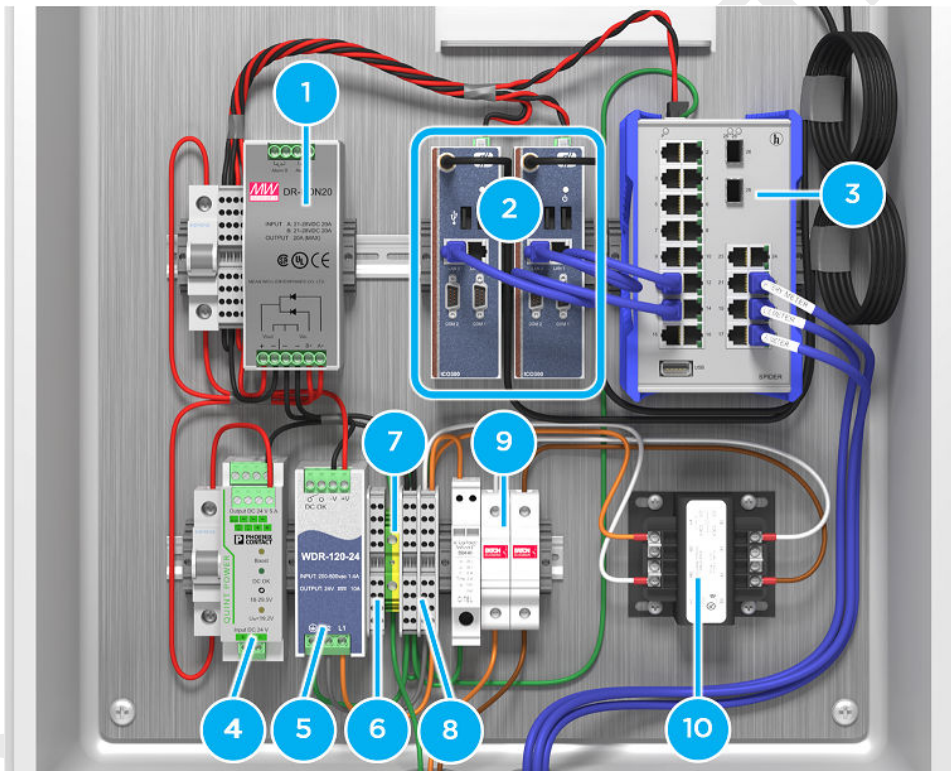
The Standard Tesla Site Controller Enclosure is provided as a pre-assembled NEMA 3R enclosure. As it weighs over 18 kg (40 lbs), use proper precautions when carrying and lifting the enclosure. Always refer to the approved plan set to confirm location, mounting, and conduit requirements.

10.1.1 Standard Tesla Site Controller Enclosure Components



NOTE: This enclosure is only delivered for sites up to 18 MW. For sites greater than 18 MW, the controller computer is delivered stand-alone as the Large Tesla Site Controller, and the customer assembles components as required for the site using the guidance in the *SCADA Design Manual*.

Figure 55. Standard Tesla Site Controller Enclosure Interior



1. Diode redundancy module
2. Standard Tesla Site Controllers (primary and secondary computers)



NOTE: Some versions of the Standard Tesla Site Controller Enclosure may not include a secondary computer. Renders in this publication may contain only one computer although you may have a dual-computer configuration.



3. Network switch
4. DC-DC converter and surge protector
5. 24 V power supply
6. Jumpstart terminals (Tesla-internal use)
7. Grounding terminals
8. Transformer terminals
9. Fuse block for incoming AC
10. Transformer (120/240/480 V AC)

10.1.2 Mounting the Enclosure

Mount the Standard Tesla Site Controller Enclosure as follows:

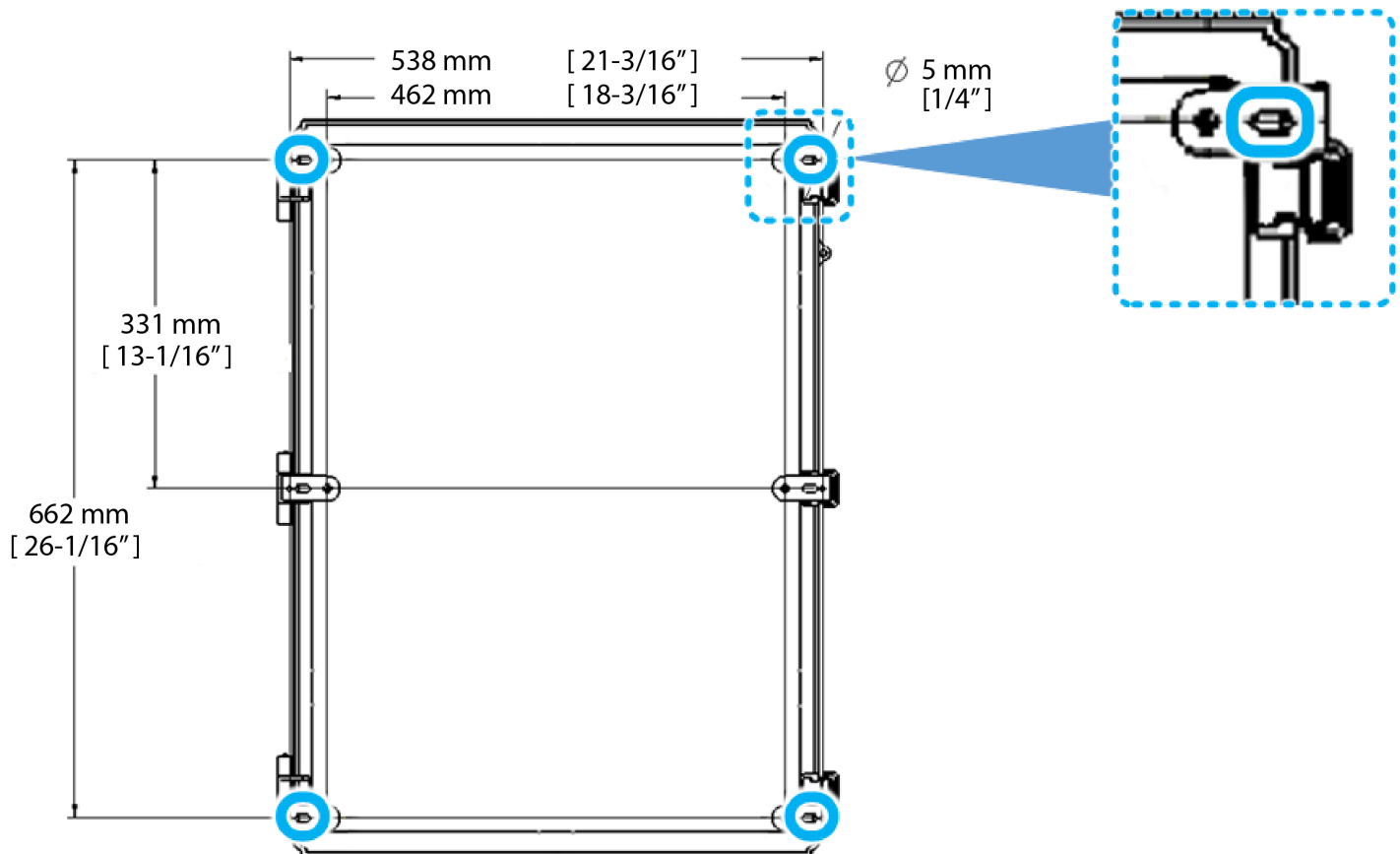
1. Use the 4 mounting holes to mount the enclosure vertically on an H-frame, rack, or wall using appropriate fasteners capable of supporting a minimum of 22 kg (48 lb).



NOTE: Mounting brackets are included with the Standard Tesla Site Controller Enclosure.

2. Ensure that the enclosure is level, plumb, and securely fastened.
3. Ensure that the enclosure has enough clearance to allow for the door to open fully, beyond 90°.

Figure 56. Standard Tesla Site Controller Enclosure Mounting Holes





10.1.3 Installing Conduit

NOTE: Conduit requirements will vary from site to site and will largely depend on the Tesla Site Controller's function on that site. Always refer to the approved plan set to confirm what is required.

CAUTION: Tesla recommends that all incoming conduits enter the Standard Tesla Site Controller Enclosure from the bottom to reduce risk of water ingress.

Drill out appropriately sized conduit entry holes at the bottom of the Standard Tesla Site Controller Enclosure and install gland nuts. Conduit should be routed such that it mates with gland nuts at the bottom of the Standard Tesla Site Controller Enclosure.



NOTE: Drill and install conduit gland couplers as needed.

Confirm that all conduits are adequately supported and securely fastened to the Standard Tesla Site Controller Enclosure.

10.1.4 Power Transformer Wiring

The Standard Tesla Site Controller must be powered from a dedicated 120-480 V AC circuit. Its internal transformer adjusts voltage as required. The engineer of record must design the means and methods of pulling this circuit.

First prepare AC power and ground terminations, then follow the wiring instructions depending on AC voltage as follows:

- [Configuring for 401 to 480 V AC on page 90](#) (default factory configuration)
- [Configuring for 231 to 400 V AC on page 92](#)
- [Configuring for 120 to 230 V AC on page 93](#)

10.1.4.1 Terminating AC Power and Ground Conductors

Terminate the AC power and equipment ground conductors before wiring the transformer.

1. Route the AC power and ground conductors from the primary power source into the Standard Tesla Site Controller Enclosure.



- Strip 15 mm (9/16 in) of insulation from the conductors.

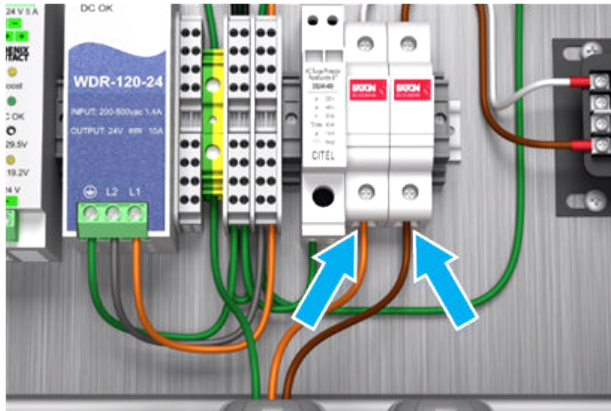


NOTE: AC power and ground conductors must have a cross-sectional area of 2.5 mm²-4 mm² (14 - 12 AWG).

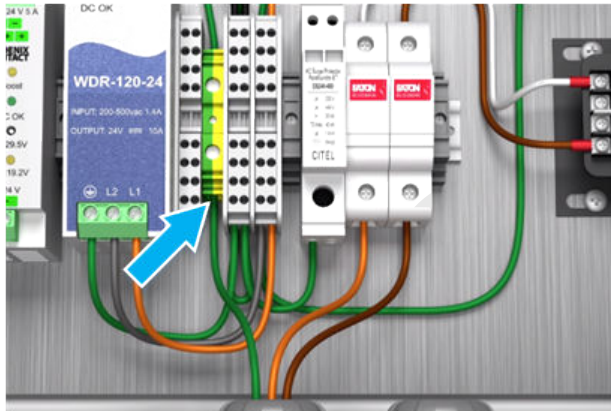
- Terminate the 2 AC conductors to the terminals at the bottom of the AC fuse block (#2) according to the label in the box.



NOTE: Ensure that the conductors are appropriately colored or labeled to clearly indicate their phase. Colors and labels may vary by region.



- Terminate the ground (green or green/yellow) conductor at the bottom of the ground block.



- To confirm that the transformer is configured for the default configuration, check that the pre-installed fuses are present and rated for the correct voltage.

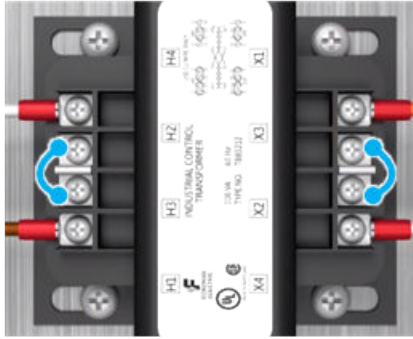
10.1.4.2 Configuring for 401 to 480 V AC

By default, the transformer is configured for 401-480 V AC. Follow these instructions if you need to reconfigure to this voltage range.

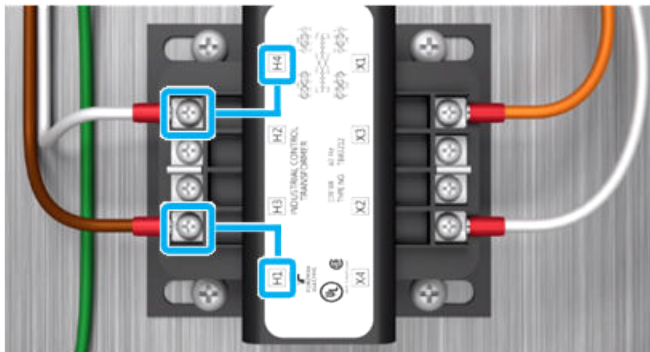
- Confirm that the pre-installed fuses in the fuse block are the default KLDR 0.5A, Tesla PN 1053860-00-A.



2. Verify that the pre-installed taps connect the middle 2 terminals (**H2-H3** and **X2-X3**).



3. On the left (*H*) side of the transformer, verify that the pre-installed *brown* and *white* wires are terminated in the outermost terminals:
 - *Brown* in *H1*
 - *White* in *H4*



4. On the left (*H*) side of the transformer, verify that the pre-installed *brown jump wire* is connecting the two innermost terminals (*H2* and *H3*).
5. On the right (*X*) side of the transformer, verify that the *orange* and *white* wires are terminated in the outermost terminals:
 - *Orange* in *X1*
 - *White* in *X2*

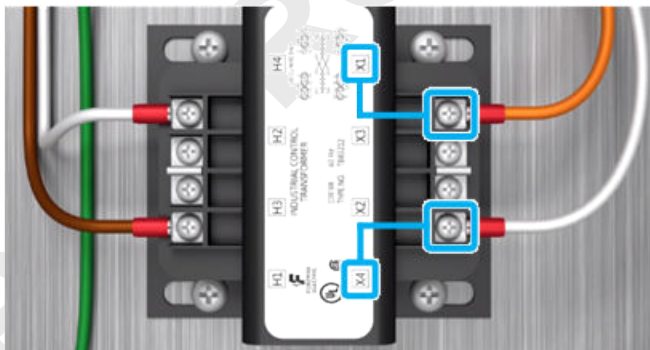
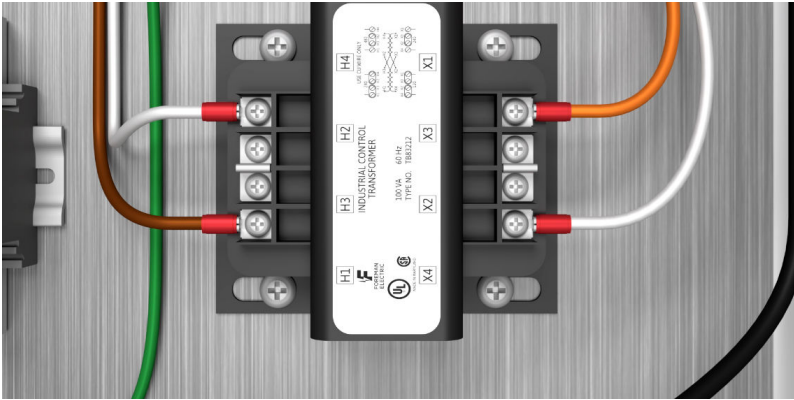


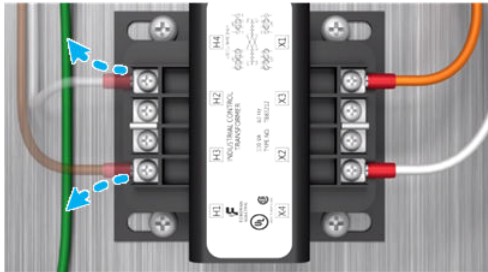


Figure 57. Transformer Configured for 401-480 V AC (default)

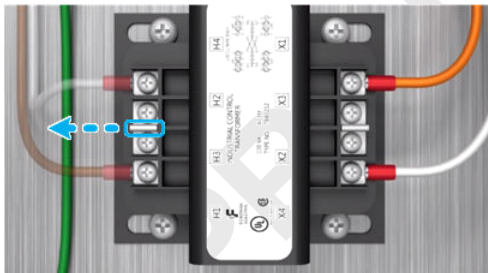


10.1.4.3 Configuring for 231 to 400 V AC

1. Replace the default fuses in the fuse block with the appropriate ones from the accessory kit: ATDR 3.5A, Tesla PN 1479411-00-A.
2. From the left (*H*) side of the transformer,
 - a. Remove the pre-installed *brown* and *white* conductors, and the *brown jump wire* (if present) connecting the middle terminals.



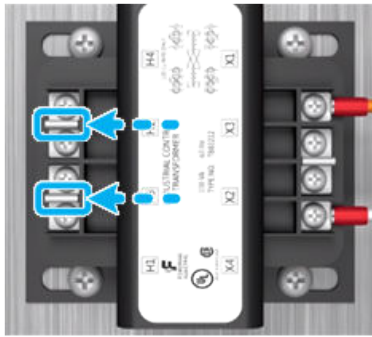
3. Remove the single tap connecting the two middle (*H2-H3*) terminals.



4. Install two taps to connect the middle to outer terminals (*H1-H3* and *H2-H4*).

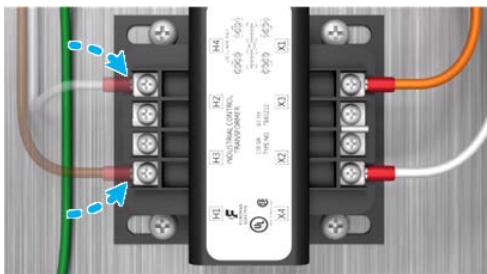


NOTE: Taps are included in the bag that comes in the Standard Tesla Site Controller Enclosure.



5. On the left (H) side of the transformer, install the *brown* and *white* conductors in the outermost terminals:

- *Brown* in H1
- *White* in H4



NOTE: Verify that the right side of the terminal has the orange conductor terminated in X1, white conductor terminated in X4, and a single preinstalled tap connecting the 2 middle terminals (X2 and X3).

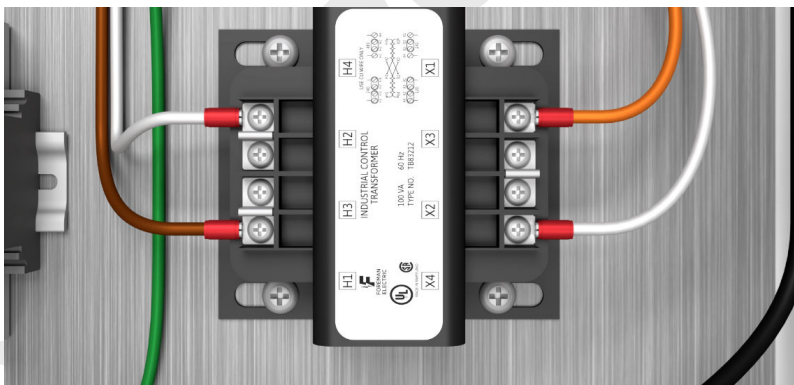


CAUTION: Do not reconnect the brown jump wire.



CAUTION: 240 V AC may only be in LINE to NEUTRAL form. Wiring as split phase can cause a short in the system.

Figure 58. Transformer Configured for 231-400 V AC

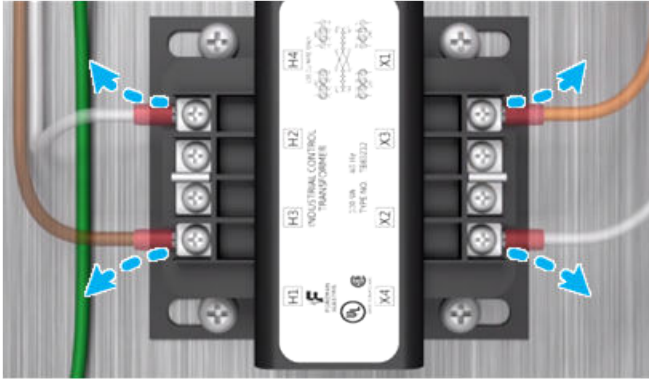


10.1.4.4 Configuring for 120 to 230 V AC

1. Replace the default fuses in the fuse block with the appropriate ones from the accessory kit: ATDR 3.5A, Tesla PN 1479411-00-A.
2. From both sides of the transformer,



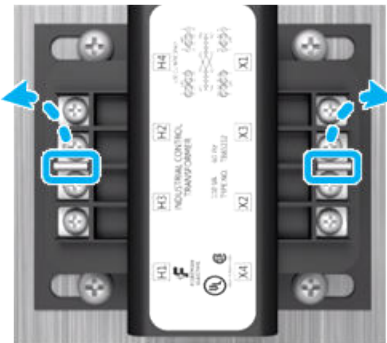
- a. Remove all wiring, including the *brown jump wire* if present.



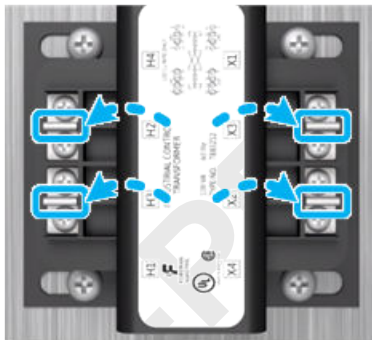
- b. Remove the single taps connecting the two middle (H2-H3 and X2-X3) terminals.



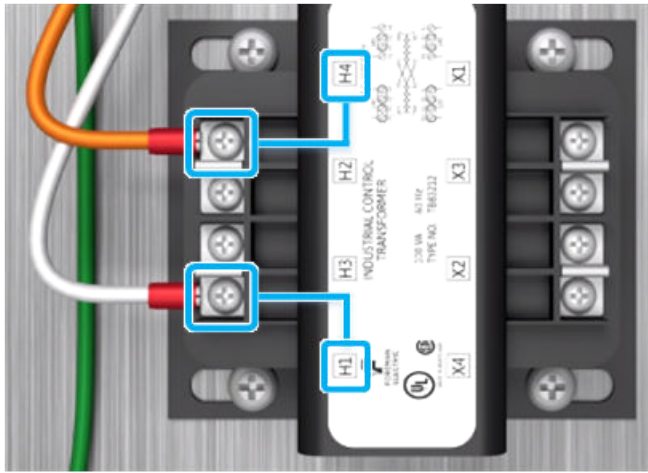
NOTE: Taps are included in the bag that comes in the Standard Tesla Site Controller Enclosure.



3. On each side of the transformer, install two taps to connect the middle to outer terminals (H1-H3 and H2-H4).



4. Relocate the *orange and white* conductor pair that were on right side of the transformer to the *left (H)* side:
 - *Orange* in the *top (H4)* terminal
 - *White* in the *bottom (H1)* terminal



5. Relocate the *brown and white* conductor pair that were on left side of the transformer to the *right (X)* side:

- *Brown* in the *bottom (X4)* terminal
- *White* in the *top (X1)* terminal

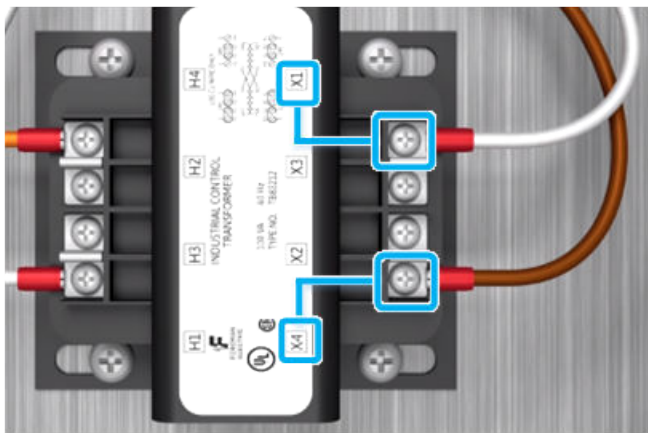
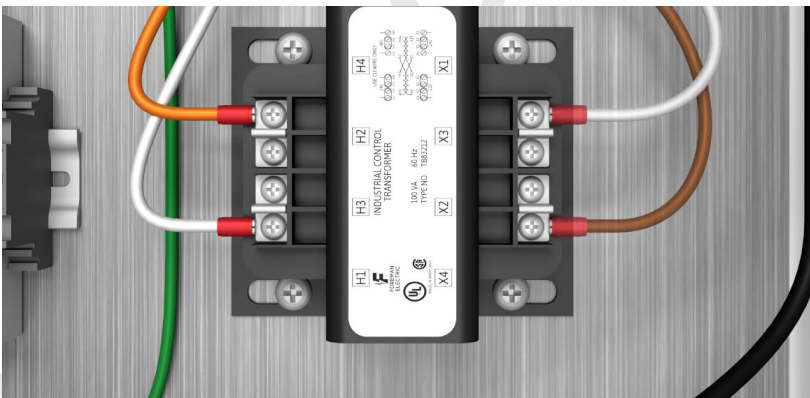


Figure 59. Transformer Configured for 120-230 V AC



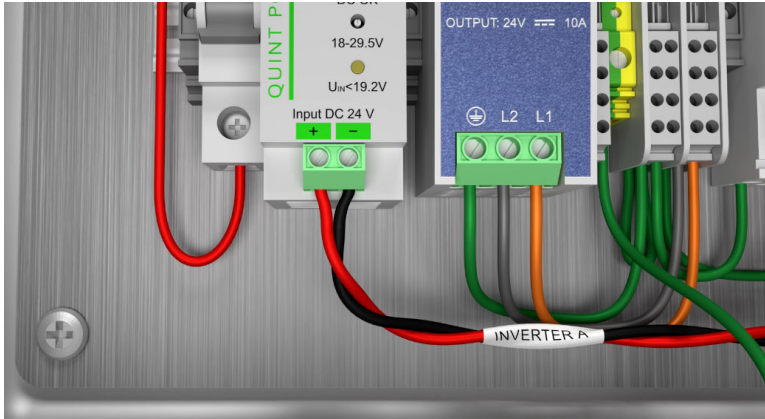
10.1.5 Wiring for Backup Power



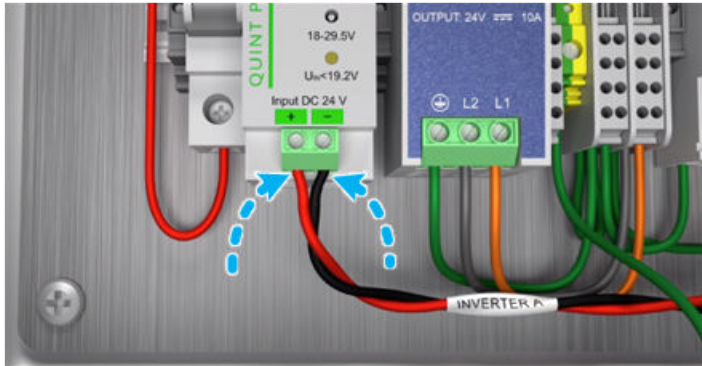
NOTE: 24 V DC backup power is required for battery systems configured as microgrids.



Figure 60. Back-Up Power Wiring (24 V DC)



1. Pull the conductors ([DC Power Requirements on page 53](#)) through the conduit from Megapack.
2. Terminate the two conductors at the auxiliary *in* and *return* terminals on the DC/DC converter in the Standard Tesla Site Controller Enclosure:



3. Connect the conductors to the Megapack backup terminals.

10.1.6 Relocating the Antenna

One antenna is provided for each Standard Tesla Site Controller, including a magnetic base and 10' (3 m) of coaxial cable. This is typically mounted to the controller's internal back plate.

 **NOTE:** If your enclosure contains two Standard Tesla Site Controllers, you will have two antennas.



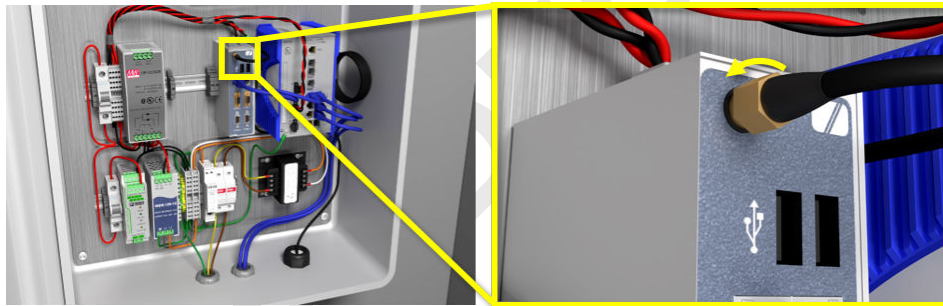
Figure 61. Antenna Mounted to Internal Back Plate (Default Position)



To improve cellular connectivity, one may relocate the antenna outside of the enclosure. Follow these instructions to relocate the antenna:

1. Open the Standard Tesla Site Controller Enclosure cabinet door.
2. Locate the old internal antenna plugged into the computer terminal and fully unscrew it at the base to remove.

Figure 62. Internal Antenna Connection

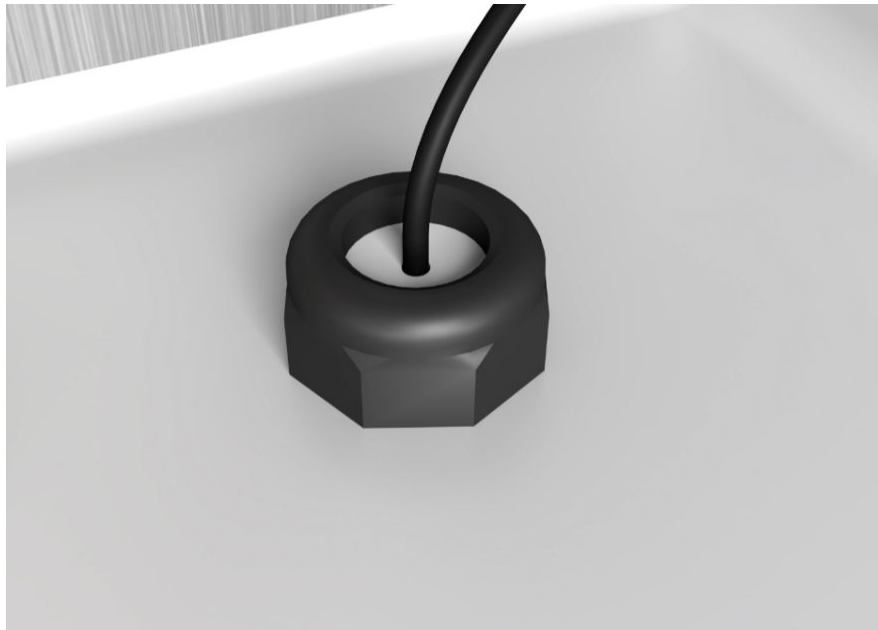


3. Use a power drill or knock-out kit to create a new hole in the bottom of the enclosure. Make sure the new hole is in line with the other conduits leading into the enclosure.



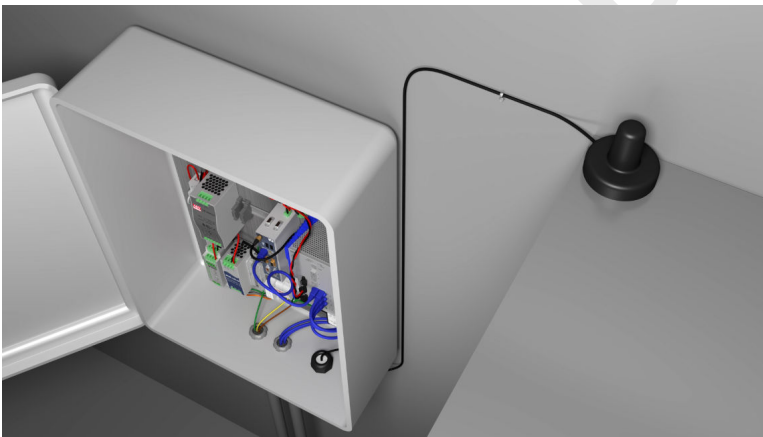
4. Install an appropriately sized water-tight cable gland to seal this new hole.

Figure 63. Coaxial Cable Gland for Antenna



5. Feed the antenna's coaxial cable through the cable gland.
6. Carefully screw the antenna's coaxial into the antenna's terminal slot in the computer.
7. Tighten the cable gland fully to prevent ingress.
8. Close the Standard Tesla Site Controller Enclosure's cabinet door.
9. Place the antenna's magnetic base on an elevated metal surface so that it stands vertically.

Figure 64. Example Relocated Antenna Placement



10.2 Communication Wiring

This section applies to both the Standard Tesla Site Controller and the Large Tesla Site Controller. For cable requirements, see [Communications Cable Requirements on page 50](#).

10.2.1 Terminating Ethernet Cables

Review the following and set up Ethernet connections and terminations as needed per site requirements.



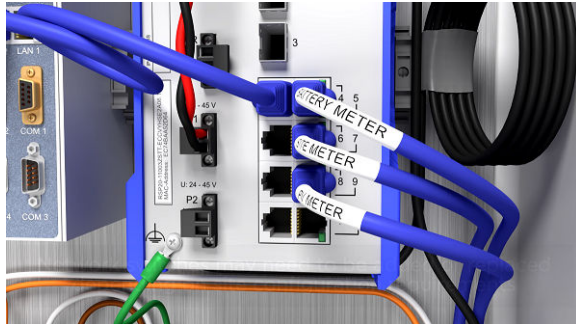
Both the Standard Tesla Site Controller and Large Tesla Site Controller provide two Ethernet ports for communication: LAN 1 and LAN 2.

Figure 65. LAN 2 and LAN 1 Ports on the Standard Tesla Site Controller



- LAN 1 is typically reserved for connection and communication over the Customer Network.
- LAN 2 is typically pre-wired to a network switch for communication on the Tesla Network.

Figure 66. Ethernet Connections to the Network Switch



For networking considerations in dual-controller configurations, see [Computer Networking on page 29](#).



11 Connecting Energy Meters

These sections provide instructions on connecting energy meters. Energy meters must be provided by the contractor.

Every site requires, at a minimum, a battery meter, but the types and number of meters used vary based on Megapack's intended use. Always verify that the meters being installed match the plan of record for the site. See [Energy Meter Design on page 20](#) for details on types of meters.

Failure to install all meters included in the approved plan of record may result in incorrect system performance or inability to meet reporting requirements.

 **NOTE:** Tesla is not responsible for the correct installation of energy meters. The battery system will not operate correctly with incorrect CT installation or wiring to the meters. This is a common point of installation failure and requires careful installer attention.

 **NOTE:** When communication to any of the meters is lost during on-grid normal operation, the system may cease to operate, depending on the control mode, until communication is resumed. If communication is lost, the values reported are the last values read from the meter until communication is resumed.

11.1 Connecting Meters to the Tesla Site Controller

Connect a direct line between the meter(s) and the Tesla Site Controller per the instructions in [Terminating Ethernet Cables on page 98](#).

 **NOTE:** All meters must be connected to the Tesla Site Controller using Ethernet cable terminated according to the T568B Wiring Standard (see [Communications Cable Requirements on page 50](#)).

 **NOTE:** If the site is using multiple external communication devices that use LAN 1 (such as a customer interface, utility monitoring device, or external-network meter), a second Ethernet switch is required. Contact Tesla for details.

11.2 Wiring Voltage Sense Inputs

For each energy meter, tap **all available bus bars**, including neutral if it exists, into the voltage inputs on the meter. Designs vary based on site-specific transformers and meters, so ensure that you follow the guidelines relevant for your site plan:

- **If using potential transformers (PTs):** You should be using a type that has the same number of outputs on its secondary side as the inputs on the primary side. If a neutral bus exists, it should connect to the PT, and a neutral line should flow through to the meter.
- **If the neutral exists (three-phase Wye):** Set the meter's voltage mode to a line-to-neutral type (*Form 9*).
- **If there is no neutral - only three active phases and ground (three-phase Delta):** Set the meter's voltage mode to a line-to-line type (*Form 5*).

 **CAUTION:** Failure to follow these guidelines will void any guarantee of accurate readings and accurate power control.



11.3 Wiring Current Transformer Inputs

To wire current transformer (CT) inputs:

1. Use core CTs only.

Figure 67. Core CT (Example)



2. Verify that the CTs used match the chosen meter's specifications and the plan of record.



NOTE: Tesla is not responsible for the correct installation of meters. The system will not operate correctly with incorrect CT installation or wiring to the meters. This is a common point of installation failure and requires careful installer attention.

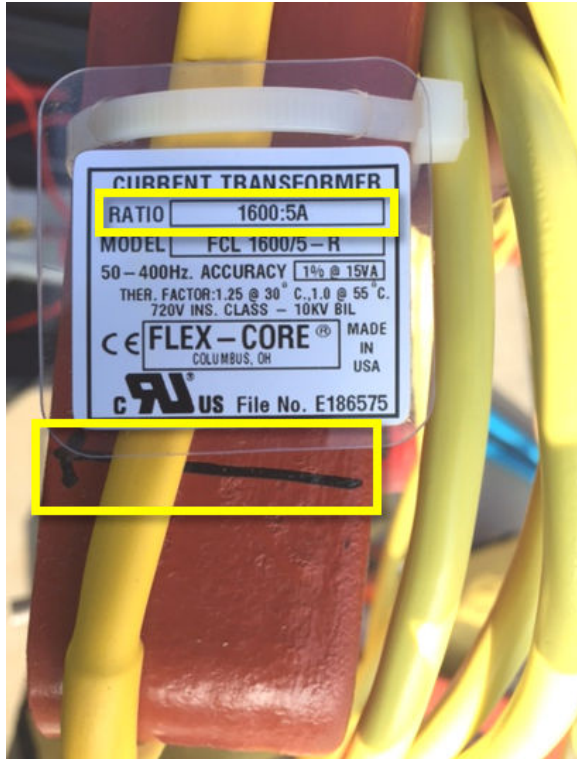


NOTE: Tesla is not responsible for the accuracy of the energy meters. Please consult with the energy meter manufacturer in the event of faulty communications from the energy meters.

3. Read the label on each CT to verify the CT ratio and the arrow that indicates the direction of flow. Record each meter's make, model, serial number, and CT ratio on the *Megapack Deployment Checklists*.



Figure 68. CT Ratio and Arrow



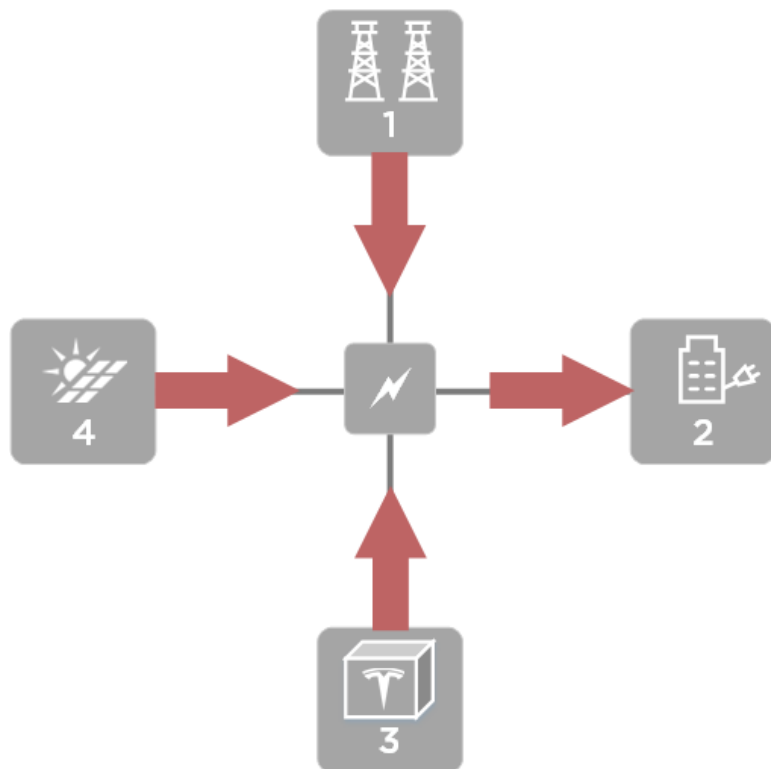
4. Ensure each phase's voltage wiring corresponds to the same phase of current wiring, and the positive and negative polarity of each phase are installed correctly to match the sign convention per the figure below.



NOTE: All meters are phase-sensitive. The battery meter & battery system phase rotations must match. The default phase rotation direction is clockwise (A,B,C) for off-grid systems.



Figure 69. Metering Sign Convention



Arrows indicate direction of flow of power. The middle indicates bus bar (feeders).

1. **Site** - Metering that monitors the feed from the utility or point of common coupling (PCC) for the entire site
2. **Loads** - (Optional) - Metering that solely monitors load power flows
3. **Battery** - Metering that solely monitors battery power flows
4. **Generation** - Metering that solely monitors solar (or wind) generation power flows
5. Install the **battery meter** wiring so that only the battery system output is measured.
6. If using a **site meter**, install wiring so that all customer loads including the battery system output are captured. Site meter wiring is typically installed adjacent to the utility meter or at the site's point of common coupling (PCC).
7. If using a **generation meter**, install wiring so that all solar generation is measured. Some sites might require more than one solar meter; refer to site drawings.
8. If using a **consumption** (facility load) **meter**, install wiring to measure the facility load only (excluding all battery system activity and generation sources).
9. For each meter, perform a resistance check between the positive and negative terminals from each CT:
 - Ensure resistance between each CT pair is ~0 Ohms.
10. On the meter UI, set the CT ratio per the installed CTs and document in the *Megapack Deployment Checklists*.



11.4 Configuring Meters

Once AC-side and Tesla Site Controller wiring is complete, use the following procedure to configure the meter or meters:

1. Turn on the meter.
2. Following the instructions in the meter manual, configure IP address parameters according to the guidelines below:

For Ethernet Using LAN 1:

Contact Tesla for IP address settings.

For Ethernet Using LAN 2:

- Assign the meters to static IP addresses. Tesla recommends that meters are assigned addresses in batches by meter type:
 - Battery = 192.168.90.201 to 192.168.90.209 (first battery meter starting at 201, next at 202, etc.)
 - Site = 192.168.90.210 to 219 (first site meter starting at 210, next at 211, etc.)
 - PV (solar) = 192.168.90.220 to 229 (first PV meter starting at 220, next at 221, etc.)
 - Busway (bus) = 192.168.90.230 to 239 (first busway at 230, next at 231, etc.)
 - Set the Subnet Mask to 255.255.255.0.
 - Set the Default Gateway to 192.168.90.1.
3. To save the IP address settings, turn the meter off, then turned back on.



NOTE: Tesla is not responsible for the correct configuration of energy meters. The battery system will not operate correctly with incorrect input parameters.



12 Starting Up and Shutting Down

12.1 Starting the System

Once Tesla has completed the commissioning and initial start-up of Megapack, the system is ready for operation. You should not need to perform extra steps to start the system.

12.1.1 Re-Energizing Megapack

If Tesla Service or a Tesla-approved entity gives the direction to re-energize Megapack after upstream maintenance, proceed with the steps below for each Megapack to be re-energized.

1. Verify that the equipment and area, including the inside of the Customer Interface Bay and AC bus bar area, is clear of tools, materials, workers, equipment, and debris.
2. Close any open doors and ensure all covers are in place.
3. Remove the lock on the Megapack AC circuit breaker, if needed.
4. Turn the enable switch on, if it was turned off.
5. Reinstate the remote shutdown contact, if it had been removed.
6. Notify operations and on-site personnel before re-energizing.
7. Turn the Megapack AC circuit breaker MOTOR switch from OFF to AUTO.
8. Close the Megapack AC circuit breaker.
9. Close the upstream external circuit breakers.

12.2 Shutting the System Down

 **CAUTION:** Do not use the enable circuit ([Safety Disconnect Features on page 59](#)) (in other words, do not open a Megapack door) as a means to de-energize the system. The enable circuit is meant as an added safety layer for incidental or accidental misuse of the system. When deliberately de-energizing the system, wherever possible always use the Tesla Site Controller to command a soft shutdown as described in the procedures in this section.

12.2.1 Shutting Down in an Emergency

 **DANGER:** If smoke or fire is visible, do not approach the Megapack and do not open any of its doors.

 **CAUTION:** External safety features such as E-stops and upstream breakers differ by region and design. Always be aware of your site's safety design and external safety features.

To shut down the system in an emergency or for unknown behavior:

1. If an external E-stop button or remote shutdown contact to Megapack is present, engage it.
2. If Megapack is serviced upstream by an external AC circuit breaker or disconnect, open the breaker or disconnect.
3. Only if safe to do so and if needed, proceed with de-energizing the Megapack ([De-Energizing Megapack on page 109](#)).
4. Contact Tesla at IndustrialStorageSupport@tesla.com to advise that the system has been shut down.

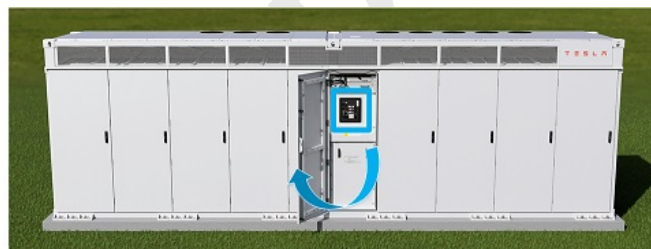
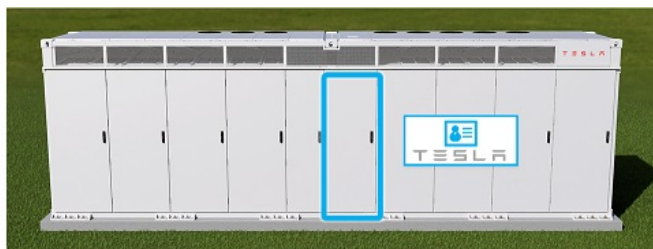


NOTE: Refer to the *Lithium-Ion Battery Emergency Response Guide* for details on response to a hazardous event.

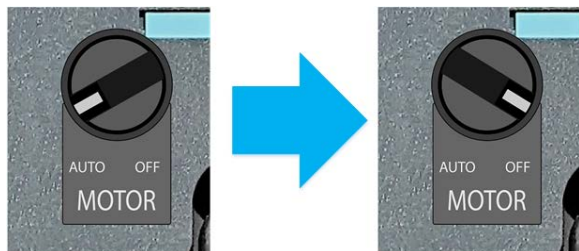
12.2.2 Performing a Planned Shutdown, Single Megapack Site

Perform this procedure at sites with a single Megapack:

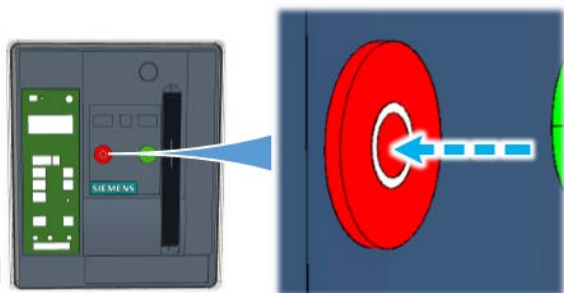
1. If possible, command a soft shutdown of the system using the software via the Tesla Site Controller by commanding a power set point of zero. If the site manager does not have the means to command power, contact IndustrialStorageSupport@tesla.com to request the power shutdown command.
2. Open the site or external circuit breaker or disconnect (if one is present).
3. Assess the steps in [Critical Door-Opening Considerations on page 71](#) and proceed if advised by Tesla.
4. Open the Customer Interface Bay door to access the Megapack AC circuit breaker:



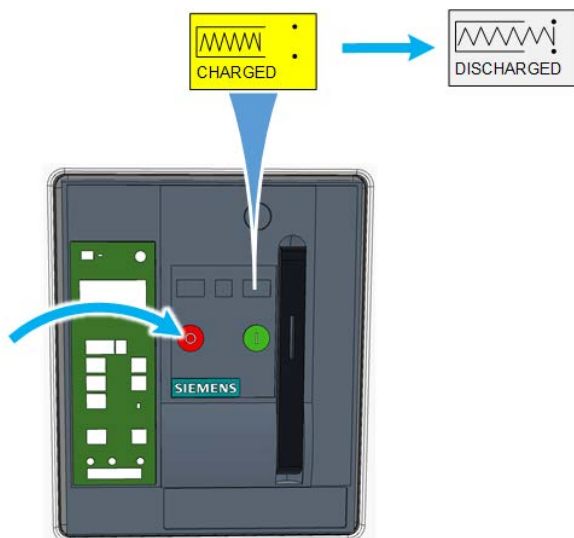
5. Switch the Megapack AC circuit breaker MOTOR from AUTO to OFF:



6. Open the Megapack AC circuit breaker by pressing the red button firmly until it clicks:



7. Verify that the Megapack AC circuit breaker is open: The *CONTACTS* window displays green and *OPEN* and the *SPRING* window displays *DISCHARGED*:



8. If planning to perform work, proceed with de-energizing Megapack ([De-Energizing Megapack on page 109](#)).

12.2.3 Performing a Planned Shutdown, Multiple Megapack Site

Perform this procedure at sites with multiple Megapacks:

1. Determine whether the entire site should be shut down, or only one Megapack.
 - **If shutting down power to the entire site**, command a soft shutdown of the system using the software via the Tesla Site Controller. If the site manager does not have the means to command power, contact IndustrialStorageSupport@tesla.com to request the power shutdown command.
 - On-grid system: command a power setting of zero
 - Microgrid system: change *Island Control Mode* to 2
 - **If shutting down power to only one Megapack**, the power commands on the Tesla Site Controller do not require a change (but a change is permitted if desired).
2. Isolate AC power at the appropriate circuit breaker:
 - **If shutting down power to the entire site**, open the site-wide external circuit breaker or disconnect to remove grid power.
 - **If shutting down power to one Megapack**, open the nearest upstream AC branch breaker (if a true breaker is present) that isolates that block.
3. Proceed with de-energizing, if required ([De-Energizing Megapack on page 109](#)).
4. **Wait** for Tesla Service or a Tesla-approved entity to arrive or provide further direction.

12.2.4 Shutting Down for Storage

This procedure, which involves removing the remote shutdown jumper in the customer I/O area, is only to be performed when a Megapack needs to be shut down for an extended period of time (more than 24 hours and up to one year).

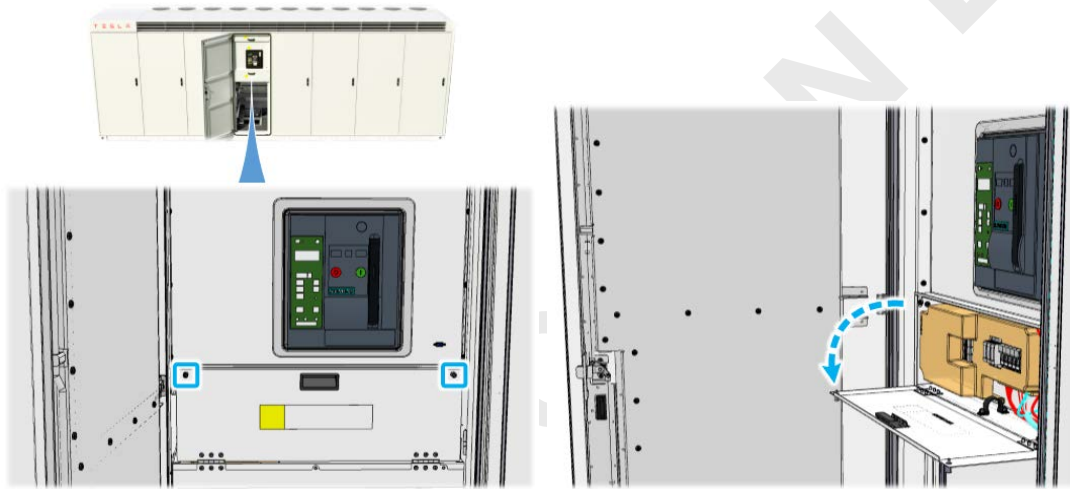
WARNING: Active work cannot be performed inside Megapack while it is in storage. If active work is required, refer to [De-Energizing Megapack on page 109](#).

1. Ensure that conditions, including duration and temperature range, will abide by the requirements in the *Megapack 2 XL Transportation and Storage Guidelines* for the duration of storage.

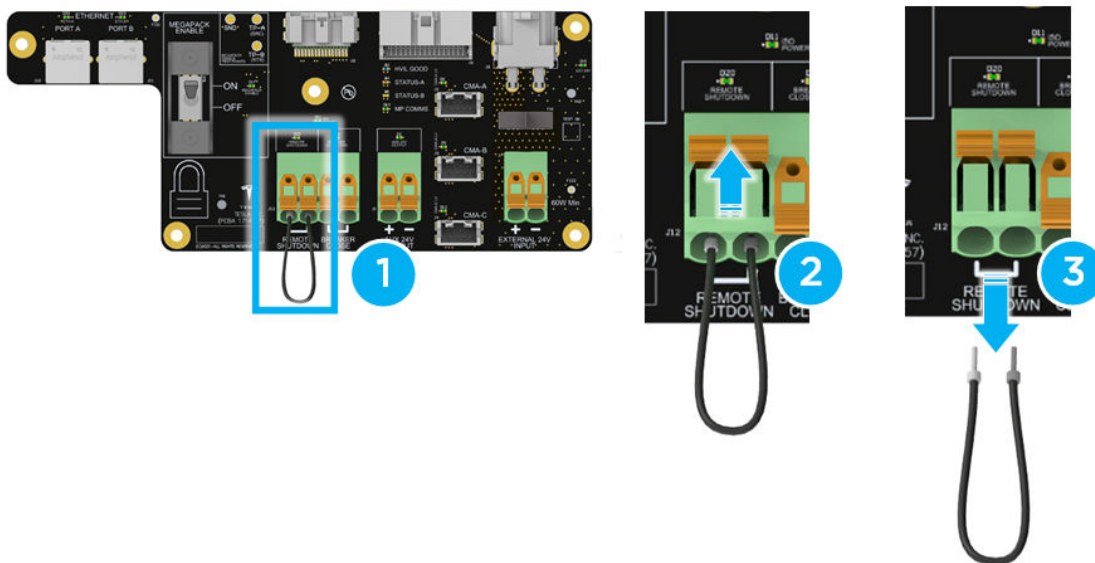


2. Charge Megapack to at least 50% state of energy.
3. Contact Tesla at IndustrialStorageSupport@tesla.com:
 - a. Mention in the email that the system will be shut down soon.
 - b. State the amount of the time the system is expected to be idle.
 - c. Request verification that the system is charged appropriately.
 - d. Obtain Tesla's approval before proceeding.
4. Shut down per the appropriate procedure (for example, [Performing a Planned Shutdown, Single Megapack Site on page 106](#)).
5. Remove the remote shutdown jumper:
 - a. Use a T30 Torx bit to remove the 2 screws securing the customer I/O area cover to the enclosure. Grab the handle and pull outward, then downward to expose the customer I/O area.

Figure 70. Opening the Customer I/O Area cover



- b. On the customer interface board, at the remote shutdown terminals (1), lift up the terminal levers (2), and pull out and remove the jumper (3):



- c. If, per site design, the jumper is not present and a remote shutdown circuit has been provided instead, remove the wires.



CAUTION: When placing Megapack into storage, do **not** turn the enable switch off.

12.2.5 De-Energizing Megapack

If you need to perform work inside Megapack, open and lock the Megapack AC circuit breaker, turn the enable switch off, and verify de-energization as described below.



DANGER: Refer to [IMPORTANT SAFETY INFORMATION on page 3](#) for full information on safety warnings and PPE recommendations before beginning any work on Megapack.



CAUTION: Do not follow this procedure to place Megapack into storage. See [Shutting Down for Storage on page 107](#).



CAUTION: Where a block of more than one Megapack is connected to the same transformer, repeat these procedures for each Megapack on the block.



NOTE: Only Tesla or a Tesla-approved entity shall perform any corrective maintenance within the Megapack.

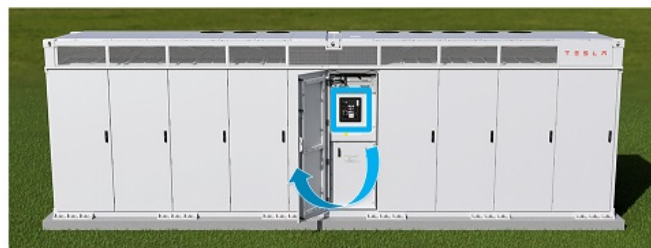
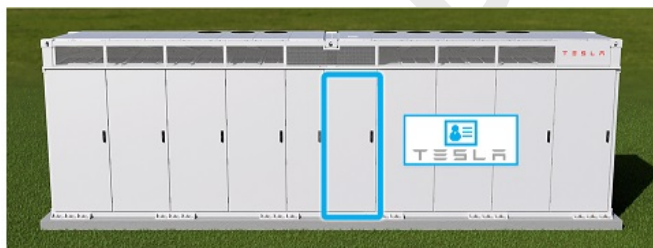
12.2.5.1 Opening and Locking the Megapack AC Circuit Breaker

You must open and lock the Megapack AC circuit breaker before proceeding with de-energizing Megapack for active work.

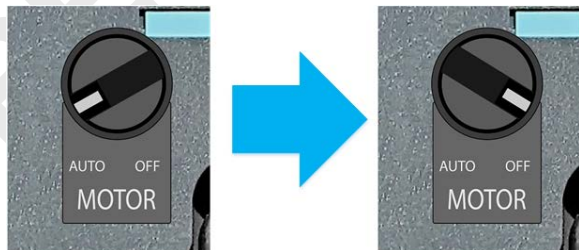


WARNING: Before performing these procedures, ensure that external AC power is removed from Megapack.

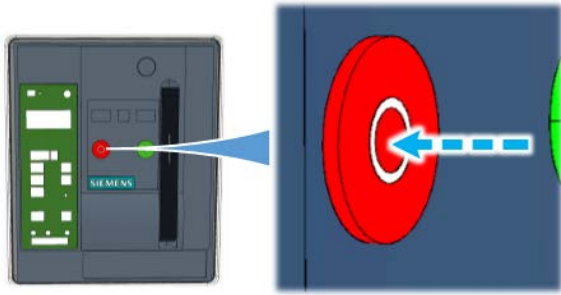
1. Assess the steps in [Critical Door-Opening Considerations on page 71](#) and proceed if advised by Tesla.
2. Open the Customer Interface Bay door to access the Megapack AC circuit breaker:



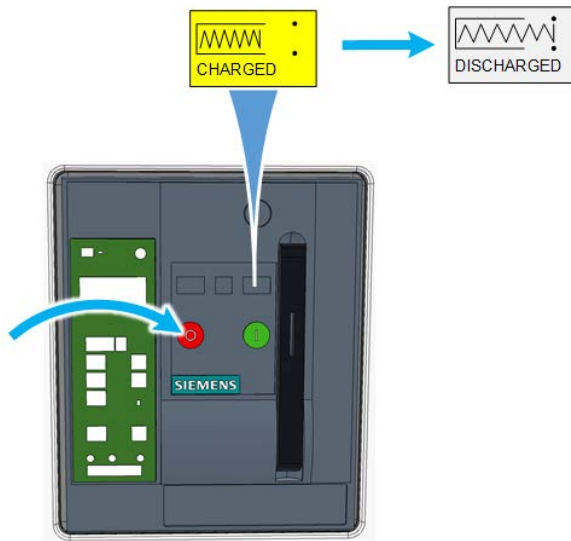
3. Switch the Megapack AC circuit breaker MOTOR from AUTO to OFF:



4. Open the Megapack AC circuit breaker by pressing the red button firmly until it clicks:



5. Verify that the Megapack AC circuit breaker is open: The *CONTACTS* window displays green and *OPEN* and the *SPRING* window displays *DISCHARGED*:



6. Push up on the spring-loaded lever (#4) to unlock the lock ring (#5):



Opened Megapack AC circuit breaker. #1: MOTOR switch to OFF. #2: OPEN button pressed. #3: CONTACTS window - green and OPEN. #4: Spring-loaded lever. #5: Lock ring.

7. Use a flat-head screwdriver to pull out the lock ring:



8. Insert the lock, lock it, and secure the AC circuit breaker:



NOTE: This procedure physically depresses the *Open* circuit button and forces the circuit breaker to stay open.

9. Proceed with **turning the enable switch off**, below.

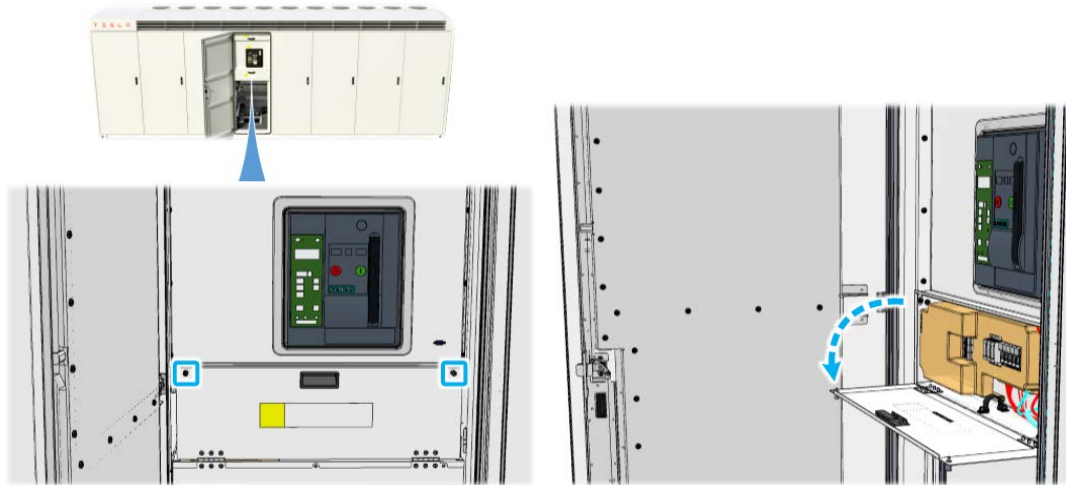
12.2.5.2 Turning the Enable Switch Off

The enable switch in the customer I/O area can interrupt the enable circuit and prevent the system from energizing. This provides a lockable isolation point for procedures that involve actively working inside Megapack.

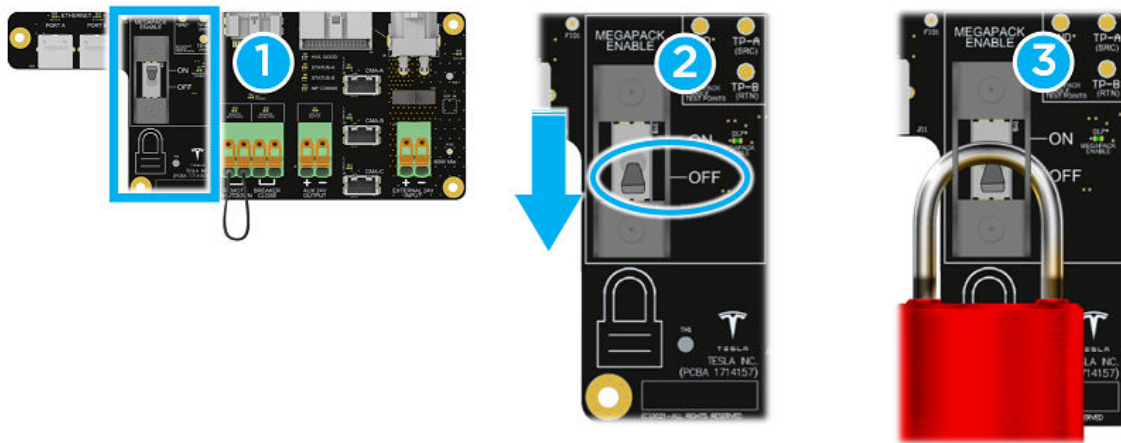


CAUTION: Do not perform this procedure to place Megapack in storage. See [Shutting Down for Storage on page 107](#).

1. Use a T30 Torx bit to remove the 2 screws securing the customer I/O area cover to the enclosure. Grab the handle and pull outward, then downward to expose the customer I/O area:



2. The Megapack enable switch is located on the customer interface board (1). Press **down** to turn the enable switch to the **off** position (2), and apply a lock (3):



CAUTION: If active work needs to pause for 1 hour or longer, turn the switch back on until active work resumes.

3. Proceed with **verifying de-energization**, below.

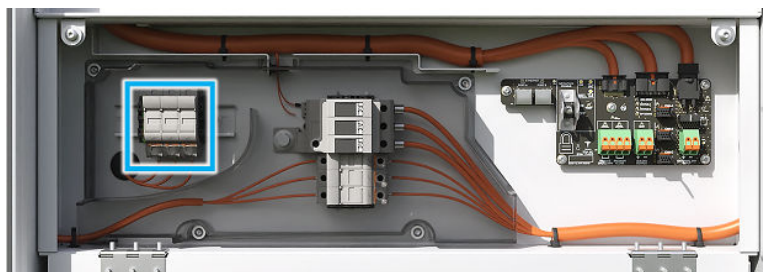
12.2.5.3 Verifying De-Energization

To confirm that Megapack has been de-energized and is safe for active work, **you must verify de-energization by measuring voltage.**



NOTE: This final check for dead should only take place if all previous steps were successfully completed.

1. Still working in the customer I/O area per above, open the fuse holder for the line-side (grid-side) tap:



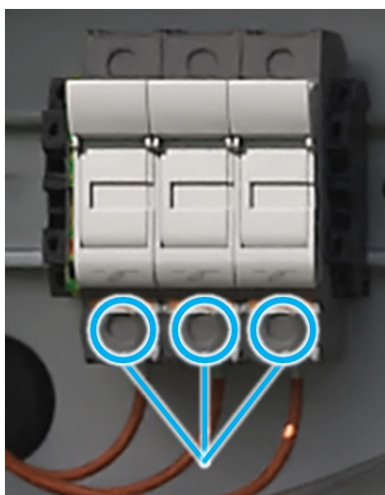
2. Check the AC voltage inside of the fuse holder:

- a. Verify a voltmeter on a known live source.



NOTE: The voltmeter should be Class IV rated for 1000 V AC.

- b. Check line-to-line and line-to-ground at the line-side (grid-side) test points as shown below.



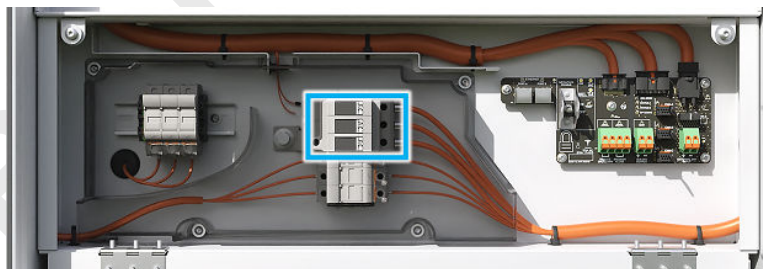
- c. Confirm that the voltage is 0 V AC.



WARNING: You may initially measure greater than 0 V AC due to residual capacitive energy. Stop and wait five minutes before testing again. **Do not proceed unless you have measured 0 V AC.**

- d. Re-verify the voltmeter on known live source.

3. Check the load-side (inverter-side) AC voltage at the MOV:

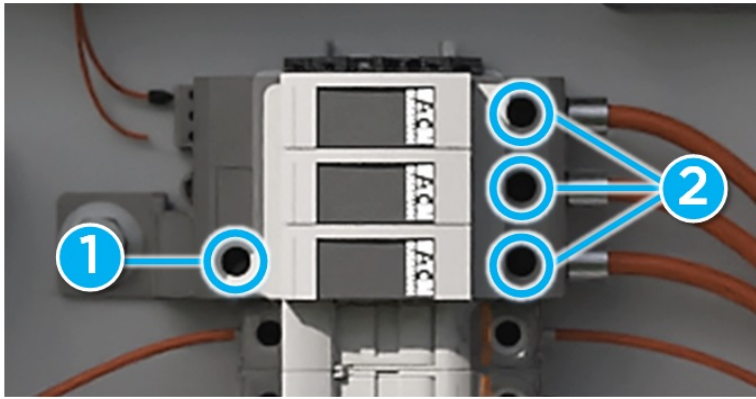


- a. Verify a voltmeter on a known live source.



NOTE: The voltmeter should be Class IV rated for 1000 V AC.

- b. Check line-to-line and line-to-ground at the load-side (inverter-side) test points as shown below.



1. Ground

2. Test points

c. Confirm that the voltage is 0 V AC.



WARNING: You may initially measure greater than 0 V AC due to residual capacitive energy. Stop and wait five minutes before testing again. **Do not proceed unless you have measured 0 V AC.**

d. Re-verify the voltmeter on known live source.

4. Repeat Step 3 on page 114, now checking for DC voltage.



NOTE: You will measure a voltage of up to 50 V DC. This is normal and expected. If voltage above 50 V DC is detected, stop and contact Tesla.



13 Regulatory Compliance

13.1 Signage Requirements

Signage must be posted in approved locations in accordance with local codes and standards.

13.2 General Compliance Information

Megapack will be compliant with the industry's most stringent global safety standards for energy storage systems, and compliant to major market grid codes. *Megapack 2 XL Compliance Packet*, a full compliance packet of completed certifications in all regions, will be available on the Tesla Partner Portal.

13.3 Environmental Compliance

Each Megapack contains coolant and refrigerant in its thermal system ([Thermal System on page 34](#)). These substances are built into the Megapack and do not need to be added at the time of installation at site. Depending on the number of Megapacks installed on a site, storage, use and handling of these substances during maintenance events may require reporting, hazard management plans, or containment procedures as required by local codes.

13.3.1 Registration, Evaluation, Authorization and Restriction of Chemicals (REACH)

The Regulation (EC) No 1907/2006 of the European Parliament and of the Council of December 18, 2006 concerning the Registration, Evaluation, Authorization, and Restriction of Chemicals (REACH) entered into force on June 1, 2007. Tesla agrees with the purpose of REACH, which is to ensure a high level of protection of human health and the environment. Tesla is compliant with all applicable requirements of REACH.

The registration requirements do not apply to Tesla, since it is neither a manufacturer nor an importer of preparations into Europe.

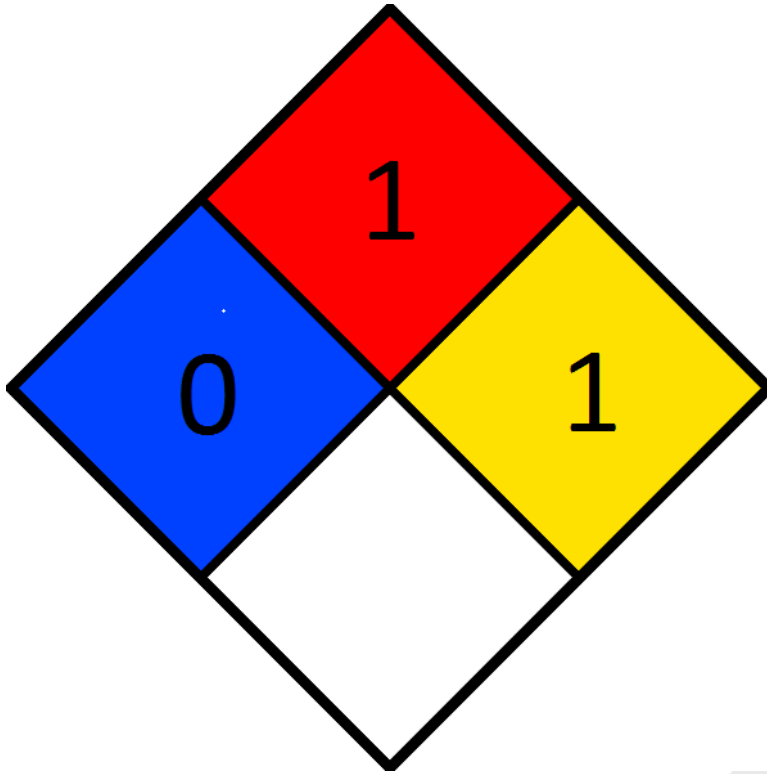
However, product (article) manufacturers or importers into Europe are obligated under Article 33 of REACH to inform recipients of any articles that contain chemicals on the Substances of Very High Concern (SVHC) candidate list above a 0.1% concentration (by weight per article). As of July 2021, Megapacks manufactured and marketed by Tesla do not contain substances on the REACH SVHC candidate list in concentrations greater than 0.1% by weight per article. Tesla continues to monitor the developments of the REACH legislation and will communicate with our customers according to the requirement above.

13.4 NFPA Compliance

For installations that require an NFPA placard, please refer to the figure below:



Figure 71. NFPA Placard



CAUTION: Water is the recommended suppressant in case of a fire involving a lithium-ion battery. Lithium-ion batteries **do not contain any solid metallic lithium** (a water reactive material); thus, the use of water is appropriate, and will not exacerbate a fire involving lithium-ion cells. For comprehensive fire response information, refer to the *Lithium-Ion Battery Emergency Response Guide*.

13.5 Electromagnetic Compatibility (EMC)

The Standard Tesla Site Controller has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case users will be required to correct the interference at their own expense.



Revision History

Revision	Date	Details
1.3	August 23, 2022	- Added isolation transformer guidance for resonance mitigation (Resonance of Devices Connected to Common 480 V AC Bus on page 28) - Modified on-site maintenance infrastructure topic to point to new app note (On-Site Maintenance Infrastructure on page 39)
1.2	August 8, 2022	- Added guidance on using line reactors (Resonance of Devices Connected to Common 480 V AC Bus on page 28) - Added water supply design reference (Water Supply on page 30) - Decreased wireway height (Wiring Interface Summary on page 49) - Provided circuit breaker option codes for default settings (Megapack AC Circuit Breaker on page 61) - Modified voltage ride through settings (Voltage Ride Through on page 54) - Updated label renders (Megapack Labels on page 15) <i>This revision was not published to Partner Portal.</i>
1.1	June 20, 2022	- Clarified support for multiple meters of the same function (Energy Meter Functions on page 21) - Clarified <i>Advanced SEL-735</i> meter requirement for projects with specific response time requirements (Energy Meter Design on page 20) - Added reference to Megapack AC circuit breaker control over DNP3 (Megapack Breaker Control on page 63) - Modified specifications and language in On-Site Maintenance Infrastructure on page 39 - Provided center of gravity location (Megapack Center of Gravity on page 32) - Provided mass calculation (Megapack Mass on page 32) - Reduced total mass of refrigerant (Thermal System on page 34) - Updated installation template part number (Installation Accessories on page 75) - Updated enable circuit example graphics (Safety Disconnect Features on page 59) - Updated example wiring interface graphic and key (Wiring Interface Summary on page 49) - Clarified that the arc flash label must be affixed to the outside of the Customer Interface Bay Door (Providing the Arc Flash Label on page 85)
1.0	April 22, 2022	Initial revision. PRE-PRODUCTION INFORMATION - SUBJECT TO CHANGE.



CONFIDENTIAL INFORMATION - SHARED UNDER NDA ONLY

Megapack 2 XL Design and Installation Manual - Revision 1.3

Last Updated August 23, 2022

PRE-PRODUCTION DRAFT